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UNIVERSITY OF SOUTHAMPTON

Ion-Implantation for Defect Engineering in Silicon

by

Martin Wearn

A thesis submitted in partial fulfillment for the
degree of Doctor of Philosophy

in the
Faculty of Engineering and Physical Sciences
School of Electronics and Computer Science

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ABSTRACT

FACULTY OF ENGINEERING AND PHYSICAL SCIENCES
SCHOOL OF ELECTRONICS AND COMPUTER SCIENCE

Doctor of Philosophy

by Martin Wearn

Over the years integrated circuits have been increasing in performance capabilities as a result of both an increasing transistor count according to the self-fulfilling Moore's law, and a continuous reduction in feature size. At such small sizes, the time delay from metal interconnections far outweighs the delay across transistor gates, restricting the global chip operating speed, suffering high power loss and low bandwidth. Photonic interconnections are a promising solution to this problem, and benefit from wavelength division multiplexing capabilities to increase bandwidth. Silicon is a favourable platform since it builds upon mature CMOS fabrication technologies and processes to reduce the economic and technological costs. It's transparency and high optical confinement, caused by the strong refractive index contrast against its natural oxide, is beneficial for signal routing but prevents generation and detection of light at communication wavelengths. This thesis focusses on a solution to the detection problem by intentionally introducing crystalline defects to extend silicon's absorption capabilities beyond the intrinsic limit.

He⁺ ion implantation has been incorporated into a free-space photodetector to introduce vacancy-type defects and shift the absorption spectrum towards longer wavelengths. It was designed with the aid of a single-defect modelling approach to incorporate the enhanced photogeneration rate into a TCAD simulation model. This implantation has caused a 3.9x increase in responsivity to a wavelength of 1550 nm in experimental results, while reducing the intrinsic response to 633 nm by 29 %. B⁺ was also investigated, but it was found to compensate existing dopants and prevent ohmic contact if the metal-semiconductor interface is exposed during implantation. Then a waveguide integrated MSM photodetector is designed and fabricated, based upon a 340 nm SOI platform. The defect-mediated photoresponse is improved from the first design after a literary analysis of ion implantations. Ar⁺ is chosen since it introduces no doping, is a larger ion, and is implanted at an energy of 300 keV and dose of $1 \times 10^{13} \text{ cm}^{-2}$ to increase the concentration of damage produced. A TiN barrier-enhancement layer is also included to increase the Schottky barrier height up to 2.07x compared to an Al-only contact, and up to 1.37x compared to a Ti/Au enhancement layer which was also fabricated. The barrier height was further increased by 74 mV by the ion implantation, and dark current has been reduced further by substrate biasing up to -20 V.

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Nomenclature

α -Si	Amorphous silicon
AC	Alternating current
APD	Avalanche Photodiode
BHF	Buffered hydrofluoric acid
BOX	Buried oxide
CF	Confinement factor
CMOS	Complimentary metal oxide semiconductor
DBR	Distributed bragg reflector
DI	De-ionized water
DLTS	Deep-level transient spectroscopy
E	Electric field
EQE	External quantum efficiency
E_c	Conduction band energy level
E_t	Trap energy level
E_v	Valence band energy level
FCA	Free-carrier absorption
FDE	Finite difference eigenmode
FET	Field-effect transistor
FIB	Focussed ion beam
GPIB	General purpose input bus
HC-NCF	Hollow-core negative curvature fibre
HDMS	Hexamethyldisilazane
HF	Hydrofluoric acid
OH^-	Hydroxide ion
InF_3	Indium trifluoride
IBC	Ion beam centre
IC	Integrated circuit
ICP	Inductively coupled plasma
IQE	Internal quantum efficiency
IR	Infra-red
KOH	Potassium hydroxide
LOCOS	Local oxidation of silicon

MMA	Methyl methacrylate
MSM	Metal-semiconductor-metal
n_{eff}	Effective refractive index
NEP	Noise equivalent power
NH ₃	Ammonia
NMP	N-methyl-2-pyrrolidone
OCV	Open-circuit voltage
PAS	Positron annihilation spectroscopy
PECVD	Plasma enhanced chemical vapour deposition
PMMA	Poly(methyl methacrylate)
PVD	Physical vapour deposition
RC	Resistance-capacitance product time constant
RIE	Reactive ion etching
SACM	Separate absorption, charge and multiplication
SiO ₂	Silicon dioxide
SiN _x	Silicon nitride
Si ₃ N ₄	Stoichiometric silicon nitride
SEM	Scanning electron microscope
SNR	Signal-to-noise ratio
SRH	Shockley-Read-Hall
SRIM	Stopping range of ions in matter
SUB	Substrate
SUSPRE	Surrey University Sputter Profile Resolution from Energy deposition
SWIR	Short wavelength infra-red
TCAD	Technology computer-aided design
TE	Transverse electric
TM	Transverse magnetic
V2	Silicon divacancy lattice defect
VOA	Variable optical attenuator
WDM	Wavelength division multiplexing
ZrF ₄	Zirconium tetrafluoride

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Chapter 1

Introduction

1.1 Silicon photonics in the IC industry

Optical fibres have been the backbone of global communications since the invention of the optical fibre amplifier in the 1980s, since they provide a much higher speed and bandwidth than coaxial cables used previously [1]. Due to a limited operating frequency incurred by the resistance-capacitance product (RC) time constant of copper interconnections in on-chip circuitry, there is a drive to extend optical links to within integrated circuits (ICs). This approach is necessary since reducing the resistance of copper wires by increasing their cross-sectional area is ineffective due to the current density concentrating on the surface [2]. Additionally, optical fibres incur no electro-magnetic interference and have non-conductive cables which simplify isolation from ground [2]. While transmission loss does not depend on modulation rate, allowing scaling of operating speed and length [2]. Optical connections currently stretch from global communications to the kerbside, with a goal of extending further into homes to incur lower losses and a reduction of the bottleneck from converting between optical and electrical domains [3].

Silicon does present some issues with use in optical applications. It's indirect bandgap makes stimulated photon emission difficult, preventing laser light sources, while it's transparency between wavelengths of around 1 and $8.5\text{ }\mu\text{m}$ prevents an intrinsic detector at telecommunication wavelengths [4, 5]. Second-order (Pockels) nonlinearity is prevented by the inversion symmetry in the crystalline structure of silicon, preventing the electro-optic effect commonly used for modulators [4, 6]. While two-photon absorption is significant at telecommunication wavelengths, reducing the efficiency of third-order (Kerr) nonlinear processes [4, 6]. However, this is still high enough to allow the creation of four-wave mixing devices within a much shorter distance than would be needed in silica fibres [4, 7]. These issues have previously driven photonics towards III-V semiconductors such as InP since it is a direct-gap material [8].

Even with these effects, silicon is still a favourable platform for implementing photonic integration. The mature infrastructure available for design and processing of complementary metal-oxide-semiconductor (CMOS) circuitry in silicon can also be used for photonics using much of the same processes, reducing the cost of developing and implementing this technology, and allowing monolithic integration of electronic and photonic ICs, without separate production of discrete components. The high refractive index of silicon compared to its native oxide (SiO_2) enables tight optical confinement and nanometer-scale dimensions, allowing for the dense integration of devices and tight bending radii for routing of waveguides as well as a reduced on-chip signal latency [2]. Optical waveguides also have lower power transmission compared to copper wires, and signal density is higher by means of wavelength division multiplexing (WDM), which allows multiple signals to be transmitted along the same line simultaneously by sending each at a different wavelength.

Silicon photonic technology requires a number of devices for generation, transmission, detection, modulation and manipulation of optical signals [8]. Due to the indirect and wide band gap of silicon, lasers typically use III-V compounds such as GaAs or InP, however quantum confinement or rare-earth doping have enabled the creation of silicon-based light sources [8]. Detection converts an optical signal into an electrical one by the absorption of incoming photons to generate electron-hole pairs, as discussed further in Sections 2.1 and 2.2. Transmission is achieved through waveguides which use the concept of total internal reflection to confine and guide light along a channel, discussed further in Section 2.3. While different types of modulator can control the properties of a propagating signal, such as mach-zehnder interferometers for measuring changes in a signal relative to a reference, or electroabsorption modulators for controlling intensity [9].

1.2 Moving towards longer-wavelength devices

Applications for photonics in the wavelength range beyond visible light are quite varied. One application comes from the field of telecommunications, where fibre optics already form the backbone of international data communication [10]. The major increase in broadband connections over the past 15 years combined with an ever increasing demand for higher speeds is pushing a need for larger capacity networks [11]. This is further increased by progression towards a more connected and networked society [12]. Extending communication bands towards longer wavelengths will allow for more channels to become available beyond the L band (1565 - 1625 nm) [14].

The choice of wavelengths used in communication systems are driven by the optical fibres used to transport the information. The three primary telecommunication windows used today are around 800 - 900 nm, 1260 - 1360 nm (O-band) and 1530 - 1565 nm (C-band), illustrated Figure 1.1 [14]. The first comes from a minimum attenuation of optical fibres

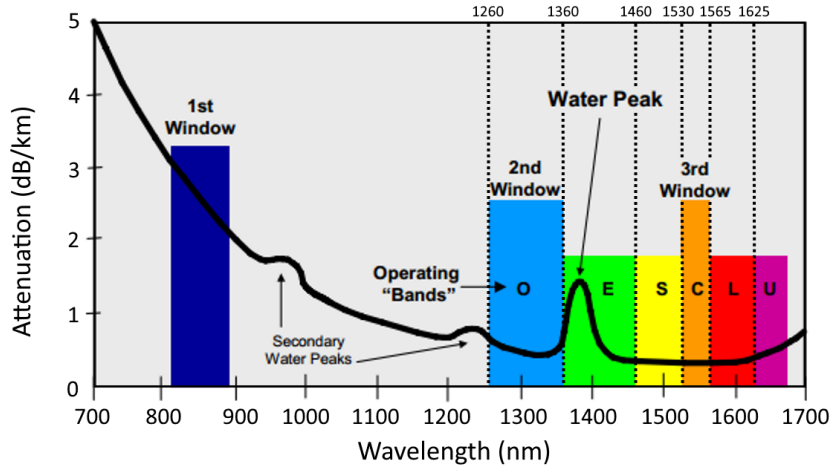


FIGURE 1.1: Attenuation in silica fibres, indicating optical communication wavelength bands. Adapted from [14,15]

used in 1970s - 1980s, the second from a region of zero fibre dispersion in silica fibres and has a lower attenuation than the first, while the third offers the lowest attenuation of about 0.26 dBkm^{-1} but is the most costly and experiences some dispersion [1]. This final minimum arises from the combination of Rayleigh scattering, which is inversely proportional to the fourth power of wavelength, and the IR absorption tail. Peaks occur as a result of OH^- contamination from water, such as that around 1385 nm , but modern techniques have significantly reduced these effects [1].

Recent research has progressed development further into the short and mid wavelength infrared bands (SWIR $1.4 - 3 \mu\text{m}$, MWIR $3 - 8 \mu\text{m}$) [16]. Hollow-core negative curvature silica fibres (HC-NCF) have been explored within the spectral range of 800 nm to $4.5 \mu\text{m}$, showing a minimum attenuation of 24.4 dBkm^{-1} at a wavelength of $2.4 \mu\text{m}$, as shown in Figure 1.2a [17,18]. Longer wavelength fibres can be produced using ZrF_4 or InF_3 for operation at $2 - 3.7 \mu\text{m}$ and $3.3 - 4.2 \mu\text{m}$ respectively, as shown in Figure 1.2b. While thulium-doped fibre lasers can provide a light source at $1.8 - 2.1 \mu\text{m}$ [19], and on-chip routing has been achieved at $2.9 - 3.9 \mu\text{m}$ using photonic crystal waveguides [20]. Some other devices demonstrated around this range are: Mach-Zehnder interferometers ($3.7 - 3.8 \mu\text{m}$) [21], Si waveguides ($3.8 \mu\text{m}$) [22] and GaSb-based photodetectors ($2 - 4 \mu\text{m}$) [22]. These show a strong potential for photonic systems operating within these regions which allow for a broader range of applications.

Such applications could include chemical and biological sensing for lab-on-a-chip integrated sensors, due to the strong vibrational absorption of some important molecules [4,24,25]. This includes double-bond functional groups between $5 \mu\text{m}$ and $8 \mu\text{m}$ ($\text{C}=\text{O}$, $\text{C}=\text{N}$, $\text{C}=\text{C}$) [26]. This would be beneficial for gas detection systems, bio-agent and medical sensing and environmental monitoring [4,25]. The Low-cost and compact sensors this provides would enable a more widespread adoption of the technology [4]. Optical detection is possible here as chemicals change the refractive index of their host liquid,

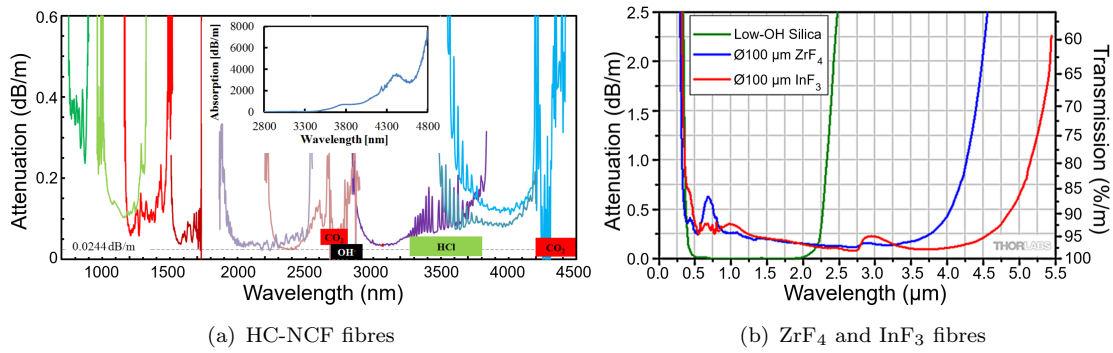


FIGURE 1.2: a) Attenuation curves for the fundamental mode of first transmission band in different HC-NCF silica fibres. Coloured bands indicate absorption features for the molecules indicated. Inset shows material absorption of F300 synthetic fused silica. b) Attenuation curves for 100 μm core ZrF₄ (blue) and 100 μm core InF₃ (red) as compared to a low OH⁻ silica fibre (green). Sourced from [17] and [23] respectively.

and incorporating this as cladding material will change the effective index of a propagating optical mode, allowing detection of chemical concentration [24]. This approach has been demonstrated by S T Fard et. al., A Densmore et. al. and K De Vos et. al. [8, 24].

Common detection methods use material compounds such as InGaAs or HgCdTe which are sensitive from about 0.7 - 1.7 μm and 2 - 16 μm respectively for commercially-available devices, but may require additional cooling to achieve the intended spectral range [27]. Improved CMOS compatibility can be achieved with germanium, which has an intrinsic absorption of 0.8 - 1.8 μm (up to 40 μm in applications allowing very low temperature operation [27]). It can be epitaxially grown on silicon and the band gap can be controlled with tensile stress, while SiGeSn compounds can be used for 2 - 2.5 μm detection with band gap tunability by material composition [4, 22]. However, it requires multi-stage fabrication and a high thermal budget for reduction of dislocation defects at the growth interface, which causes a high dark current [28]. One drawback of current longer wavelength systems is a lack of complete integration, since they are made up of numerous discrete components [4]. This makes them bulky, expensive, and in need of complex fabrication techniques for the mix of materials used. A benefit of silicon photonics is the ability to create fully-integrated chips containing all the essential optic components using processes already available in mature CMOS foundries, making a more compact and cheaper solution [29]. Co-integration with electronic processing into the same chip will further enhance this.

1.3 Photodetection in silicon

Detection of photons in any material can occur when incident light has sufficient energy to promote an electron from its resting state in the valence band (E_v), to a more energetic state in the conduction band (E_c), generating an electron-hole pair in the

process. In a photoconductor this reduces device resistance, altering the voltage across it, while in a photodiode an electric field across a p-n junction draws them apart to create a photocurrent in external circuitry [30]. This junction is formed by an abrupt interface between different regions of silicon doped to be p-type (majority carriers are holes) on one side, and n-type (majority carriers are electrons) on the other side, with the electric field contained within the carrier-depleted region between them. The forbidden gap between the energy bands set the minimum required photon energy, which in silicon is 1.12 eV. Outside of this electric field, the majority of pairs generated cannot travel far enough to reach the contacts before recombining and dissipating the energy as a phonon [31]. Therefore devices need to restrict the creation of absorbing regions to the volume where an electric field exists, to reduce losses and promote a good signal-to-noise ratio (SNR).

During operation, a device is considered photovoltaic when only the built-in voltage across the junction is used to generate the field, whereas under application of a reverse bias the device becomes photoconductive. The first case experiences an increase in voltage across the contacts under illumination, such as in solar cells, while the second case induces a current flow, such as in photon detectors. Photovoltaics exhibit lower noise due to near zero-current operation but have a slower response, whilst photoconductive devices have a faster and linear response at the expense of a minimum current flow termed the dark current [32]. To aid extraction of carriers, devices are often used in photoconductive mode to increase the internal field. An exception to this is in solar cells since they are used as energy sources and therefore shouldn't require the application of a voltage to function.

An alternative photodetector structure can be created using the depletion region formed at the interface between a metal and semiconductor, with the field, and consequential absorption, occurring within the semiconducting material. Since there are two terminals, there are two depletion regions. These require a sufficient bias to be applied to expand these regions enough to bridge a conducting path between the contacts and allow current flow. The depth of this region is also much shallower than in a p-n junction, allowing for a lower capacitance and therefore a potentially higher operating speed. Generated carriers are extracted by the field as before, but dark current flow is controlled by the Schottky barrier height, making the interface quality and material choice important.

Since silicon will intrinsically only absorb light up to a wavelength of 1.1 μm , it will not operate at telecommunication wavelengths or into the mid-infrared regime. Extrinsic absorption can be incorporated into silicon by creating defects in the crystal lattice which introduce mid-gap energy levels within the material. This alleviates the issues associated with integrating foreign materials by careful selection of ion species, and has provided detection at 1.1 - 1.75 μm [33], 1.9 μm [34] and 2 - 2.5 μm [35]. This approach benefits from being both CMOS compatible and having a much simpler fabrication with a lower thermal budget.

1.4 Aims and objectives

The aim of this thesis is to investigate the nature of point defects within the silicon crystal lattice and how they can be used to extend the material absorption beyond the intrinsic detection limit. Upon designing a microfabrication process which allows the creation of these defects at a sufficient quantity, it is added into an existing process flow to demonstrate the integration approach needed. The consequential ion implantation process is performed using either He^+ or B^+ since a low energy is required to reach an optimal depth. It has been suggested that; the first passivates the vacancies created as a part of the implantation damage to make them more thermally stable; while the second has been shown to work in waveguide-integrated detectors in literature.

The first device structure is a free-space illuminated photodetector, which benefits from a high fabrication tolerance (due to its large dimensions) and coupling wavelength insensitivity (due to the lack of grating couplers) to allow for comparison of the intrinsic and extrinsic response. Once fabricated, a fibre laser operating at a wavelength of either 633 nm or around 1550 nm is scanned over the surface for visualisation of the effect of defect-mediated carrier generation across the surface area, as compared to an undamaged device. Further analysis then allows determination of the strength of trap-assisted current response relative to the band-to-band response. Since these devices are exposed to environmental lighting conditions during operation, further investigation of the degradation of the signal-to-noise ratio due to background radiation is also carried out.

To both remove the influence of environmental lighting and increase optical power density travelling through the absorbing region, a subsequent device is designed and fabricated using a waveguide-integrated photodetector. This incorporates a metal-semiconductor-metal junction across a rib waveguide, which allows longer devices to be created to offset the low absorption coefficient of defect-mediated detection, without the significant capacitance that would be seen in an equivalently sized p-i-n structure. Since this design would suffer in dark current, a barrier-enhancement layer with a higher work function than the metal contact is added to reduce this. For this work, titanium nitride is used for this characteristic, whilst also being tunable by adjustment of the material stoichiometry and deposition thickness, however this is not demonstrated here. A literary review and stopping range of ions in matter (SRIM) simulation comparison of implantation parameters is also performed to improve the efficiency of defect creation within the useful absorbing volume of the detector. This focusses on Group IV elements and inert gases to avoid unintentional doping of the device by the implanted ions.

1.5 Scope of thesis

The range of photon energies which can be absorbed by silicon intrinsically is limited by its band gap energy of 1.12 eV, this restricts detector applications of this material to the visible and ultraviolet spectrums. Silicon photonic technology benefits from tight optical confinement and established fabrication processes, but uses telecommunication wavelengths of 1.3 μm and beyond. This thesis explores a method of enabling detection in silicon by intentionally creating crystalline defects within a region intended for photon absorption. Chapter 2 covers the background theory of waveguiding (Section 2.3), absorption of photons (Section 2.1), detector structures for extracting the generated carriers (Section 2.2) and finishes with reviewing some of the defects found in silicon and how they can be created by high-energy ion implantation (Section 3).

The use of defect-mediated absorption is first investigated in a free-space photodetector in Chapter 4, and is intended for operation at a wavelength of 1.55 μm . Structurally, this device is a p-n junction diode formed between an inner and outer ring, with an unmasked doping at the start to create an additional vertical junction to increase the volume of depleted silicon, as described in Chapter 4. A pre-existing set of photolithography masks are used to demonstrate the ease of integrating the additional processing steps into device fabrication, as discussed in Chapter 5. The current response of the devices under illumination by light at wavelengths of 633 nm and 1550 nm is demonstrated in Chapter 6. This includes: heat maps of current flow with and without high-energy implantation, demonstrating the increase in extrinsic response to 1550 nm light towards a similar level as the intrinsic 633 nm response, and finally evaluating the influence of background lighting conditions on the signal-to-noise ratio.

Next, Chapter 7 covers the design, fabrication and electrical characterisation of a metal-semiconductor-metal photodetector integrated into a SOI-based rib waveguide. After an initial description of the design, a cross-section based upon a SOI platform with 340 nm thick silicon active layer is illustrated using Lumerical MODE Solutions software in Section 7.1.3. This is followed by the simulation of different ion implantations in this waveguide structure using SRIM calculations in Section 7.1.4, which are compared to additional simulations performed to reflect structures used in literature. Then the use of TiN and Ti/Au as barrier enhancement layers is discussed in Section 7.1.5, prior to fabrication in Section 7.2. The dark current performance is then measured and evaluated in Section 7.3. Starting with the fundamental IV curves, an asymmetric increase in Schottky barrier height is obtained for an angled high-energy Ar^+ implantation, and then the dark current is further reduced by application of a negative substrate bias.

Chapter 2

Photodetection theory

2.1 Optical absorption and detection

Optical absorption involves the excitation of an electron by an energy equal to the incident photon energy $h\nu$ to promote it into a higher energy state, where h = Planck's constant (4.136×10^{-15} eVs) and ν = frequency of propagating light [36]. Absorption of this photon causes the generation of an electron-hole pair which travel towards opposing contacts to create a photocurrent in an external circuit, as illustrated in Figure 2.1.

The total energy absorption is the sum of its constituent parts, namely: intrinsic, extrinsic, free-carrier and propagation. The first is excitation across the material band gap to allow generation of free carriers, whereby the absorbed energy $E = h\nu = E_c - E_v$

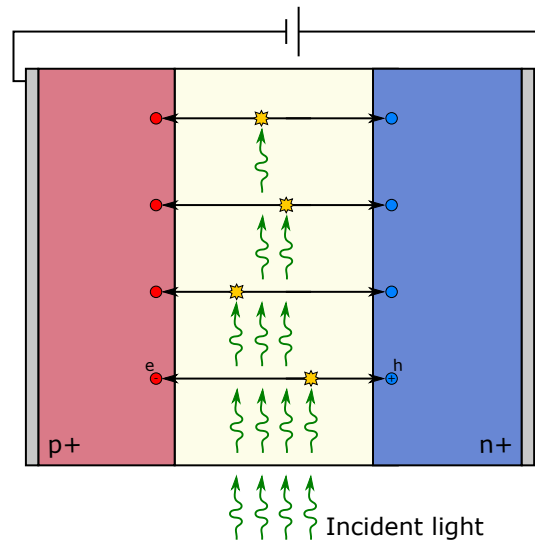


FIGURE 2.1: Photon absorption to generate electron-hole pairs in a simple p-i-n photodetector structure, also illustrating decreasing optical intensity with distance into absorbing medium

with E_c the energy of the conduction band edge and E_v the valence band edge. The second uses an intermediate energy level to aid transfer across the band gap. While free-carrier absorption (FCA) stimulates an electron to a higher energy level in the conduction band, not generating any additional carriers but still causing a loss in power from the optical mode. While propagation losses for an integrated photodetector come from other material and structural losses as found in a photonic waveguide, comprising mostly of scattering.

Absorption is characterised by the reduction in light intensity from the surface of a medium upon which an optical beam is incident in Equation 2.1 according to the Beer-Lambert law [31]. It will reduce or completely remove optical power from a propagating signal depending on the strength (α) or distance (x) of absorption.

$$I(x) = I_o \exp^{-\alpha x} \quad (2.1)$$

Where I_o = incident optical power intensity, α = absorption coefficient and x = distance through medium. The absorption coefficient is dependent on both material and wavelength, and quantifies the fraction of power absorbed by a given medium, whereby the penetration depth α^{-1} is the distance over which 63 % of light intensity is absorbed [31].

These effects are used in photodetectors for monitoring the presence of particular photon energies in sensors, or the conversion from the optical to electrical domain in communications systems prior to CMOS processing. An understanding of absorption processes present within a device is necessary to minimize those which are not desired, to improve both the intended response and efficiency. Additionally, reversal of these effects in a direct band gap material can recombine electron-hole pairs to generate and emit photons for conversion of signals back to the optical domain [31]. Absorption processes are also useful in attenuators for reducing optical power, such as in variable optical attenuators which use an applied forward bias to control the amount of FCA present in the device and therefore the percentage of transmitted power [37].

2.1.1 Intrinsic photon absorption

Intrinsic absorption is the excitation of an electron from a relaxed state in the valence band, the outer-most shell of an atom, across the material band gap and into the conduction band. This internal photoeffect leaves behind a hole, and both carriers are then free to flow around the material. This band-to-band absorption is illustrated in Figure 2.2, and is the basis for most modern photoconductive devices since the generation of electrons within a material will increase its electrical conductivity [39]. By exposing these carriers to an electric field, they are drawn apart before recombining and can be used to generate a photocurrent in external circuitry. The simplest way of achieving

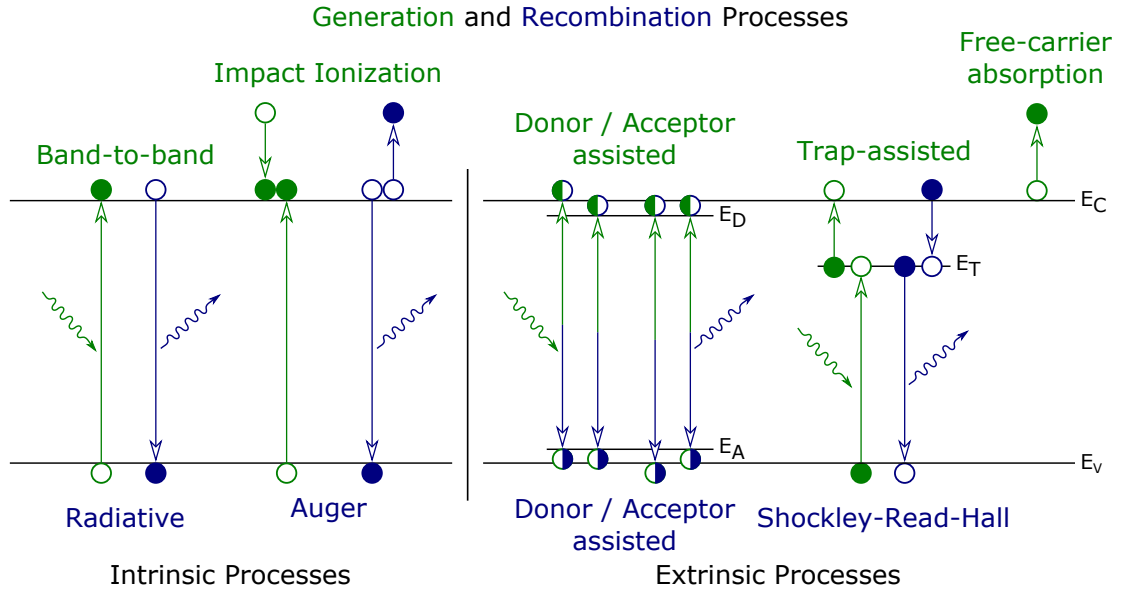


FIGURE 2.2: Generation and recombination processes in intrinsic and extrinsic semiconducting material between conduction (E_C) and valence (E_V) bands, including assistance from acceptor (E_A), donor (E_D) or trap (E_T) intermediate energy levels. Transitions can incur with the absorption or emission of a photon or phonon to account for the change in carrier energy. Adapted from [38]

this is with a p-n junction, whereby electrons are drawn towards the n-type end and holes towards the p-type end. This transition has a high absorption coefficient (around 10^{2-4} cm^{-1} for 500-900 nm wavelengths) due to the large number of possible energy states in each band, but can only absorb photons with energy greater than or equal to the band gap [10]. The reverse of this effect is the loss of energy of an electron as it falls from the conduction to valence band, losing its energy to the emission of a photon or phonon by means of radiative recombination. However, this is not common in indirect semiconductors such as silicon since a change in electron momentum is also needed to transfer between bands [40]. In silicon the band gap of 1.12 eV corresponds to $1.01 \mu\text{m}$ light, therefore it cannot absorb the standard communications wavelength of $1.55 \mu\text{m}$ intrinsically. By comparison, germanium has a gap of 0.66 eV and can absorb up to $1.87 \mu\text{m}$. This varies with temperature by $\pm 2.5 \text{ meV}$ for a $\pm 10^\circ\text{C}$ change from room temperature (300 K), and with impurity doping concentration by -22.5 to -225 meV for $10^{18-20} \text{ cm}^{-3}$ in silicon [41]. Temperature dependence is caused by an increased inter-atomic spacing, decreasing the average potential seen by electrons and reducing the size of the band gap. While the doping concentration dependence arises from the overlapping of electron wavefunctions bound to the impurity. As the average density increases with concentration, an energy band forms near the closest intrinsic material band edge rather than a discrete level, effectively reducing the band gap for shallow impurities.

If the electric field used for carrier extraction is very high, an electron can gain sufficient energy to collide with another electron in the valence band to impact ionize it into a higher energy state, creating additional carriers [42]. This causes the effect of gain

whereby one photon can generate a flow of multiple electrons in an avalanche effect [42]. In contrast, there is an inverse process called auger recombination, which involves the loss of energy of one electron by passing it onto another and allowing the first to fall across the band gap and combine with a hole [43]. The energy passed onto the second electron excites it higher in the conduction band, but some can be converted to a change in momentum which allows the transfer across an indirect band gap [43]. This process dominates in regions of higher carrier concentration or high injection, with a cubic dependence on free carrier concentration [40]. These processes, are illustrated in Figure 2.2 along with the extrinsic processes described in the next section.

2.1.2 Extrinsic photon absorption

When a photon does not have sufficient energy to exceed the band gap, it can still be absorbed extrinsically by use of an intermediate energy level within the gap. This can use shallow donor or acceptor levels from impurity doping for a transition to either band, or between impurities as shown in Figure 2.2. With high dopant concentrations a similar behaviour to radiative recombination with an adjusted recombination coefficient occurs [44].

A trap level positioned deeper in the band gap introduced by further impurities or defects within the material can also aid a transition between bands if the photon has at least the impurity ionization energy [45]. These facilitate the absorption of lower energy (longer wavelength) photons, with the remaining energy needed to complete the transition between bands supplied by thermal or other effects [46]. Either transition between the trap and a band can involve a photon, but since the remaining distance needs additional energy, defect-mediated detectors may struggle to operate at very low temperatures [47]. These processes have a low absorption coefficient (around $15 - 60 \text{ cm}^{-1}$) due to a much lower number of available states, with a linear relation to carrier concentration [10]. Trap-assisted absorption is the approach used in later chapters, and its inverse is known as Shockley-Read-Hall recombination which is the primary form found in indirect gap materials such as silicon for carrier concentrations below 10^{18} cm^{-3} [40]. This is also true when defect concentrations are high, such as near the material surface or interfaces and needs to be minimized to improve the efficiency of devices [38]. Environmental exposure during fabrication can cause this high density at surfaces/interfaces [38]. The use of extrinsic absorption for photodetection has been demonstrated in a number of published results [2, 24, 35, 46, 48–52].

The drawbacks of extrinsic detection include a lower responsivity and absorption coefficient than intrinsic devices, requiring a longer device to absorb all of the signal power [53]. However this does allow their use in power monitoring applications, as only a portion of the signal is lost for detection [53]. Optimal performance in this case needs an improved efficiency, which can be enhanced using resonant cavities [53]. Alternatively, different

types of defect may have a more favourable efficiency or absorption coefficient, such as self-interstitials and divacancies respectively [34]. Unfortunately, whilst there are plenty of reports on infrared absorption, and some which link defect concentration to photocurrent, there is limited knowledge on controlling the quantity and type of defects created [29]. This is primarily dependent on the substrate temperature and ion flux during implantation which is rarely reported, preventing a complete theoretical model of defect formation, and only the availability of some proven experimental results [29].

2.1.3 Free carrier absorption

Photons can still be absorbed if they do not have sufficient energy for band-to-band transitions either with or without an intermediate trap state by means of FCA. This process allows an electron to absorb energy and momentum within the conduction band by excitation into a higher energy state within that band (similarly for holes and a lower state in the valence band), providing more energy for pre-existing carriers instead of creating new ones [40, 54]. FCA is represented using the classic Drude model by Equation 2.2 so long as the chosen wavelength does not coincide with impurity or lattice absorption lines which create additional absorption [40, 54]. It also affects the refractive index as shown in Equation 2.3 [54].

$$\alpha_{fc} = \frac{e^3 \lambda^2 N}{4\pi^2 \epsilon_0 c^3 n (m^*)^2 \mu} \quad (2.2)$$

$$\Delta n_{fc} = \frac{-Ne^2}{m^* \epsilon_0 \omega^2} \quad (2.3)$$

Where e = electron charge (1.602×10^{-19} C), λ = wavelength (μm), N = free carrier concentration (cm^{-3}), ϵ_0 = free space permittivity ($8.85 \times 10^{-12} \text{ Fm}^{-1}$), c = speed of light ($3 \times 10^8 \text{ cm.s}^{-1}$), n = refractive index, m^* = effective mass and μ = mobility of majority carriers [55]. FCA reduces the output power of an optical signal propagating through the device, becoming the more dominant cause of optical absorption at longer wavelengths, since lower energy photons can't use intrinsic absorption, and is strongly influenced by high carrier concentrations [40]. For photodetectors it causes device degradation, with absorption increasing with the square of the wavelength. It also reduces optical confinement due to its reduction of the refractive index in a waveguide's core [40]. At 350 K, FCA approximates to $\alpha_n = 1 \times 10^{-18} \lambda^2 n$ for electrons in n-type Si and $\alpha_p = 2.7 \times 10^{-18} \lambda^2 p$ for holes in p-type Si (wavelength in cm) [40].

FCA becomes significant when there is heavy doping or injection present, reducing detector efficiency since power is absorbed without increasing photocurrent. This can occur at the edges of a propagating optical mode if it partially overlaps with heavily doped contact regions. Such losses can be effectively removed by reducing optical mode

area, by using a shorter wavelength or by increasing the distance from contacts at the expense of transit time. For a detector integrated into a rib waveguide structure, the height of the wings could also be reduced to enhance horizontal mode confinement since this reduces the effective refractive index of the slab on each side of the waveguide core. The FCA can be measured by comparing optical loss from a rib waveguide with doped contacts against one without, and allows determination of doping concentration if calculated for each contact separately at a specific wavelength using Equation 2.2. The absorption present in defect-mediated devices may vary from that calculated using intrinsic material data since the introduction of defects influences carrier mobility and refractive index, as well as the carrier concentration in the contact regions if the defect-mediated region overlaps with it.

2.2 Photodetectors

2.2.1 Operation of p-i-n photodiodes

The simplest detector uses a p-n junction operating under reverse bias and illuminated from free-space. This junction creates a depleted region within which an electric field can extract generated electron-hole pairs, as shown in Figure 2.3a. This however is only capable of detecting photons within this very small volume, as any carrier generation further than one diffusion length from the edge of this region will recombine. Extending the depletion region width without increasing bias, is achieved with a $p^+ - i - n^+$ structure as shown in Figure 2.3b, where it spans the full width of the intrinsic region. This also lowers junction capacitance, increasing response time until a point where carrier transit time becomes the limiting factor, since drift velocity will saturate at high electric fields [31, 42]. Therefore, the design of intrinsic region width has a trade-off between greater photon absorption and lower capacitance, against lower carrier transit time [42].

Waveguide integration typically uses a $p^+ - i - n^+$ doping profile across a rib structure, with the centre rib being near-intrinsic and the two sides being oppositely heavily-doped as can be seen in Figure 2.4a [53, 56–59]. The dimensions of the taller rib in the centre are set to provide good optical confinement for the waveguide at the intended operating wavelength, with light propagating perpendicular to the plane of the page. The horizontal component of confinement also determines the height of the shorter "wings" on each side, which are heavily doped to generate a strong electric field for carrier extraction. In silicon photonics, this structure is typically formed on an SOI wafer to confine the propagating mode vertically using the buried oxide layer and reduce parasitic losses to the substrate [47, 53, 56, 60]. Consideration is needed to prevent the optical mode overlapping with the high impurity wings as this would cause excess losses by FCA. As such some designs add additional lower-doped regions between rib and wings, as in Figure 2.4b. This maintains the low electrical resistance needed for ohmic contacts

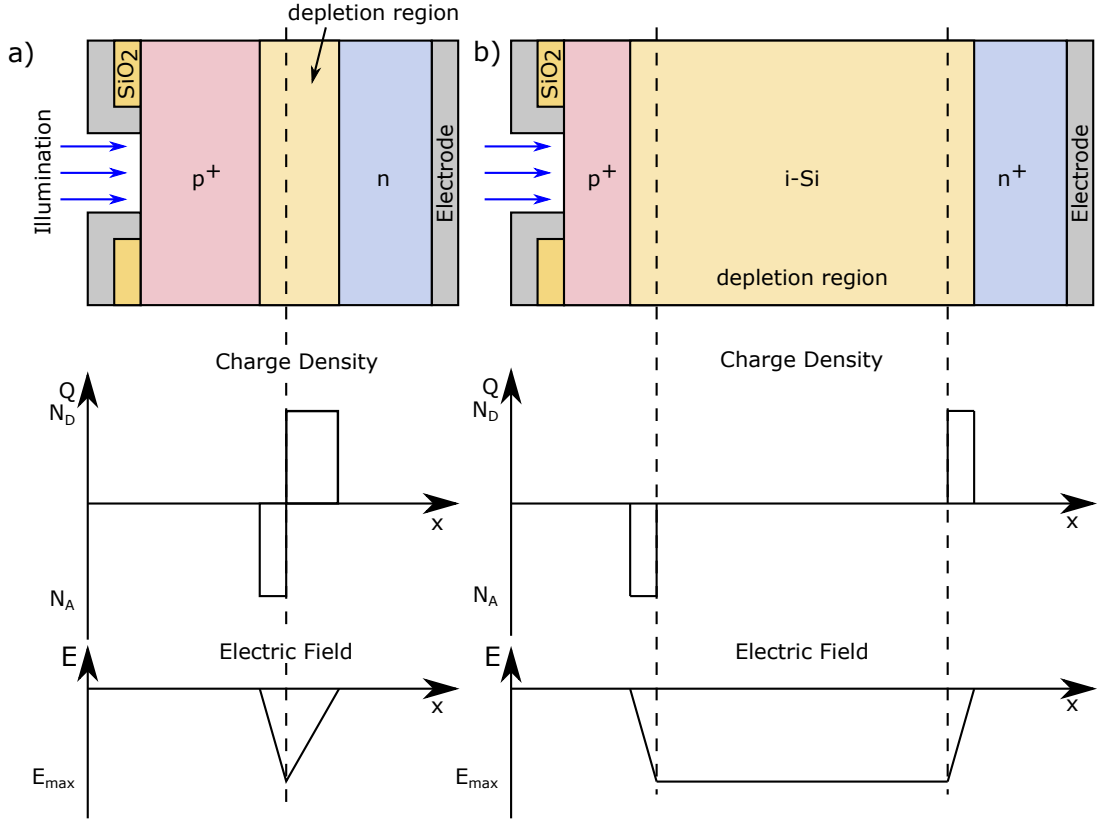


FIGURE 2.3: Cross-section, charge density and electric field distribution diagrams for a) p⁺-n and b) p⁺-i-n⁺, adapted from [31]

(p⁺/n⁺) and high field for carrier extraction, whilst reducing FCA loss as well as allowing slightly longer depletion on each side of the intrinsic region (p/n) [61]. If impurities or defects are incorporated to adjust absorption characteristics, they can be included after passivation and before metallization over the rib as indicated in Figure 2.4a [56, 62].

An important aspect to consider during operation is the influence of absorption noise in the system. This includes thermal excitation which causes some electrons to gain sufficient energy to excite into the conduction band and shot noise from fluctuations in the incoming signal [42]. Shot noise cannot be improved by detector design, while thermal

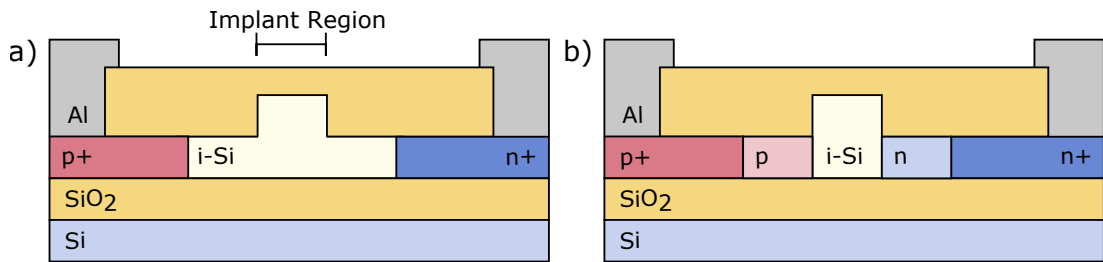


FIGURE 2.4: Cross-sectional diagrams for an integrated Si photodetector; a) a simple p⁺-i-n⁺ structure reproduced from [56], b) a p⁺-p-i-n-n⁺ structure reproduced from [34].

excitation can be reduced by keeping detector temperature far below the bandgap energy since the same ionization energy is needed as with photoexcitation [45]. However, some thermal energy may need to be present for extrinsic absorption to satisfy the remaining distance across the band gap.

2.2.2 Avalanche photodetectors and phototransistors

An expansion upon the p-i-n structure is an avalanche photodiode (APD) [31]. This adds additional doping between the front contact and intrinsic region, as shown in Figure 2.5. Carrier generation occurs within the intrinsic region as before, but when electrons reach the n-type region they experience a larger electric field and accelerate. This provides the carrier with enough kinetic energy to impact-ionize some additional Si atoms and release more electron-hole pairs [31]. If the field is strong enough both the original electron and the newly generated one can cause additional ionization, leading to an avalanche multiplication effect [31]. As such these devices obtain a quantum efficiency higher than 100 % [31]. This ionization is also achievable in the previously discussed junction devices with a strong enough reverse bias. Since the device only boosts the energy of one carrier type, ionization is typically restricted by design to the one with highest efficiency, which in Si are electrons. The proportion of electrons to holes

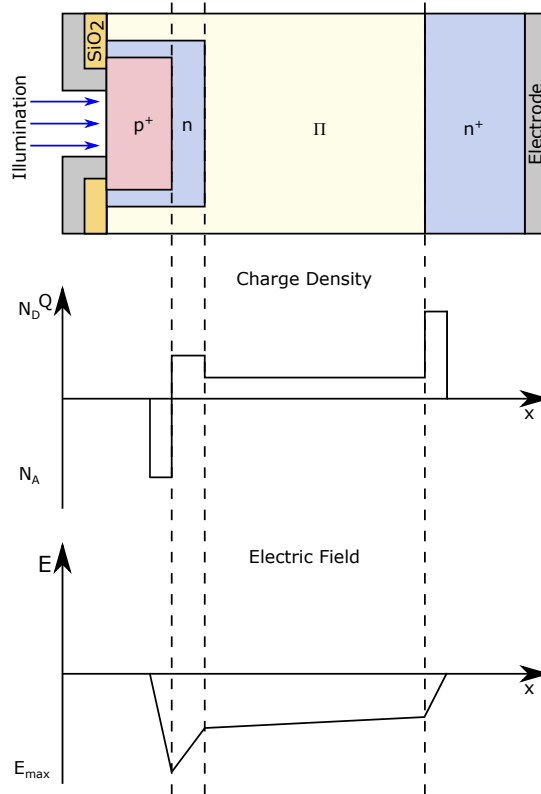


FIGURE 2.5: Cross-section of an avalanche photodetector structure, showing the charge density and electric field across the device, adapted from [31]

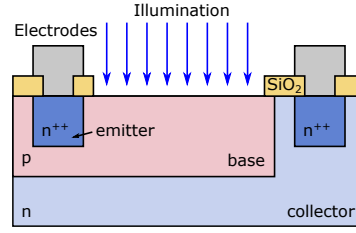


FIGURE 2.6: Cross-sectional view of a n^+ -p-n phototransistor, under free-space illumination from above

being multiplied is known as the ionization ratio, which should be maximized to reduce noise and increase bandwidth [28]. Although this structure allows detection of very few photons, the statistical multiplication creates excess noise as a result of fluctuations in carrier generation, increasing the dark current. Response time is slow due to additional avalanche build-up and the time it takes for the last hole to cross the intrinsic region, but can create a faster circuit due to not needing additional electronic amplification [31].

An alternative structure is a phototransistor, formed of n^+ -p-n (emitter-base-collector) regions as shown in Figure 2.6, which acts as a current amplifier upon illumination [31]. Carriers are generated by photon absorption within the base-collection junction whereby electrons drift to the collector and holes drift to the base [31]. This causes the injection of electrons from emitter to base and since the recombination time is much longer than the diffusion time, most electrons reach the collector amplifying the original electron-hole pair [31]. Additionally, illumination can be through the emitter side by use of a heterojunction structure, where the emitter is transparent to the incoming wavelength and absorption occurs in the base and collector [31].

Further enhancement of quantum efficiency and responsivity can be achieved by using a photonic crystal structure, as shown in Figure 2.7. Here a photonic band gap restricts the wavelength of light which can propagate through it to a narrow band which is designed

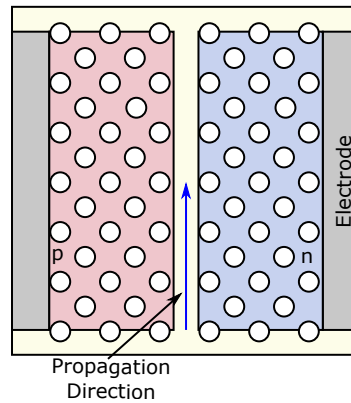


FIGURE 2.7: Surface view of a p-i-n diode built into a defect-line photonic crystal waveguide, adapted from [64]

to coincide with the capabilities of the photodetector [63]. Omitting a line of holes in the periodic structure to create a defect-line waveguide allows the routing of light [63].

2.2.3 Germanium and heterogenous photodetectors

Germanium devices have had a lot of work in developing photodetectors due to the material's smaller band gap. This allows longer wavelength detection as well as higher carrier mobility and absorption coefficient [28]. The photoresponse can also be improved by use of tensile strain to reduce the band gap energy [25,28]. It also makes the material's direct gap dominant since this gap shrinks faster than the indirect one [25,28]. A responsivity of up to 1.1 AW^{-1} and bandwidth of 32 GHz has been achieved for such a waveguide integrated design [28]. These devices have developed a lot from an early need for cryogenic cooling to reduce the inherently high dark current, to the ability to epitaxially grow it on top of Si by use of a SiGe buffer layer (or two-step thermal growth) to overcome the 4.2% lattice mismatch between Ge and Si [28]. This mismatch is prone to causing a high number of dislocation defects at the interface, which increases dark current and lowers photocurrent due to carrier recombination [25,28]. The minimum dark current density is limited to 0.01 mAcm^{-2} at room temperature by thermionic emission of carriers across the interface [28]. Published results for the dark current of p-i-n devices have ranged from $0.035 - 0.79 \mu\text{A}$ for surface-illuminated designs, or $0.2 \text{ nA} - 3 \mu\text{A}$ for waveguide integrated designs [28]. A comparison of typical characteristics for some published silicon and germanium photodetectors is given in Table 2.1.

TABLE 2.1: Comparison of Si and Ge photodetector structures, showing common characteristic parameters. These are: detectable wavelength range, wavelength of peak response, responsivity at this peak, detector gain, rise time and dark current respectively. Sourced from [31].

Structure	λ_{range} (nm)	λ_{peak} (nm)	R at λ_{peak}	Gain	t_r (ns)	I_{dark}
Si p-n	200 - 1100	600 - 900	0.5 - 0.6	< 1	0.5	0.01 - 0.1 nA
Si p-i-n	300 - 1100	800 - 900	0.5 - 0.6	< 1	0.03 - 0.05	0.01 - 0.1 nA
Si APD	400 - 1100	830 - 900	40 - 130	10 - 100	0.1	1 - 10 nA
Ge p-n	700 - 1800	1500 - 1600	0.4 - 0.7	< 1	0.05	0.1 - 1 μA
Ge APD	700 - 1700	1500 - 1600	4 - 14	10 - 20	0.1	1 - 10 μA

SiGe has also found use when combined with a Si APD, as a separate absorption, charge and multiplication (SACM) structure can be formed [28,60]. This uses intrinsic Ge for photon absorption (shorter absorption length, higher electron mobility and faster response) and intrinsic Si for carrier multiplication with an intermediate p-type Si charge layer [28]. This allows independent control of photon absorption and efficiency, and multiplication gain to optimise noise and speed [28,60].

InP and InGaAs can also be incorporated into a SACM structure by using the difference in bandgap to restrict absorption to the InGaAs region [31,65,66]. A superlattice APD

can also be used to restrict impact ionizations to one carrier type by only allowing a gain in energy sufficient for impact ionization to only be experienced by electrons [31, 65, 66]. When epitaxial deposition is used, InP can detect wavelengths up to $2.2\ \mu\text{m}$, while InGaAs will go up to $2.6\ \mu\text{m}$ [7]. The tertiary alloy HgCdTe is one structure currently used for detection in the mid-infrared due to its wide detectable range of $1\ \mu\text{m}$ to $30\ \mu\text{m}$, by alteration of its composition for bandgap tunability [66]. This also creates small changes in lattice constant, which allows the growth of high quality layers and heterostructures with a broad operating temperature range [66]. Due to the different materials used and a weak Hg-Te atomic bond, fabrication of these detectors is difficult and prone to uniformity and yield issues [66]. It has however found widespread use within focal plane arrays (FPAs) with their only major competitor being thermal detectors [66].

An alternative compound detector uses a GaSb-based epitaxial layer bonded on top of a Si waveguide [22]. Use of this material has achieved detection for the range $2 - 4\ \mu\text{m}$ with a responsivity of at least $1.4\ \text{AW}^{-1}$, at the expense of a higher cost due to complicated fabrication [22]. GeSn has also been experimented with since the addition of tin to germanium causes a reduction in the bandgap, thus allowing detection of longer wavelength signals up to approximately $2.5\ \mu\text{m}$ [22].

PbTe-based detectors have been shown to achieve high responsivity ($25\ \text{VW}^{-1}$) in the $3 - 4\ \mu\text{m}$ wavelength range under a bias current of $0.1\ \text{mA}$ at about -60°C [26]. These have been used for their good chemical stability compared to other compound detectors, easy deposition and tunability of the band gap energy when alloyed with SnTe, for detecting wavelengths from $6 - 12\ \mu\text{m}$. This material has been used as a very thin layer within Distributed Bragg Reflector (DBR) stacks since a thinner film enhances responsivity and reduces the signal to noise ratio (SNR) [26]. While the DBR increases quantum efficiency due to resonant cavity enhancement, increasing responsivity to $800\ \text{VW}^{-1}$ at -55°C under $0.06\ \text{mA}$ bias for the same wavelength range [26].

It has been shown that the high cost of integrating compound semiconductors onto Si waveguide circuits can be mitigated by replacing epitaxial growth with wet deposition of nanoparticles of PbS and HgTe [22]. This approach benefits from multiple methods of depositing the solution (e.g. spin coating) and efficient use of the materials used. Whilst wavelengths of $2.4 - 2.8\ \mu\text{m}$ have been shown operable, with the potential of $5\ \mu\text{m}$ and a good responsivity [22]. However this process is still in its infancy and needs further research.

The heterostructures discussed here are not generally CMOS compatible as a result of using incompatible materials, while germanium has been integrated into silicon photonic multiple project wafer (MPW) flows based on a CMOS pilot line [53, 67]. However fabrication is still more complex than using purely silicon-based materials, making them a more costly approach [53].

2.2.4 Device characteristics

Photodetection devices are typically characterised by a number of performance measures such as quantum efficiency, responsivity, bandwidth or response time and system noise. Each is optimized based on the intended application; sensors need a high responsivity to ensure detection of incoming photons without the need for high speed; whereas communication networks need high speed and bandwidth, so efficiency can be allowed to decrease.

An additionally important design condition for waveguide integrated detectors is the overlap of optical mode with the depleted region. Since increasing reverse bias increases the depletion region width, it also enhances this overlap and the device responsivity until saturation occurs due to increasing beyond the optical mode size [68]. Device design and operating conditions should optimize this overlap for maximum response without unnecessary power consumption and high voltages.

2.2.4.1 Quantum efficiency

Quantum efficiency measures how well photons are converted to electron-hole pairs. The number of electron-hole pairs generated per incident photon entering the device is defined as the internal quantum efficiency (IQE), limited to unity for devices without gain [31]. While the proportion of pairs collected by the contacts per photon incident on the device is external quantum efficiency (EQE), described by Equation 2.4 [31].

$$\eta = \frac{\text{number of free electron-hole pairs generated and collected}}{\text{number of incident photons}} = \frac{I_{ph}/e}{P_{opt}/h\nu} \quad (2.4)$$

Where I_{ph} = photocurrent, e = elementary charge (1.602×10^{-19} C), P_{opt} = incident optical power, $h\nu$ = photon energy (h = Planck's constant, ν = frequency of light). Some ways of improving efficiency are; reducing surface reflections, increasing photon absorption at the intended wavelength and preventing carrier recombination or trapping [31].

2.2.4.2 Responsivity

Responsivity ($\mathfrak{R}$) is defined as a measure of photo-generated current per watt of incident light, calculated as in Equation 2.5 [39].

$$\mathfrak{R} = \frac{\eta e}{h\nu} = \eta \frac{\lambda_o}{1.24} = \frac{I_{ph}}{P_{opt}} \quad (2.5)$$

Where λ_0 = free-space wavelength (in microns). It is wavelength dependent as photon detectors are more sensitive to the flux of incoming photons rather than the optical power. Plotted against wavelength it provides the spectral response of a detector. Increasing with wavelength for improved performance, until η counteracts it by becoming incredibly small [31]. It provides an insight into detectable wavelengths and the magnitude of response, and for an ideal efficiency of 100 % it plots a straight line with gradient 1.24^{-1} until the cut-off wavelength as denoted by the material bandgap. Devices which do not exhibit gain will have a response below this line, otherwise greater than 100 % efficiency is achieved through carrier multiplication effects.

2.2.4.3 Bandwidth

The operating speed of a device is described by either the bandwidth or response time and measured as part of the AC or transient response. The time taken for a carrier to drift from where it was generated to an electrode is termed the transit time, as shown in Equations 2.6 [31].

$$t_e = \frac{W - x}{\nu_e} \quad t_h = \frac{x}{\nu_h} \quad \nu_e = \mu_e E \quad \nu_h = \mu_h E \quad (2.6)$$

Where W = depletion region width, x = distance from edge of W , $\nu_{e/h}$ = drift velocity, $\mu_{e/h}$ = carrier mobility, E = electric field and subscripts e and h correspond to electrons and holes respectively. The maximum transit time occurs when $x = 0$ since photocurrent is determined by electron flow [31]. Faster operation results in a larger bandwidth, which for photonic systems can be > 30 GHz for a 30 ps response time (Table 2.1). The theoretical maximum speed is bound by the carrier transit time, but a given device will also have a RC time which is likely to limit performance more [68]. This is the product of junction capacitance and the sum of series and load resistances. For the case of a p-i-n diode under reverse bias, the junction capacitance is equal to depletion layer capacitance, given in Equation 2.7.

$$C = \frac{\epsilon_o \epsilon_r A}{W} \quad (2.7)$$

Where ϵ_o = vacuum permittivity ($8.85 \times 10^{-12} \text{ Fm}^{-1}$), ϵ_r = relative permittivity and A = junction area [31, 69]. It is also dependent upon doping concentration on each side of the junction since this affects the depletion width.

2.2.4.4 Detection noise

The ability of a detector to distinguish between the presence and absence of a signal is determined by the difference between those two states. This is described as the signal-to-noise ratio (SNR) and is a ratio of the signal power relative to noise (given in dB), as shown in Equation 2.8 where i_n = noise current [31].

$$SNR = \frac{I_{ph}^2}{i_n^2} \quad (2.8)$$

Most noise in photodetectors comes from random generation of carriers by thermal excitation or tunnelling, this dark current flows when there is no illumination and is therefore independent of it [31]. It experiences shot noise due to the discrete nature of electron charge, and causes power leakage and a reduction in the system SNR, creating a need for higher photocurrent generation [31]. Additionally, photocurrent experiences quantum noise and the pair form the main sources of noise in p-n and p-i-n devices, totalling i_n as in Equation 2.9 [31].

$$i_n = (2e(I_d + I_{ph})B)^{1/2} \quad (2.9)$$

Where I_d = dark current, I_{ph} = photocurrent and B = bandwidth. A higher dark current can be tolerated in some applications so long as the generated photocurrent is sufficient to maintain a high SNR. Another means of quantifying the amount of noise is the noise-equivalent power (NEP), this is the minimum input optical power needed to generate a photocurrent equal to the noise at a given wavelength with a bandwidth of 1 Hz [31]. Defined in Equation 2.10, it is the minimum detectable power as limited by system noise [31].

$$NEP = \frac{P_{opt}}{\sqrt{B}} = \frac{1}{\Re} \sqrt{(2e(I_d + I_{ph}))} \quad (2.10)$$

Its inverse is known as the detectivity, which if multiplied by the square root of the detector area and noise bandwidth, the specific detectivity, which provides a means of comparing devices independent of size or bandwidth [70].

2.3 Optical waveguides

The basic component for integrated photonic systems is the waveguide for transporting optical signals around a chip. It is constructed as a medium of high refractive index (RI) core surrounded by a medium of low RI, referred to as the core and cladding respectively.

When light propagates through the core it reflects off the core-cladding interfaces, and if the angle formed between the propagating beam and the normal to the surface is greater than the critical angle, total internal reflection occurs guiding the light along the length of the waveguide [39].

Each transverse electromagnetic (TEM) wavefront can be polarized for either transverse electric (TE) or transverse magnetic (TM) orientation. These are characterised by the total electric or magnetic fields being perpendicular to the direction of propagation respectively, with the favoured orientation controlled by both the incoming polarization and waveguide geometry. Upon propagation of these wavefronts within a waveguide, if the bounce angle reflection is such that a double bounce reproduces the original wavefront, self-consistency is maintained and a waveguide mode is formed [39]. Multiple angles can produce this effect creating different modes, and each maintains the same transverse distribution and polarization along the waveguide [39]. The number of single polarization modes is represented by Equation 2.11 for both a planar and rectangular dielectric waveguide respectively, with the numerical aperture in Equation 2.12 [39].

$$M \doteq \frac{2d}{\lambda_0} NA \quad M \approx \frac{\pi}{4} \left(\frac{2d}{\lambda_0} \right)^2 (NA)^2 \quad (2.11)$$

$$NA = \sqrt{n_1^2 - n_2^2} \quad (2.12)$$

Where d = slab height and rectangle width, λ_0 = free-space wavelength, $n_{1/2}$ = refractive index of core or cladding respectively (where $n_1 > n_2$), as illustrated in Figure 2.8 [39]. The rectangular case assumes both height and width are equal, and an average spacing of π/d between wave vectors of adjacent modes in both directions along the plane perpendicular to the direction of propagation [39]. The symbol $\doteq$ indicates the right-hand side is increased to the nearest integer, and the total number of modes is twice that calculated above since there are two polarizations (TE and TM). As a result of the dispersion relation reducing the propagation constant for longer wavelengths, a larger cross-sectional area or higher RI contrast between core and cladding is needed to confine them [39]. To reduce modal losses, careful design of waveguide geometry is needed to maintain single mode operation for longer wavelength propagation ($M < 1$).

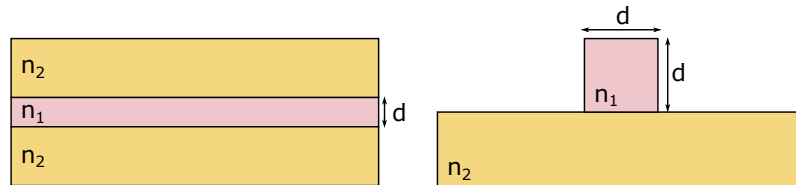


FIGURE 2.8: Slab and strip waveguide cross-sections, illustrating parameters used in Equation 2.11

The power distribution of the optical mode propagating within the waveguide forms a Gaussian shape, with a peak in the waveguide centre and the tails propagating within the cladding [29]. The proportion of power within the core can be increased by using a higher refractive index contrast, shorter wavelength, or larger cross-section. Whilst confining more power within the core leads to lower propagation losses, if the cross-section becomes too large it will allow confinement of non-fundamental modes [71]. The power propagating in the cladding is the evanescent field and determines the distance between neighbouring waveguides needed to prevent crosstalk, and can be used for coupling between structures [71]. An alternative style of waveguide uses the evanescent field to confine light within a slot of low-index material formed between two high-index material strips [29].

2.3.1 Sources of loss

As an optical signal propagates down a waveguide it succumbs to various sources of power loss which exponentially reduce how far it can travel before the signal is lost [31]. These arise from scattering, absorption, modal power exchange and an imperfect transport medium with structural variations along its length. Minimization of these effects are key to good device performance in passive components, while active devices may enhance one of these effects for manipulation or conversion of the signal.

Scattering arises from imperfections within the waveguide, deflecting light such that it hits the core-cladding interface below the critical angle allowing it to escape and power to be lost [71]. This can be caused by impurities within the bulk of the material, and is often negligible for undoped passive structures [71]. Fabrication will introduce some side wall roughness due to the statistical particle collisions and chemical processes involved in deposition and etching. It can be reduced by improving the directionality of ion bombardment in plasma etching of side walls, or by optimising deposition chemistry for a more uniform material layer without formation of localized islands. This roughness also introduces reflections and phase perturbations that are wavelength dependent, altering the peak amplitude and width in WDM systems [71]. An upper bound on scattering loss by surface roughness is given in Equation 2.13.

$$\alpha_{max} = \frac{\sigma^2 \kappa}{k_0 d^4 n_1} \quad (2.13)$$

Where σ = root-mean-square roughness, k_0 = wave vector of light in a vacuum, d = half-width of core, n_1 = effective index of Si slab with same thickness as the core, and κ depends on geometry and statistical distribution of roughness [29]. Assuming a single-mode waveguide, κ is 0.48 for exponential statistics and 0.72 for Gaussian [72]. An SOI platform based on 500×220 nm waveguides typically achieves 2 - 3 dBcm⁻¹ scattering loss, which needs a root-mean-squared roughness of around 1 nm or less [29,71]. Moving

into the mid-infrared reduces the effect of scattering loss due to the larger dimensions needed and an inverse dependence of loss on core width (as in Equation 2.13). The lower limit on loss is caused by Rayleigh scattering, which is a linear effect from scattering centres which are much smaller than the wavelength of propagating light [73]. Its effect in crystalline silicon is minimal since it is caused by random fluctuations in material density, however SiO₂ cladding does experience it [73]. The effect is greater for shorter wavelengths and can become more significant than other losses, but is principally lower for longer wavelengths until infra-red absorption takes over [73].

Power is also lost by the absorption of photons, and is most significant for passive devices if the material system used is not transparent at the propagation wavelength [71]. Since intrinsic absorption occurs, and is most noticeable at ultraviolet wavelengths [71]. Otherwise, further absorption can occur assisted by free carriers or sub-surface trap states, and additionally by surface states if waveguides are not sufficiently passivated [71]. For the case of active devices, heavily doped regions or metal within the boundaries of the optical mode will create further free carrier or parasitic absorption respectively, for the 500×220 nm SOI platform at a wavelength of 1550 nm the loss to metal can be around 1.8 dBcm⁻¹ [71]. For very long wavelengths, infrared absorption can occur due to the loss of energy to vibration of chemical bonds between atoms in the propagation medium. Losses by absorption creates modal perturbations and is characterised by the imaginary part of the refractive index, the absorption coefficient or the extinction coefficient.

Most rectangular waveguides are operated in the fundamental TE₁₀ mode as this has the highest effective refractive index and signal strength. Waveguide design prevents confinement of higher order modes to prevent a reduction in signal power by modal loss. This can occur as a result of a waveguide's cross-sectional area being larger than that needed for single mode operation, geometric variation causing unintended modal dispersion, or nonlinear effects. Additionally, the attenuation coefficient is mode-dependent, with higher order modes incurring greater absorption losses making the fundamental mode more favourable.

For routing waveguides around a chip, they are required to bend which causes additional loss by radiation and modal mismatch from transitioning between straight and bent sections. Radiative loss of highly confined TE-polarized strip waveguides are typically small, but it is non-negligible for TM-polarized or rib waveguides with $< 5 \mu\text{m}$ bend radius [71]. Most bending loss is modal mismatching from an imperfect overlap between straight and bent waveguide sections creating scattering at the transitions [71]. A similar effect is seen in splicing loss when aligning two guides end-to-end [71]. This effect can be decreased by using a lateral offset between straight and bent sections for better overlap, or continuously varying curvature along the bend which also prevents excitation of higher-order modes [71]. For the 500×220 nm SOI platform, loss from a 90° bend roughly matches normal propagation loss if the radius is at least 10 μm [71].

Global waveguide losses can be reduced further by using very wide multimode guides since the primary source of loss is scattering at the core-to-cladding interface, which is reduced according to Equation 2.13. An example implementation of these wider-core waveguides achieved a propagation loss of 0.27 dBcm^{-1} using $2 \mu\text{m}$ wide, 250 nm high rib waveguides [71].

2.3.2 Waveguide structures

Some common waveguide cross-sections are illustrated in Figure 2.9. The slab waveguide only provides one-dimensional optical confinement and therefore has limited practical use. Strip waveguides confine light in two dimensions and are used as a standard guiding platform. The embedded type is created by diffusion of a higher RI material into the substrate, and the non-embedded type deposited and patterned on top of the substrate. [39]. Whilst rib or strip-loaded waveguides confine light vertically and allow only a small modal expansion horizontally, these are useful for creating electrical connections for detecting or manipulating the optical signal.

The planar layout of waveguides can vary from a straight structure to a number of configurations for different uses or routing as shown in Figure 2.10. S bends shift the signal sideways, while a Y branch can split or combine signals. A Mach-Zehnder interferometer can be used for sensing the difference between measurement and reference arms, or for use as an optical switch [74]. Multiple waveguides positioned very close together are able to exchange power and form directional couplers, which is also the basis for ring resonators. While perpendicular intersections allow the crossing of photonic channels without significant interaction between the two, a feature not available in traditional electronic routing techniques.

An additional structure that can be used for confining and propagating light is a two-dimensional photonic crystal [39]. This is formed from an array of equally spaced holes with a period tuned to create a specific photonic band gap of disallowed frequencies, with a line of missing holes through which light can propagate [39]. An air-slab interface, or similarly low RI material cladding, above and below the crystal plane expands confinement the vertical dimension [63]. If the optical frequency falls within this band gap, it is prevented from travelling through the periodic hole structure and therefore

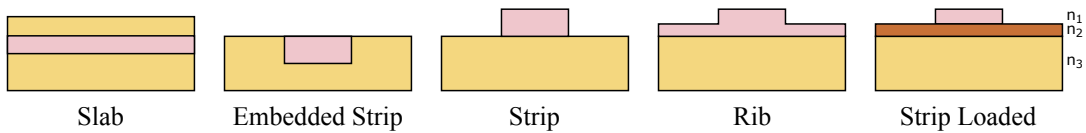


FIGURE 2.9: Different waveguide cross-sections, where refractive index $n_2 > n_1 > n_3$, reproduced from [39]

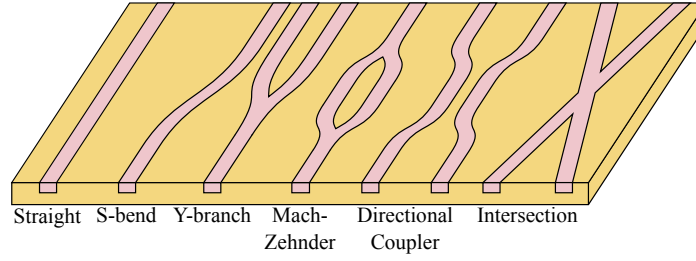


FIGURE 2.10: Different configurations for embedded strip waveguide structures, reproduced from [39]

is guided along the channel [39]. This approach benefits from the ability to create 90° bends without significant loss, but only operates within a certain small frequency range.

When coupling light into a waveguide from an optical fibre, common approaches include end-fire coupling in the plane of the waveguide, or surface gratings which allow propagation out of plane, above the chip [29]. Coupling mechanisms need to ensure good spatial modal matching to obtain the desired optical mode as well as efficient transformation of the propagation vector into the new guiding medium, as additional losses, scattering and back-reflections can occur [29]. End-fire coupling usually makes use of an adiabatic taper to gradually convert between modes, with its length determined by the magnitude of the modal change and the distance between neighbouring modes, to reduce radiation losses or conversion into parasitic modes [29]. Gratings use diffraction effects from in-phase scattering of light from periodic “teeth” to create constructive interference, and if the scattered field is tuned to the receiving waveguide’s mode, coupling can occur [29]. This grating can change the direction of the propagating mode allowing coupling at an angle to the surface or into the opposite direction [29].

An alternative approach is the use of multimode/multipath interference, such as using the evanescent field in a directional coupler. Whereby two closely spaced waveguides share light distributed between their even and odd supermodes which can be coupled from one to the other, tunable by the interaction length [29]. If this is added into a resonant structure, a small amount of coupled power at the resonant wavelength can be amplified to efficiently couple between input and output waveguides [29].

2.4 Summary

This chapter has provided a theoretical understanding of photon absorption processes within a semiconductor. It includes the role of energy bands for intrinsic absorption; shallow or deep levels within the band gap for extrinsic absorption; then the importance of free-carrier absorption in heavily doped regions. It then covered how this process can be used for the generation of an electrical photocurrent in a few different devices and

material systems. Also included is the theory of photonic waveguiding structures, upon which a photodetector can be added for integrated circuit designs. Later chapters will reference some of this theory and specifically use the characteristic parameters within performance analysis. For comparison between the induced trap-assisted absorption and the intrinsic capability of silicon, performance will be measured at both wavelengths of 1550 nm and 633 nm which provide photons of energy 0.80 eV and 1.96 eV respectively.

Chapter 3

Defect engineering

The presence of defects in a crystalline semiconductor material leads to extrinsic absorption of photons as discussed in Section 2.1.2. Defect-mediated designs use this process as an advantage, intentionally creating defects using high-energy bombardment fabrication processes to change the absorption characteristics within a particular region of a material. For example, this allows silicon photodetectors to operate beyond visible wavelengths and the lower transparency limit of $1.1\text{ }\mu\text{m}$, allowing applications in data communications. It also opens up the possibility of monolithic integration of photonic devices in an all-silicon platform.

Small-scale variations from a perfect crystalline structure are called lattice defects and are made up of one or more point defects in the form of missing atoms (vacancies), replacement foreign atoms occupying lattice sites (substitutional impurities) or additional foreign atoms positioned between sites (interstitial impurities) [75]. They can arise in the bulk material or at boundaries creating interface losses, such as the dislocation defects that can be found at Si-Ge boundaries due to the 4% lattice mismatch [76]. Through diffusion effects and heavy damage, more complex structures can form whereby a number of smaller defects have grouped together to form larger, extended defects [77]. Point defects and some of these extended defects are illustrated in Figure 3.1.

3.1 Structure and creation of lattice defects

Impurities can be introduced by traditional doping methods such as diffusion or ion implantation, and form interstitials or complexes with pre-existing defects or impurities [80]. While additional defect creation needs a high-energy process capable of damaging the crystal lattice [80]. Experimental devices have used high energy ion-implantation with, for example: boron [24, 35, 56], phosphorous [56], argon [52, 81, 82] or silicon [10, 33, 49, 57, 59, 83, 84] followed by annealing to control the distribution of majority defect types within the region of interest.

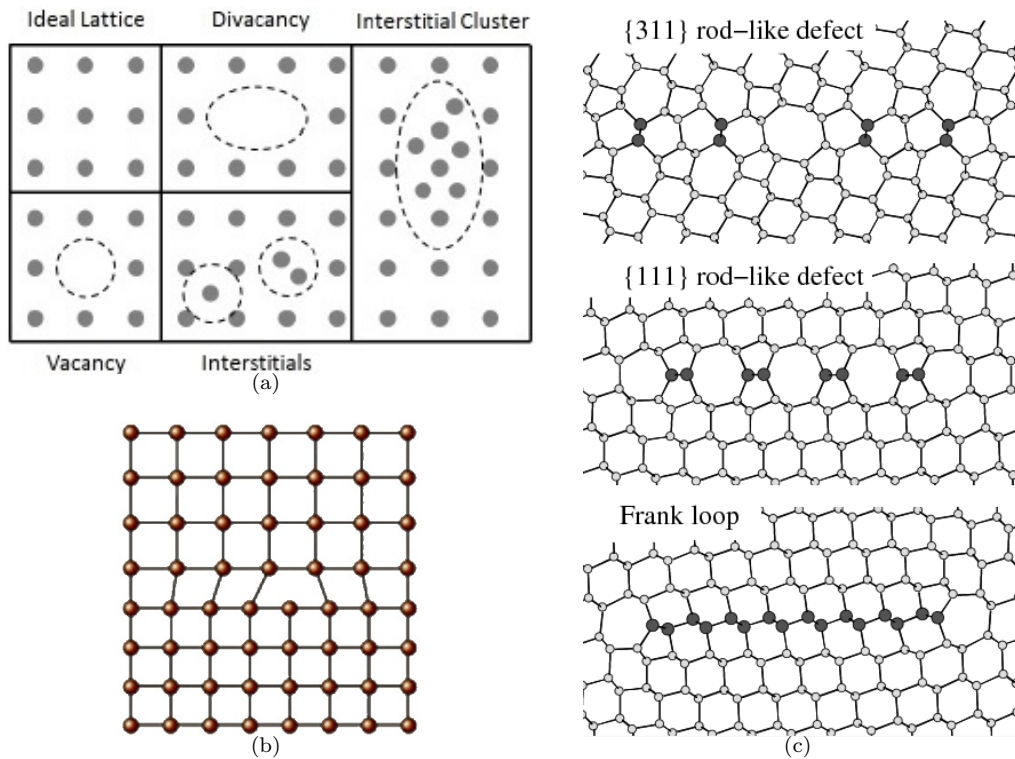


FIGURE 3.1: a) Two-dimensional square lattice illustrating different defect types; vacancy (single missing host atom), interstitial (single extra atom), divacancy (two adjacent vacancies), and an interstitial cluster, adapted from [47]. b) Dislocation defects at Ge-Si interface occurring as a result of different lattice constants, adapted from [78]. c) Some extended defect complexes, adapted from [79].

Each defect has a number of dangling bonds which determines how many electrons it can hold, and therefore its charge state [80]. This is influenced by the quantity and type of charge carriers in the material, characterised by the Fermi level position [80]. If the charge state is negative the defect acts like an acceptor, while if positive it acts like a donor [80].

N-type bulk material causes a defect to become more negatively charged due to an increased number of donor electrons, with the opposite effect for p-type material [85]. This suggests that it is possible to control which charge state a defect has by controlling the background doping concentration. Additionally, each charge state has a different energy level position within the host material band gap [46]. This is caused by an increase in Coulomb repulsion energy for each added electron, followed by a lowering in the gap as the lattice relaxes during broken bond reconstruction [80]. If the charge state can be changed whilst maintaining an ability to generate carriers, these positions could be incorporated into a photodetector design to absorb different wavelengths. In one study, donor concentration has been used to control the number of occupied indium centres, therefore influencing the total absorption of an optical modulator [50].

By use of single-defect type modelling, it has been determined by D. Logan et. al.

that n-type Si is favourable for defect-mediated detection which incorporates the divacancy [68]. In this case the defect will become occupied by free electrons provided by the donor atoms, moving into the negatively charged state as shown in Figure 3.2, when it will act to cancel out the positive charge of the donors [68]. This doping compensation enhances depletion, reducing device capacitance [68]. Conversely, lightly doped p-type Si will maintain the neutral charge state of the divacancy, where it is unable to accept a significant number of majority carriers [68]. This results in there being no way of balancing the ionized acceptors, giving the opposite effect [68]. Under illumination, the generated electrons occupy defects and cause the net static charge to become more negative, in n-type Si this causes a reduction in the depletion region for high defect concentrations [68]. The effect of this is an increasing photocurrent with defect concentration until a characteristic peak, which then decreases as more defects are introduced (this peak is observed at higher concentrations when the contact spacing is very small) [68].

Defects can be created by any process that impacts high-energy particles upon a wafer, such as plasma etching, however most only significantly effect the surface [86]. Commonly, ion implantation is used for intentional defect creation, as this is a process currently used in CMOS fabrication lines and allows control of energy and dose in addition to the implanted species [86]. The initial spread of defects with depth is controlled by the energy, increasing to drive ions deeper into the target, while ion dose controls the extent of damage or impurities at a given depth and energy [87]. Ion species affects both the depth and extent of damage, since larger atoms enhance effects nearer the surface, while smaller atoms travel further into the target material creating a longer spread [87]. The species also influences potential defect compounds created by the choice of element introduced or majority charge state of sites due to doping effects [87].

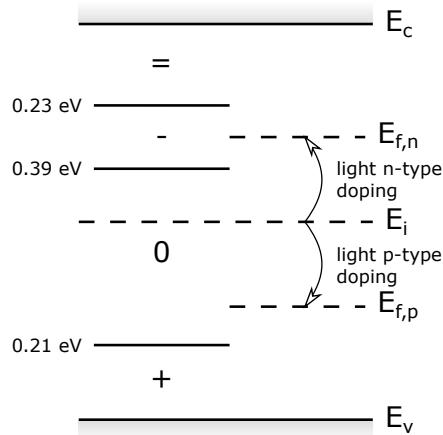


FIGURE 3.2: Charge state transition energies for the Si divacancy, illustrating the change in Fermi level with light n/p-type doping. Indicates states are: double acceptor ($=$), single acceptor ($-$), neutral (0), single donor ($+$). Energies are given relative to nearest band edge.

The concentration of different defect types is initially controlled by the ion implantation parameters. Subsequent annealing causes smaller defects to diffuse into more complex aggregates or out of the material, as determined by their activation energy [80]. These larger complexes are more stable at elevated temperatures since more energy is needed to trigger diffusion [10]. This has been shown by experimental results of ion-implantation [46,61,88], and defect annealing [80,88]. As anneal temperature increases, responsivity improves around the current majority defect's sensitive energy range up to a certain maximum temperature, after which it decreases as the concentration of that defect declines [46]. If these defects transform to create a new electrically active majority species, similar characteristics could occur but with a lower response and higher peak concentration temperature [46]. The response is lower due to the recombination of Frenkel pairs and partial loss of defects from the material. Since these larger defects each use more of a finite number of point defects available in the material, the new concentration will be lower than the original defect species. This would be seen in the spectral response as a shift to longer wavelengths with a reduction in peak response and integrated area under the graph. The peak would also broaden. This effect could be alleviated by a greater dose and energy during implantation to induce more damage, or use of an additional implantation step. However, consideration is needed to prevent this damage occupying more than 10% of lattice sites since the material would become amorphous (α -Si) [46]. Whilst α -Si does have an extended absorption edge compared to crystalline silicon ($1.25\ \mu\text{m}$ vs $1.1\ \mu\text{m}$), the divacancy absorption band ($1.8\ \mu\text{m}$) disappears completely once the material is completely amorphous [87]. This defect, or similar species, are needed for absorption around a wavelength of $1.55\ \mu\text{m}$.

3.2 Known defect species in silicon

Common defects introduced in Si by implantation are the divacancy, complexes between vacancies and impurities and Si interstitials [68,91]. The Vacancy-Oxygen complex in particular does not contribute to generation at a wavelength of $1550\ \text{nm}$, but has a larger capture cross-section than the divacancy and so may dominate carrier recombination instead [68]. This suggests that an ideal absorbing volume would only contain the defect type of interest as others will cause excess noise (smaller energy gap) or increase carrier recombination (larger gap). Some known defects are listed in Table 3.1, indicating species, key temperatures and energy levels where known. Various energy levels have been observed which are of unknown origin and require further studies before they can be used in photodetector devices.

A lot of work has been conducted with emphasis on using the Si divacancy over other defect types [34,56,61,68,88,92]. This is because it is the dominant defect present after high-energy implantation for absorption around $1550\ \text{nm}$, using a characteristic $1.8\ \mu\text{m}$ absorption band [68]. The divacancy has 4 charge states, these are V_2^- (double acceptor),

V_2^- (single acceptor), V_2^0 (neutral) and V_2^+ (single donor), trapping 4, 3, 2 or 1 electrons respectively [80]. Due to differing measurement methods and sensitivity of equipment available, there have been a range of different mid-gap energy levels measured for the transition between each of these charge states. The range of values are; $E_c - (0.2 - 0.4)$ eV (V_2^-), $E_c - (0.36 - 0.57)$ eV (V_2^-) and $E_v + (0.115 - 0.32)$ eV (V_2^0) [75]. However most experiments agree on values of 0.23, 0.39 and 0.21 eV for each transition respectively, with all of these measured using deep-level transient spectroscopy (DLTS) [75]. Figure 3.3 shows the atomic structure and energy levels for a number of vacancy-related defects in silicon, with the divacancy shown on the right hand side (f). Defects along the top row

TABLE 3.1: List of impurities and defects determined in silicon and the effect of temperature, sources indicated. V_2 = divacancy, subscripts i/s = interstitial/substitutional atom, subscripts 0 – 4 = number of occupying electrons. Positive (negative) value in energy column corresponds to transition above (below) conduction (valence) band, with charge level or transition indicated. (=) = doubly negative, (-) = singly negative, (0) = neutral, (+) = singly positive, (++) = doubly positive. E_A = diffusion activation energy, E_L = transition energy from nearest band edge, E_c = charge state levels. Fz = floating zone, Cz = Czochralski grown. ρ = resistivity. A dash indicates unknown data.

Defect	Energy	Energy Type	Disappearance with Temperature	Ref
(VP) ⁻	0.43 eV	E_A	127 °C	[80]
(VAs) ⁻	0.47 eV	E_A	168 °C	[80]
(VSb) ⁻	0.47 eV	E_A	186 °C	[80]
V_2	-0.23 eV(=/-), -0.41 eV(-/0), +0.2 eV(+/0)	E_L	225 - 275 °C (Cz), 300 - 350 °C (Fz)	[80]
	1.3/1.25 eV	E_A		[80, 88]
	0.39 eV(=), 0.54 eV(-)	E_c	-, 168 °C	[88]
(V_2) ⁻	0.43 eV, 0.45 eV	E_A	-	[80]
VSn	0.32 eV(+), 0.07 eV(++)	E_c	205 °C	[80]
V_2O	0.233 eV, 0.467 eV	E_A	350 °C	[80]
VOH	0.32 eV			[80]
VH_{0-4}	-	-	-, 223 °C, 223 °C, 527 °C, 509 °C respectively	[80]
(VO) ⁻	0.163 eV/ 0.17 eV	E_A	350 °C	[80]
(PV- V_2) ⁻	0.453 eV (Fz)	-	-	[80]
V	0.18±0.02 eV (n-type), 0.32±0.02 eV (p-type), 0.45±0.04 eV (high- ρ)	E_A E_A E_A	-203 °C (n-type) -123 °C (p-type) -73 °C (high- ρ)	[80]
	0.18 eV (=), 0.32 eV (++), 0.45 eV (0)	E_c E_c E_c		
Si_i	1.2 eV(0/+), 0.4 eV(+/++)	E_L E_L	-133 °C to -98 °C (Diffuses)	[80]
C_i	-0.1 eV(-/0), +0.28 eV(0/+)	E_L	50 °C	[80]
B_i	-0.13 eV(0/+), -0.37 eV(-/0)	E_L	-35/25 °C (charge state dependent)	[80]
Al_i	0.17 eV(+/++)	E_L	225 °C	[80]
Si_i-O_i	-	-	50 °C	[80]
B_sV	+0.31 eV(0/+), +0.37 eV(0/+), +0.11 eV(unknown)	E_L		[89]
Zn_s	+0.33 eV(++), +0.58 eV(++)	E_L		[46]
Si:Se	-0.34±0.04 eV, -0.525±0.016 eV	E_L		[90]

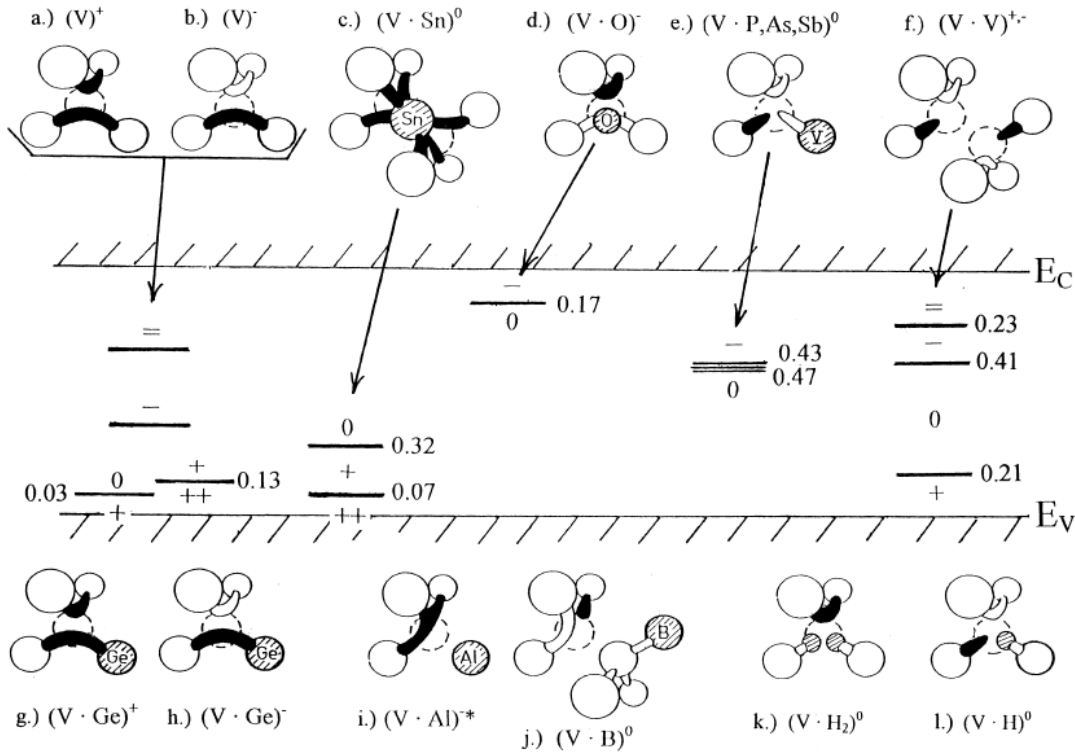


FIGURE 3.3: The structure of trapped vacancies and vacancy-impurity pairs, where a - f) have had energy levels (in eV) identified and g - l) have not. Conduction (E_C) and Valence (E_V) bands are indicated, along with the charge state each defect is in when the Fermi level is between neighbouring energy levels, sourced from [80].

have been structurally identified along with their energy level transition energies, and along the bottom row have only been structurally identified [80].

The 0.23 eV and 0.21 eV levels have been suggested to produce three infrared absorption bands in Si at $1.8 \mu\text{m}$, $3.3 \mu\text{m}$ and $3.9 \mu\text{m}$. Of these, the first has been shown to be usable for photodetection at $1.55 \mu\text{m}$, suggesting the others may have the potential for use [47, 93]. It is believed that the divacancy achieves a lower IQE than the interstitial cluster, but it benefits from a 5 - 10x higher absorption coefficient, this offsets the low IQE and allows for shorter devices [10]. One study has however needed a high reverse bias (25V), suggesting the need for improvements in carrier extraction [10].

Since multiple different defects are present at any one time, work is needed in determining how to create a favourable concentration of a desired defect structure whilst reducing the creation of others, since these will act as recombination centres and reduce device efficiency. Additionally, the silicon growth method used for the starting wafer influences the concentration of impurities in the material before processing. Which has been shown to effect the temperature at which defects will transform [80]. For example, in Czochralski grown silicon, the divacancy anneals out at $225 - 275^\circ\text{C}$ whereas in float-zone silicon it anneals out at $300 - 350^\circ\text{C}$ [80]. This effect could be a result of the higher-purity of float-zone silicon, which has a much lower oxygen content in the

starting wafer as compared to $1 \times 10^{18} \text{ cm}^{-3}$ in Czochralski grown [94]. This also results in a larger concentration of vacancy-oxygen complexes, which may dominate over divacancy absorption due to their larger capture cross-section [68]. It has been suggested that after annealing out divacancies with a 300°C anneal, silicon interstitials become the majority defect which are stable up to 600°C with a higher responsivity but lower absorption coefficient [82].

For extending detection into the mid-infrared region ($2 - 6 \mu\text{m}$) use of smaller energy transitions are needed, and whilst two-photon absorption is not present in silicon beyond $2.1 \mu\text{m}$ [4], other losses from nonlinear effects need to be considered [25]. Detection at $2.2 - 2.4 \mu\text{m}$ has been demonstrated using an energy transition from the double-acceptor Zn trap state level to the conduction band in a silicon $p^+ - p - i - n - n^+$ waveguide integrated photodetector [10]. Whilst detection out to $3.3 \mu\text{m}$ or $3.9 \mu\text{m}$ has been predicted for samples containing divacancy defects using transitions from the doubly negative charge state (-0.21 eV) to the conduction band, and from the valence band to the single positive charge state ($+0.25 \text{ eV}$) respectively [47,95]. From Table 3.1, there may also be potential for detection in the $3.5 - 3.8 \mu\text{m}$ wavelength range ($0.352 - 0.326 \text{ eV}$) using B_sV , VO, Si:Se or VS_n if the associated charge states are electrically active.

3.3 Device integration

Defect-mediated absorption can be integrated into detectors by high-energy ion implantation into a region of silicon through which an optical signal propagates. For a free-space detector this will be over the entire device area, while a waveguide-integrated structure would benefit from a “window” in the passivation layer to restrict defects to the waveguide core. Heavily doped p and n-type regions provide an electric field and allow connection to electrical contacts for carrier extraction, creating a photocurrent in an external circuit. The low absorption coefficient of defects can lend to applications which need to measure an optical signal without a significant reduction in power, such as power or wavelength monitoring, or thermal tuning [10]. Control over the tapped proportion of the signal is achievable by varying the concentration of defects introduced in the detector region, or changing the interaction length of the device. Unfortunately, the distance needed can become relatively large if a significant proportion of the signal is needed. For the divacancy this can require over $100 \mu\text{m}$ for a p-i-n junction across a waveguide, and operating speed becomes limited by the higher capacitance, but can be improved by using a metal-semiconductor-metal structure, as in Chapter 7 [47,83].

Defect-enhanced devices have a peak in external quantum efficiency at a different concentration to responsivity, towards higher concentrations efficiency decreases as a result of carrier recombination [68]. This suggests an optimal concentration exists whereby generated carriers from a given defect site are more likely to be extracted by the electric

field than to recombine at a different site. It has been suggested that this falls between 1×10^{16} and $2 \times 10^{18} \text{ cm}^{-3}$ [24,68]. Away from the optical mode, no generation can occur and all defect sites act as recombination centres reducing the proportion of absorbed power which can be extracted as photocurrent [68]. This could also occur when there is insufficient energy from the combination of an incoming photon and thermal stimulation to complete the electron promotion. Therefore an optimal device would only contain the defect species which absorb the intended wavelength, constrained to both the volume of the optical mode for carrier generation and depletion region for carrier extraction, with all remaining area masked during ion implantation [96,97].

This creates the need to optimize both the window width in the implantation mask and contact separation for control of operating speed, where the window controls overlap between mode, electric field and defective volume. If the optical mode covers a larger cross-sectional area than the defective volume, a lower proportion of power can be absorbed requiring a longer device [62]. Whilst if larger than the depletion region, power will be lost to FCA in the heavily doped regions, reducing photocurrent [47]. A wider depletion region, assuming the other two overlap optimally, will extract the maximum amount of carriers but may succumb to speed limitations due to a higher than necessary transit time, as well as using more chip area. Whilst if the defective volume is larger than the other two, recombination rather than generation would occur, increasing dark current [68].

Overlap with doped regions can be tolerated if the gain from overlapping optical mode, defect-volume and depletion region outweighs the additional losses from compensation of defects by heavily-doped regions [68]. It has been shown in both n-type and p-type silicon that a reduction in doped-region separation can cause an increase in photocurrent [47]. In n-type Si, photocurrent maximises when region separation is similar to the waveguide rib width, but slightly narrower than the implantation window [47]. This suggests that for this device, the extra FCA that occurs from a small overlap of the optical mode with the heavily-doped contact regions is non-detrimental, and is overcome by an increased extraction probability when generation becomes very low at the tails of the optical mode. Although it does mean that a non-optimal quantum efficiency is achieved [47].

Additionally, bandwidth has been shown to be better in n-type silicon for low defect concentrations up to $5 \times 10^{16} \text{ cm}^{-3}$, after which p-type performs better [68]. This occurs because vacancy-type defects act as acceptor levels, which compensates the electron concentration in n-type silicon, and adds to the hole concentration in p-type. With relatively few defects, the majority carrier concentration in n-type is higher than that in p-type, leading to a greater reduction in depletion region width and higher electric field. Beyond $5 \times 10^{16} \text{ cm}^{-3}$, the compensation in n-type has reduced the free electrons to less than the free holes in p-type and so the field is now higher in p-type silicon.

3.4 Defect-enhanced detectors in literature

Previous experimental studies into defect-enhanced detectors have primarily focussed on creating divacancies or interstitial clusters as the primary defect. They are commonly the majority species remaining after high-energy ion implantation, and allow absorption of wavelengths between $1.55\ \mu\text{m}$ and $2.3\ \mu\text{m}$. Post-implantation annealing was used to control the majority species concentration, with the divacancy expected to diffuse out beyond around $300\ ^\circ\text{C}$ [98].

A waveguide-integrated photodetector developed by J. Ackert et. al. has achieved a responsivity of $4.7 \pm 0.5\ \text{AW}^{-1}$ under $40\ \text{V}$ reverse bias and $1.55\ \mu\text{m}$ light. It used B^+ implantation at $500\ \text{keV}$ with a dose of $5 \times 10^{12}\ \text{cm}^{-2}$ and $200\ ^\circ\text{C}$ isochronal annealing [56]. Since this device is using a high voltage and achieves $> 1\ \text{AW}^{-1}$, it is operating in an avalanche regime and therefore will suffer speed limitations (see Section 2.2.2); Alternative work by B. Souhan et. al. has achieved $9.9 \pm 2\ \text{mAW}^{-1}$ at $5\ \text{V}$ with $2.2 - 2.3\ \mu\text{m}$ light, using Si^+ implantation at $190\ \text{keV}$ with a dose of $1 \times 10^{13}\ \text{cm}^{-2}$ and a $200\ ^\circ\text{C}$ anneal [49]. Here a larger ion species and higher dose is used to increase the defect concentration and allows for absorption at a longer wavelength. The transition between one of the intrinsic energy bands and a mid-gap level created by the divacancy concentration is attributed as the cause of absorption here, however it is not known which charge state it was in, which is influential on the magnitude of this transition [49, 80]. For an equivalent device, a larger ion species needs a higher energy to reach the same waveguide depth, however it was reduced here due to the lack of a thick ($1\ \mu\text{m}\ \text{SiO}_2$) passivating layer. Further investigation also used Si^+ , but at an energy of $195\ \text{keV}$ (same dose) and wavelength of around $1.9\ \mu\text{m}$, achieving a responsivity of $31.8\ \text{mAW}^{-1}$ at $5\ \text{V}$ [34]. This suggests the defects present respond more to the shorter wavelengths within the 1.55 to $2.3\ \mu\text{m}$ range.

Absorption has also been achieved using zinc ions implanted into silicon, which creates two energy levels at $0.3\ \text{eV}$ and $0.58\ \text{eV}$ above E_v by occupying substitutional sites within the lattice [46]. The transition between $E_v + 0.58\ \text{eV}$ and E_c is the only one to provide photon absorption, allowing detection of $2.2\ \mu\text{m}$ wavelength photons, while the other enhances re-population by thermal excitations [46]. R. Grote et. al. used the structure shown in Figure 3.4 with Zn^+ implantation at an energy of $260\ \text{keV}$ and a dose of 10^{12} or $10^{13}\ \text{cm}^{-2}$, followed by annealing at increasing temperatures up to $350\ ^\circ\text{C}$. However, since this ion is metallic it can create short-circuits or a localized Schottky barrier in thin layers if the location of implantation is not tightly controlled.

In another experiment, an increase in excess loss from 75 to $300\ \text{dBcm}^{-1}$ had occurred by increasing the Si^+ dose from 1.25×10^{12} to $1.75 \times 10^{14}\ \text{cm}^{-2}$, at an energy of $700\ \text{keV}$, as illustrated in Figure 3.5 [2]. Subsequent annealing of the devices shows out-diffusion of the primary defect after $225 - 250\ ^\circ\text{C}$, which is attributed to the divacancy [2]. This is

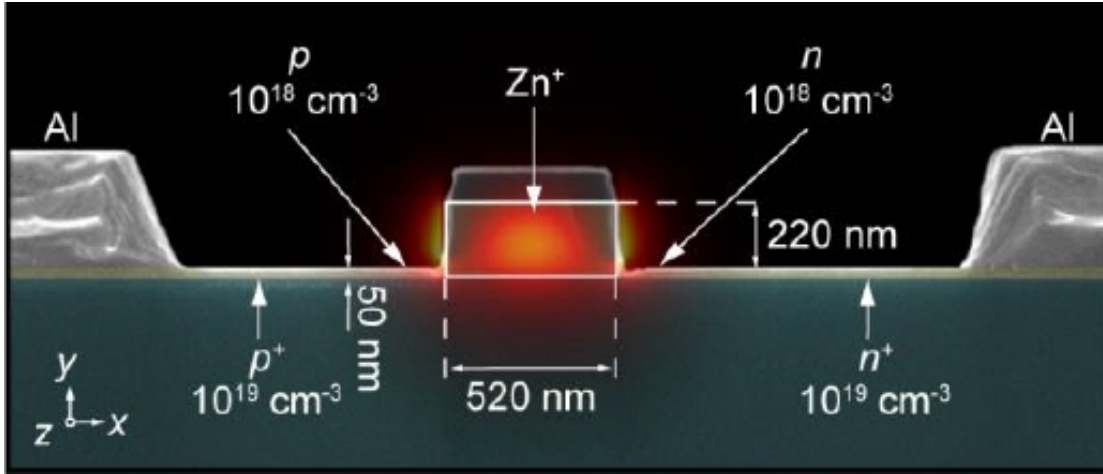


FIGURE 3.4: Cross-Section of a waveguide integrated photodiode, doped with 10^{12} or 10^{13} cm^{-2} Zn^+ ions in the centre rib region. Sourced from [46].

likely different from the previously mentioned value of 300°C , due to the starting wafer growth method being Czochralski grown rather than float zone [92]. Another benefit of using silicon ions is that there is no doping or foreign species introduced, and as such the response is purely due to the defects that have been created. Alternative ion species which do not dope the substrate may also provide only defect-mediated absorption, with the change in size or energy effecting the concentration and spread of defects.

The above rib waveguide structure is not the only one that has been used for defect-mediated integrated detector designs. An alternative has used a LOCOS (Local Oxidation of Silicon) process, as shown in Figure 3.6 [96]. Here, each side of the optical

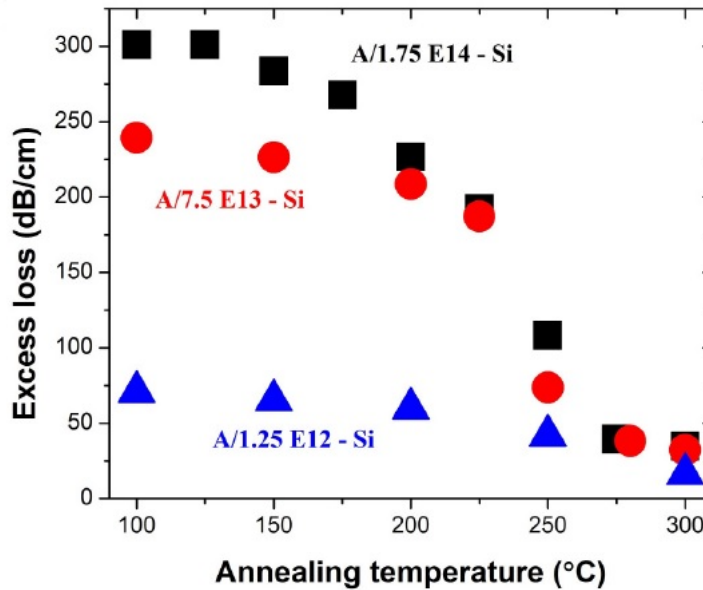


FIGURE 3.5: Excess optical loss of racetrack resonator devices as a result of defects implanted with silicon at 700 keV and various ion doses (1.25×10^{12} , 7.5×10^{13} and $1.75 \times 10^{14} \text{ cm}^{-2}$) after 5 min $100 - 300^\circ\text{C}$ anneals, sourced from [2]

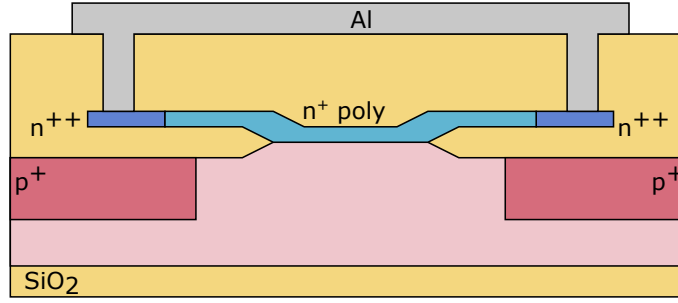


FIGURE 3.6: Integrated photodetector structure based on LOCOS process (p^+ regions join to an Al contact), redrawn from [96]

mode are both p-type, while space above forms the n-type contact region. This is a standard CMOS process, and provides low side wall scattering loss since selective oxide growth creates a smooth interface between Si and SiO_2 using nitride as a mask [96]. Defects were introduced to this design using a 4 MeV Si^+ implantation at a dose of $1-5 \times 10^{13} \text{ cm}^{-2}$ after metallization and through a photoresist-defined window [96]. It achieved a responsivity of 47 mA W^{-1} to $1.55 \mu\text{m}$ light, with the waveguide having a low transmission loss of $2.9 \pm 1.1 \text{ dB cm}^{-1}$.

3.5 Defect concentration measurement

Direct measurement of which electrically-active defect levels are present within a material can be performed with Deep-Level Transient Spectroscopy (DLTS) [99], or a more modern Laplace DLTS [92]. This technique uses a lock-in amplifier to scan through frequencies whilst observing the transient capacitance and varying the decay time constant of the defect present by varying the sample temperature, to produce a peak in the DLTS spectrum [100]. This is the difference in capacitance across the scanned rate window against temperature [100].

Plotting the decay time constant (τ_e) against temperature (T) as $\ln(\tau_e T^2)$ vs T^{-1} provides an Arrhenius plot with a gradient equal to the defect energy level. This and the DLTS peak location are unique for different defects allowing identification, and the magnitude of the peak allows concentration measurement [100]. The Laplace DLTS approach additionally converts the transient response from the time domain to frequency domain, which allows for a more distinct representation of defects in the sample even between very close lying energy levels [100]. DLTS is the most common approach used for deep-level trap characterisation, but impurity identification is not always easy and it is difficult to characterise high resistivity samples [100].

An alternative approach uses Positron Annihilation Spectroscopy (PAS), which uses the detection of gamma rays emitted from a sample which has been subjected to a ray of positrons which annihilate with electrons in the sample [100]. The angle and energy

of the two emitted gamma rays, along with the positron lifetime, provide information about open volume type defects (vacancy complexes) in the sample due to their altering of the local electron density [100]. Identification of defect species and concentration can be inferred from the lifetime, or from doppler broadening of the gamma rays as characterised by the S parameter, which is the number of counts in the centre of the energy peak over the total number in the peak [100]. Measuring this parameter against annealing temperature allows determination of vacancy creation during implantation and subsequent destruction [100]. Whilst varying the energy of the positron beam provides information about the depth of defects in the sample [100]. This approach is unfortunately limited to void-like defects and requires very elaborate equipment [100].

3.6 Influence on optical properties

The introduction of defects into a waveguide results in a localised increase in absorption coefficient, and therefore an increase in the refractive index due to the Kramers-Kronig dispersion relations [101]. This therefore changes the single mode distribution of the propagating optical mode [62]. If the waveguide is designed to operate at the upper limit of single mode confinement, this effect could result in modal radiation along the length of the photodetector [62]. If the ion implantation used is below the amorphization threshold, the change in refractive index is minimal and can be ignored for non-resonant devices [62].

For resonator-enhanced devices, initial ion implantation causes a small red shift in the transmitted power and a lowering of the quality factor, returning back to the pre-implanted state after annealing as shown in Figure 3.7a, and has a greater shift in

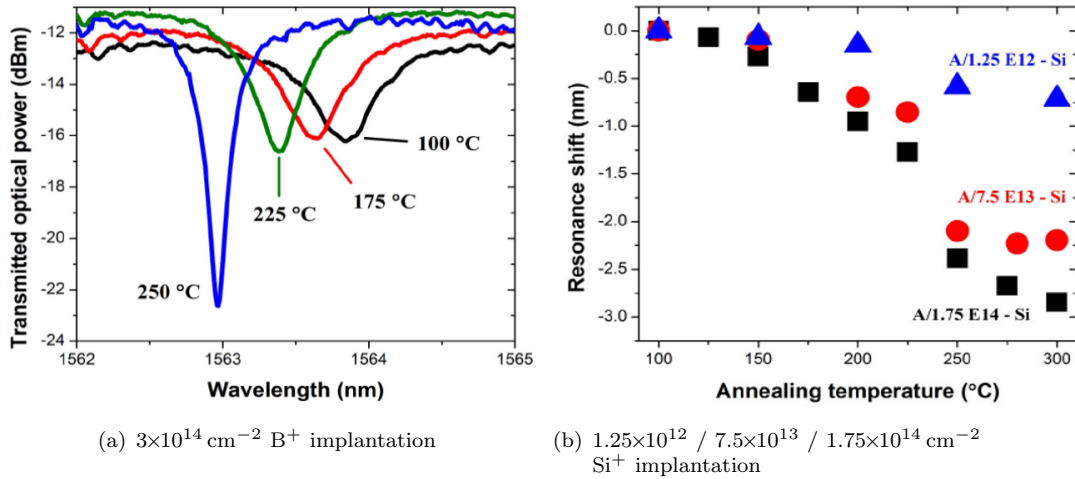


FIGURE 3.7: Optical transmission for a racetrack resonator after implantation (100 °C) and subsequent annealing, showing resonance shift and increase in Q factor with temperature and increased shift with dose, sourced from [2]

resonance for higher doses as shown in Figure 3.7b [2]. Assuming a change in effective refractive index for the waveguide is caused solely by a change in refractive index in the silicon (neglecting the effect on cladding oxide) an effective index susceptibility of 1.177 ($\delta n_{\text{neff}}/\delta n_{\text{Si}}$) has been calculated by J. Ackert [2]. From this, the change in refractive index was determined to be 0.005, 0.016 and 0.021 for resonance shifts of 0.7, 2.2, and 2.9 nm respectively for the increasing doses used in Figure 3.7b [2]. This effect can be used as a means of tuning the resonant wavelength and absorption loss post-fabrication as an alternative to a thermo-optic tuner, or the application of an external electric field to use electroabsorption and electrorefraction via the Franz-Keldysh effect or injection of free-carriers, when the addition of extra electrical contacts is not desired [2, 101].

3.7 Summary

This chapter has provided an explanation of what crystalline material defects are, how they occur, and what species are known to develop in the main active region material used within this thesis, silicon. An insight into how defect-mediated absorption can and has been used in detectors is highlighted, and it has been shown that post-implantation annealing aids to control, or completely remove, the majority defect species concentration. The primary defect occurring is that of the divacancy, and is believed to have a significant majority such that concentration analysis can assume this to be the only species and that it primarily holds a single charge state as determined by the Fermi level of the host material. Direct measurement of defect concentration is not reasonably possible with the available equipment, so has to be indirectly calculated from empirical equations. Subsequent device design used in the following chapters incorporates high-energy B^+ or He^+ implantation to stimulate the creation of defects. The first ion is known from literature to induce divacancies, while the second may act to passivate the vacancies/divacancies it creates, causing an enhanced thermal stability. Both of these are small ions, and as such require a lower energy to reach a given depth compared to larger elements.

Chapter 4

Defect-mediated absorption in a free-space photodetector

Growth and development of photonic systems is heavily influenced by cost and availability of the production processes that are available. The most promising platform uses silicon as the core substrate material, and incorporates ion implantations to create doped regions within widespread CMOS-compatible technologies. This process can be adapted to create a favourable defect species, as discussed in Chapter 2, offering a viable option for extending the range of photonic device capabilities in silicon. This would allow greater integration, lower costs, and both easier and higher-yield fabrication of advanced microelectronic systems. This chapter demonstrates the integration of such a process into a free-space silicon-based photodetector, with a detection wavelength extended to 1550 nm.

4.1 Device description

The detector described here-in adds a high-energy B^+ or He^+ ion implantation to an existing fabrication process, using a photolithography mask set kindly provided by I. Sari. It is a free-space detector illuminated from above, as illustrated in Figure 4.1a, with metal contacts forming the shape of a pair of concentric rings. The outer ring forms the cathode and the inner ring forms the anode, with each connected to a metal contact pad for electrical probing as shown in Figure 4.1b. Both contacts have a high concentration doped region below to ensure good ohmic contact for diode-like IV characteristics. The also enhance the electric field strength to improve the speed and likelihood of carrier extraction for useful photocurrent. The dimensions labelled A, B and C vary within each die according to the list in Figure 4.1, however it is expected that only C will have an effect on device performance, so only this dimension is considered during analysis. To reduce cross-talk and the loss of carriers to neighbouring devices when under a

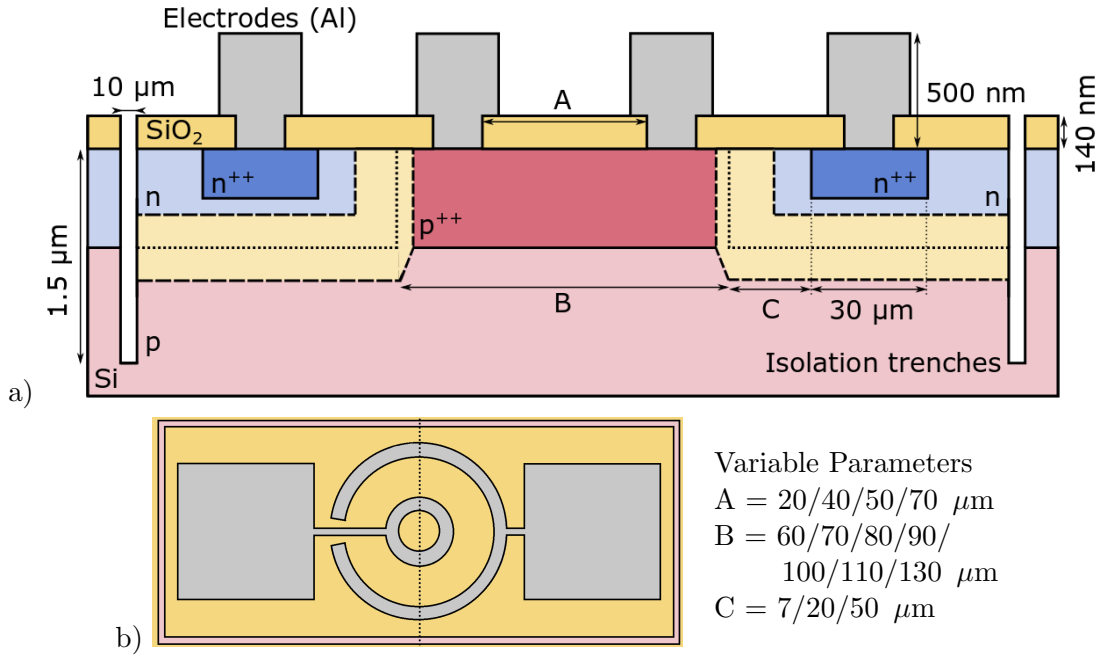


FIGURE 4.1: Diagrams showing free-space detector design, where a) cross-section with pale yellow colouring indicating depletion regions and b) surface view showing metal contacts with the cross-section position illustrated by a dotted line. Variable device dimensions are indicated by a letter and listed.

high reverse bias, isolation trenches are incorporated to prevent lateral expansion of the depletion regions formed with the bulk substrate. This also allows for better control of collected charge distribution, preventing a contribution of random carriers generated from regions of the substrate not illuminated, reducing device noise.

An additional expansion of the original design adds a blanket phosphorous implantation to create an upper n-type surface layer roughly 400 nm deep. This creates a horizontal depletion region in addition to that which forms between contacts rings, therefore increasing the absorbing region for incoming light. The lower doping contrast across the junction between this new layer and the substrate also enables a greater depletion width, compared to that between the n^{++} region and the substrate. It also enables the unmasked high-energy implantation to create more carrier generation centres where defects form within the depletion region, rather than recombination centres in undepleted silicon. The size of the depletion region is controllable by the strength of the reverse bias applied between contacts, allowing for more defects to act as generation centres by increasing the voltage. At high voltages this region can extend deep into the substrate, significantly increasing the total optical power which can be absorbed. As a result of adding this layer, the dose and energy for the B^+ implantation used to form the p^{++} region needed to be increased to ensure dopant inversion and successful bridging of the gap between contact ring and the substrate.

B^+ and He^+ ions are used to induce defects since they are light, and as such need a lower implantation energy to reach a desired depth than larger elements. The first ion

has been used in a number of published results and was shown to introduce divacancies as the majority defect species [24, 35, 56, 102, 103]. The second ion has been used by Liu et. al. for an in-line power monitor and is believed to introduce passivated vacancies, divacancies and He-vacancy complexes, transforming into voids for larger ion doses [59, 104]. Helium passivation of vacancies stabilises them and reduces their mobility, causing them to remain in the sample at room temperature [104]. Otherwise they would anneal out at very low temperatures (up to -73°C , see Table 3.1) [104]. Whilst implantation energy is discussed in the following section, both ions are implanted with a dose of $5 \times 10^{12} \text{ cm}^{-2}$, which was determined in studies by D. Thomson et. al. to be near-optimal for B^+ ions [35]. It provides a good trade-off between increasing absorption with defect concentration, and a reduced quantum efficiency due to excess recombination centres which would reduce the electrical device characteristics. At the upper limit, the amorphization threshold needs to be avoided to maintain simple defect creation and avoid de-crystallization of silicon.

Implantation energy is designed to force the ion range and maximum concentration to a depth far below the isolation trenches. This prevents a significant quantity of ions remaining in the upper active region where they would influence the overall doping concentration. Using stopping range of ions in matter (SRIM) calculations, B^+ energy was tailored so the ion concentration was at 0.1 % of the maximum value at a depth of $1.5 \mu\text{m}$ to align with the bottom of the isolation trenches, giving an energy of 1800 keV. For the He^+ ions, a higher concentration can be tolerated since this ion does not introduce any doping to the substrate, as such 3 standard deviations (0.4 %) below the maximum is used and gives a value of 600 keV. Another benefit of pushing the ions this deep into the substrate is an increased total amount of energy to transfer to the production of vacancies.

4.2 Simulations

The simulations performed throughout this section use Silvaco's Athena technology computer-aided design (TCAD) software for process-level simulation, while Silvaco's Atlas TCAD is used for the device-level [105]. These were available for use on the Iridis4 supercomputer cluster, maintained by the Computational Modelling Group, at the University of Southampton [106]. Additional Monte Carlo simulations were performed using (SRIM) calculations for ion and vacancy concentration profiles against depth, in software developed by J. F. Ziegler [107].

4.2.1 Implantation profiles and trench depth

Initially, implantation profiles for each doped region (n , n^{++} and p^{++}) were simulated using Silvaco's Athena TCAD software to obtain desired depth and surface concentrations [105]. The unmasked n -type region was intended to be around 400 nm deep, with a sufficient dose to overcome compensation with the substrate doping. While the p^{++} region needs to extend slightly further than this depth to ensure it bridges the gap between the metal contact ring and p -type substrate, allowing for some fabrication tolerance. The n^{++} region was then designed to be around a quarter of the n -type depth, so that it is a sufficient distance from the p - n junction below to allow greater expansion of the depleted region. A surface concentration of over $1 \times 10^{19} \text{ cm}^{-3}$ for the heavily doped regions will ensure a good ohmic contact is formed with the electrodes, whilst also enhancing the electric field strength to improve carrier extraction (EQE). Depth is defined as the position at which net doping has dropped to 10% above the background substrate concentration, or the n -type surface concentration for the n^{++} region depth. The final depth and surface concentrations were; 438.7 nm and $2.99 \times 10^{18} \text{ cm}^{-3}$ for the n -type, 113.0 nm and $4.95 \times 10^{19} \text{ cm}^{-3}$ for the n^{++} then 516.1 nm and $6.80 \times 10^{19} \text{ cm}^{-3}$ for the p^{++} regions.

To simulate the expected performance of these devices, a model was built using Silvaco's Atlas TCAD software. This began with a device which does not include any defect modelling to establish the isolation trench depth. With experimental results operating at a maximum reverse bias of 10 V, this depth was designed to be the minimum needed to prevent the depletion region extending beyond this distance. A reverse bias sweep was performed for an unilluminated device, extracting the electric field and calculating the depletion region edge depths formed between the upper n -type layer and the p -type substrate at a range of voltages, with the results plotted in Figures 4.2 and 4.3 respectively. The depth at which the electric field at 10 V has dropped to 10 % of its

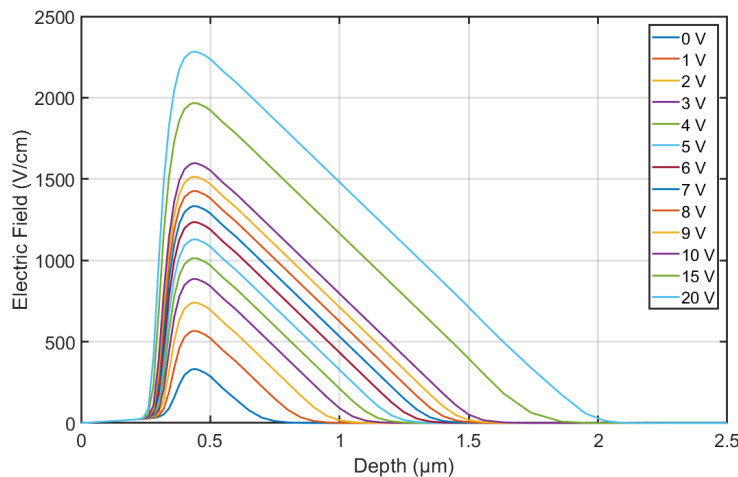


FIGURE 4.2: Electric field vs depth at a range of reverse bias voltages for device without defect-inducing implantation to establish the isolation trench depth

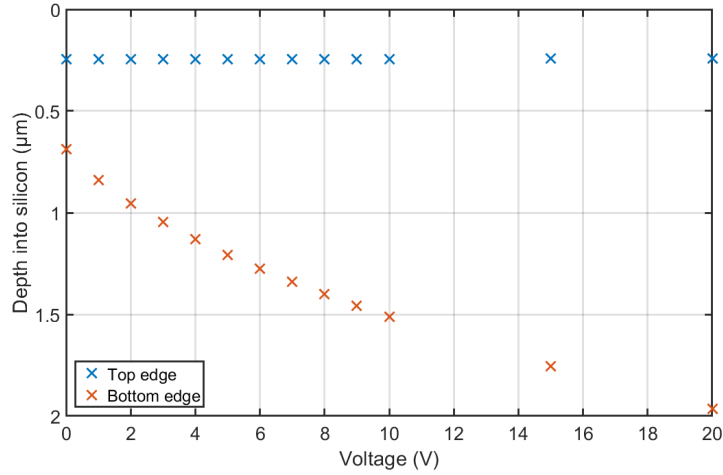


FIGURE 4.3: Depth of depletion region edges vs applied reverse bias for device without defect-inducing implantation to establish the isolation trench depth

maximum value is $1.45 \mu\text{m}$, while the bottom of the depletion region is $1.51 \mu\text{m}$ from the silicon surface. This gives a target trench depth of $1.5 \mu\text{m}$. Operation beyond this voltage will extend the depletion deeper, but lateral spreading will still be severely reduced, especially in the upper 400 nm where this device primarily operates. The depth at which the strongest response to induced defects will be seen is where the electric field is strongest, which according to Figure 4.2 is at around 400 nm , and as can be seen in Figure 4.1a is the junction between upper n-type layer and the substrate. Additionally, the depletion region is able to freely extend further into the substrate as reverse bias voltage increases, allowing a greater proportion of generated carriers to be extracted. This is beneficial until the depth at which all the incident power has been absorbed, or until the maximum depth of implanted defects for trap-assisted absorption, after this point only thermal noise will be added. Suggesting photocurrent will increase until a peak voltage and rather than saturating it will slowly decrease while noise takes a greater majority of terminal current.

4.2.2 Single-defect modelling

It is assumed that the majority of defects present after high-energy ion implantation are in the form of the divacancy, and as such a single-defect modelling approach is used in this work. As a result, it is expected that this model will have an overestimated photo-generation rate, and a reduced spectral broadening after the introduction of defects.

Divacancy energy levels are introduced into the device using the *trap* script command. An acceptor-like energy level is added, occurring at a uniform concentration of $2.08 \times 10^{17} \text{ cm}^{-3}$ within the top $2.91 \mu\text{m}$ of the silicon substrate. Within this region the analytical defect distribution model, explained below, is greater than $1 \times 10^{-9} \text{ Vac.Ang}^{-1} \cdot \text{ion}^{-1}$ and is modelled as uniform for simplicity. This concentration is the average value of the

distribution model, and the capture cross-section of the divacancy is $3.65 \times 10^{-15} \text{ cm}^2$ and $3.6 \times 10^{-14} \text{ cm}^2$ for electrons and holes respectively [47]. The trap distribution could be improved by using a c-interpreter function to represent the concentration as a function for depth, but is not included here since these trap levels only contribute to the reduction in free carriers due to additional Shockley-Read-Hall recombination.

The effect of divacancies is included in the model by using the *beam* statement with a c-interpreter custom photogeneration rate, which when called will return an electron-hole generation rate (in $\text{cm}^{-3}\text{s}^{-1}$) based upon the (x,y) co-ordinate position within the structure. A copy of this function is included in Appendix A.1. It is based upon an analytic approximation of the spread of vacancy damage against depth, as produced by a SRIM calculation of He^+ ions at 600 keV into silicon. An ion distribution can be analytically represented by a Gaussian profile, and will model the distribution of vacancies beyond the maximum damage depth, but prior to this the cumulative effect of damage creation necessitates the use of a high-order polynomial approximation. This damage profile is then multiplied by the implantation dose of $1 \times 10^{13} \text{ cm}^{-2}$ and converted to divacancy concentration using an empirical formula by P. Coleman et. al. for vacancy-type defect concentration at half ion range remaining at room temperature, given in Equation 4.1 [62].

$$C_D = (2.79 \times 10^{10}) \phi_A^{0.63} \quad (4.1)$$

Where ϕ_A = Adjusted ion dose, calculated as the implantation dose multiplied by the number of vacancies per ion per angstrom at half ion range as calculated by SRIM simulation [62]. This concentration is scaled by the divacancy optical cross-section at a wavelength of 1550 nm ($7.7 \times 10^{16} \text{ cm}^{-2}$), to obtain the absorption coefficient between the neutral and singly negative charge states [47].

Next, the generation rate is calculated for an optical fibre incident upon the surface at an angle from Equation 4.2 [47].

$$G_{opt}(x, y, z) = I(x, z) \frac{\lambda}{hc} \alpha_t \exp(-\alpha_t y) \quad (4.2)$$

Where α_t is the divacancy absorption coefficient (cm^{-1}), and $I(x, z)$ is the optical intensity profile of the incident beam as described by a Gaussian waveform. The intensity is normalized by the total optical power to ensure the weighting factor given in the *beam* statement corresponds to the optical intensity (Wcm^{-2}). Since Atlas assumes a uniform $1 \mu\text{m}$ depth for the cross-section in a 2 dimensional simulation, there is no variation in intensity in the z-direction, reducing its profile to $I(x)$ as given in Equation 4.3.

$$I(x) = A \exp\left(-\frac{(x - \bar{x})^2}{2\sigma_x^2}\right) \quad (4.3)$$

Where x is the position along the surface of the device (in microns), $\bar{x}$ is the position of the incident beam center on the surface and σ_x is the standard deviation of the Gaussian waveform (in μm). Since the beam area is reduced from a circle of diameter $10.4 \mu\text{m}$ to a $10.4 \mu\text{m} \times 1 \mu\text{m}$ rectangle, the optical intensity increases to maintain the same power, and therefore the divacancy concentration is scaled by a factor of 8.168 to maintain the same concentration per Wcm^{-2} .

The angle of the beam is accounted for by offsetting the x position with respect to the depth y into the substrate, and is taken to be 6.3° to match the average value in the experimental results of Chapter 6. The standard deviation is taken to be $3.67696 \mu\text{m}$ to correspond to a single mode fibre with mode field diameter of $10.4 \mu\text{m}$. The optical power (P) is expressed as in Equation 4.4, which leads to A as the weighting factor given by Equation 4.5, and [47].

$$P = \int_x I(x) dx = \sqrt{2\pi}\sigma_x T \quad (4.4)$$

$$A = \frac{P}{\sqrt{2\pi}\sigma_x T} \quad (4.5)$$

Where T is the simulation depth. From this, it was determined that the optical intensity to be used in the Atlas command is 54248.9 Wcm^{-2} to correspond to an optical power of 5 mW .

To determine the dark characteristics, the calculated unilluminated current in Amperes-per-micron must be multiplied by the length of the arc to obtain the full current between contacts. The radius for this length is taken as the position of the optical fibre in relation to the centre of the circle, giving a value of $250.84 \mu\text{m}$. This approach does not account for the effects of current crowding around the edges of the smaller ring, or spreading at the outer ring edge, and as such is likely to be an overestimate of the dark current.

4.2.3 Incorporating defects into a free-space photodetector

The results of simulations for a device with a $160 \mu\text{m}$ diameter between the outer edges of the larger ring are listed in Table 4.1. For this model a defect profile for 600 keV He^+ ions at a dose of $1 \times 10^{13} \text{ cm}^{-2}$ is used. Illuminated results are recorded using a laser power of 5 mW , with the exception of the spectral response which uses a power of 1 mW to avoid avalanching effects at longer wavelengths greatly increasing the current response.

TABLE 4.1: Results from device simulation of free-space photodetector with and without a defect-inducing implantation

	Wavelength	Unimplanted		He ⁺ -Implanted	
		2 V	7 V	2 V	7 V
Dark current	-	1.16 nA	2.55 nA	6.95 nA	13.9 nA
Terminal current	1550 nm	192.1 nA	206.3 nA	1.09 μ A	1.35 μ A
Terminal current	633 nm	370.3 μ A	666.0 μ A	189.3 μ A	429.4 μ A
Photocurrent	1550 nm	191.0 nA	203.7 nA	1.09 μ A	1.34 μ A
Photocurrent	633 nm	370.3 μ A	666.0 μ A	189.3 μ A	429.4 μ A
SNR	1550 nm	44.3 dB	38 dB	43.9 dB	39.6 dB
SNR	633 nm	110.2 dB	108.3 dB	88.7 dB	89.8 dB
Responsivity	1550 nm	38.4 μ A.W ⁻¹	41.3 μ A.W ⁻¹	218.8 μ A.W ⁻¹	270.1 μ A.W ⁻¹
Responsivity	633 nm	74.1 mA.W ⁻¹	133.2 mA.W ⁻¹	37.9 mA.W ⁻¹	85.9 mA.W ⁻¹
OCV	1550 nm	0.50 V		0.45 V	
OCV	633 nm	0.75 V		0.60 V	

Firstly, an IV sweep from -10 to 10 V was performed for a device without defect inclusion under dark conditions, then repeated using a beam with a wavelength of 633 nm and then 1550 nm. These beams are a standard monochromatic source incident upon the surface half-way between the two contacts and at an angle of 6.3° off-normal, to assess the unimplanted case. The sweeps were then repeated with the 1550 nm laser replaced by the custom photogeneration model described in Section 4.2.2. To model the He⁺-implanted case the distribution of divacancy trap levels is also added for modelling electrical recombination effects. The results of these can be seen in Figures 4.4 and 4.5. They demonstrate that this particular defect distribution causes a 5.67x (6.55x) increase in terminal current at a reverse bias of 2 V (7 V) and a wavelength of 1550 nm, with the majority of this increase caused by the gain in photocurrent. Meanwhile, the response to 633 nm wavelength light has dropped by 48 % (36 %) at a reverse bias of 2 V (7 V) as a result of defect introduction. The predicted dark current for this

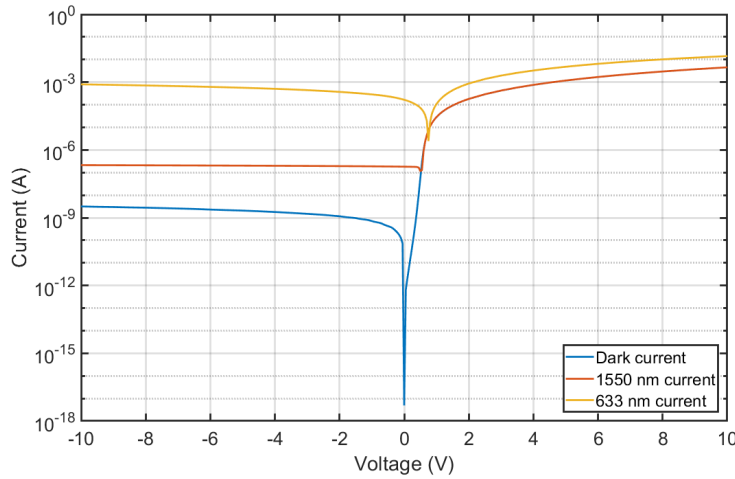


FIGURE 4.4: IV characteristics for free-space detector without ion implantation induced defects, under dark conditions and illuminated by 633 nm or 1550 nm light as described in the text

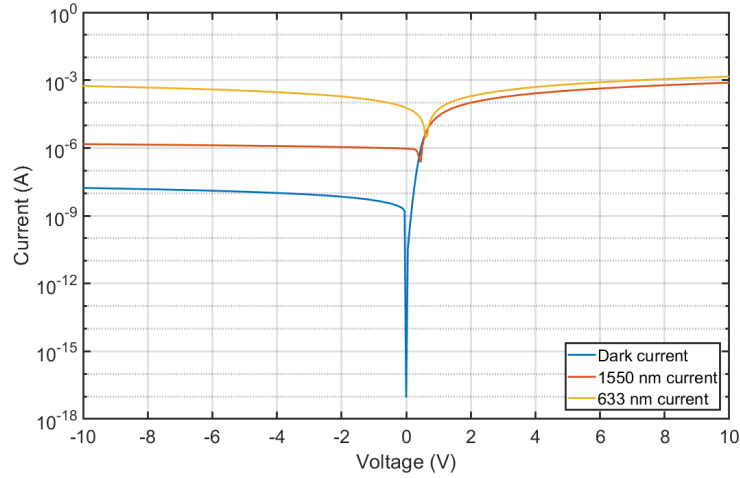


FIGURE 4.5: IV characteristics for free-space detector with He^+ ion implantation induced defects, under dark conditions and illuminated by 633 nm or 1550 nm light as described in the text

device is only a few nanometers, which will increase for the experimental devices due to fabrication imperfections, but does show how the trap levels introduce additional random recombination noise and increase current by 5.79 nA (11.35 nA) at 2 V (7 V) .

Using Equation 2.8 the SNR is calculated for each device, and shows an increase under 1550 nm wavelength light only at the higher voltage in Table 4.1. This is likely due to how deep the introduced defect concentration is into the substrate, with a projected ion range of $2.23 \mu\text{m}$ resulting in defects with a concentration below $0.005 \text{ A}^{-1} \cdot \text{ion}^{-1}$ up until a depth of $1.35 \mu\text{m}$. From Figure 4.2, it can be seen that the electric field at 7 V is able to extract carriers generated at the depth of maximum defect concentration, while the field at 2 V is insignificant beyond a depth of around $1.1 \mu\text{m}$. This graph also shows that over 20 V would be needed to reach the full span of defects as modelled by a depth of $2.91 \mu\text{m}$ in Section 4.2.2. Meanwhile the SNR against 633 nm wavelength light is around 0.7 % (1.4 %) after defect introduction at 2 V (7 V) due to the significant loss in photocurrent. Additionally the OCV point is determined, and is expected to shift slightly towards a higher positive bias upon illumination and do so to a greater extend for a higher photogeneration rate. This behaviour is seen between 1550 nm and 633 nm wavelengths since the intrinsic process will create more carrier pairs, however the defect introduction causes both voltages to reduce. These suggest that while defects can be used to aid the generation of photocurrent in a photodetector, it may not be suitable for operation in a photovoltaic device since a lower fill factor may result. Illustrated in Figure 4.6, is the responsivity of each device against applied reverse bias voltage. The defect introduction causes the response to 633 nm light to roughly half, while the response to 1550 nm light increased up to 6.9x from the unimplanted device.

The final part of simulations evaluates the spectral response of the device without defects under a reverse bias of 5 V, to determine the normal variation with wavelength and intrinsically low response to the desired 1550 nm wavelength, as shown in Figures 4.7

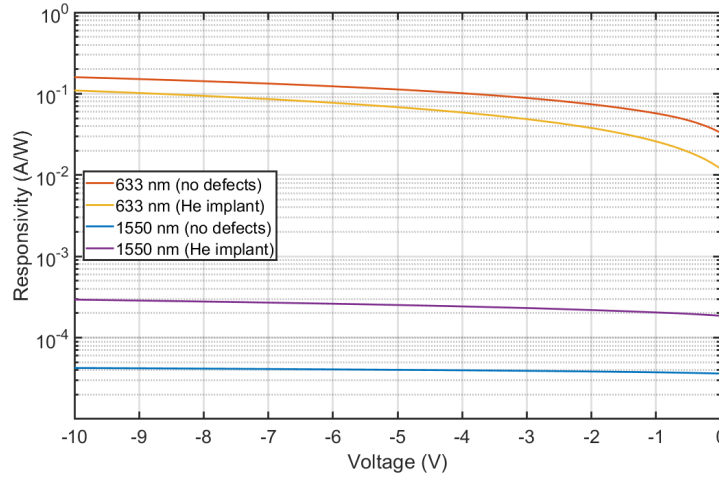


FIGURE 4.6: Responsivity of free-space detector illuminated by 633 nm and 1550 nm light, with and without He ion implantation induced defects

and 4.8. The source responsivity is from the power of the incident laser, increasing with wavelength due to the lower energy of each photon resulting in a higher number available for detection. The available responsivity is the current produced within the device due to the internal conversion efficiency of photons to electron-hole pairs, while the terminal response is that seen at the contacts as a result of the external quantum efficiency. It can be seen that there is no response beyond the material band gap energy of 1.12 eV ($1.11 \mu\text{m}$), as expected. The peak responsivity occurs at a wavelength of around 740 nm and value of $472 \text{ mA}\cdot\text{W}^{-1}$ since crossing the band gap in an indirect semiconductor needs additional energy to account for the change in momentum between the highest energy state in the valence band and the lowest state in the conduction band. Furthermore, the quantum efficiency is similarly restricted by the material band gap, but peaks at a lower wavelength of 500 nm and 63.4%.

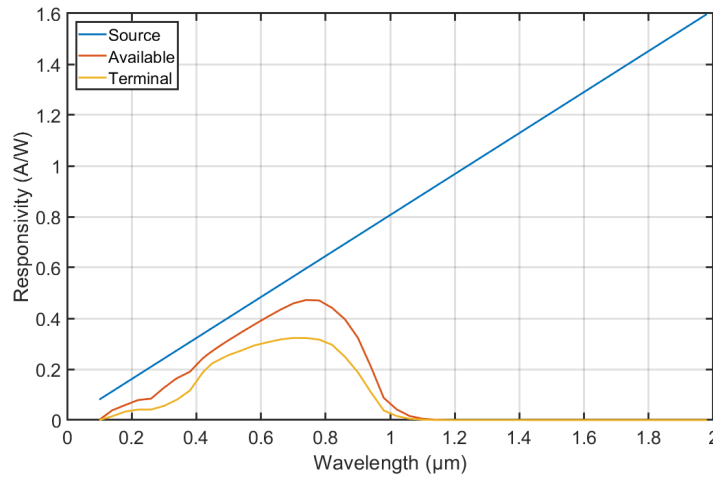


FIGURE 4.7: Spectral responsivity of device without defect modelling, illuminated by a monochromatic beam at a power of 1 mW

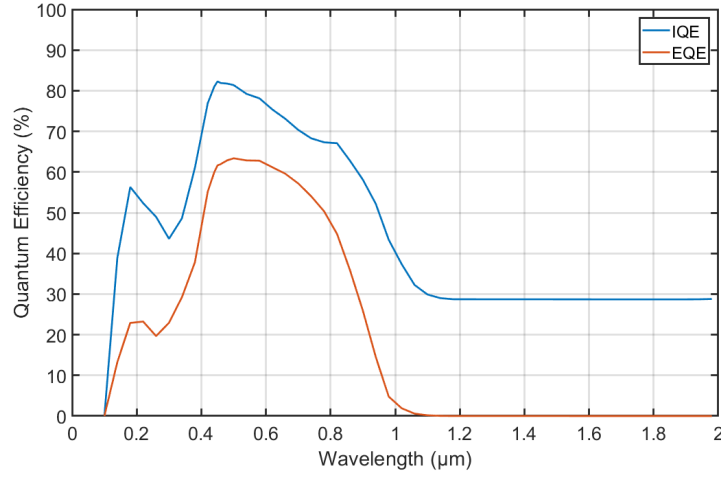


FIGURE 4.8: Quantum efficiency against wavelength of device without defect modelling, illuminated by a monochromatic beam at a power of 1 mW

4.3 Summary

This chapter has described the design of a free-space silicon photodetector using high-energy ion-implantation to induce the creation of defects and extend the absorption towards longer wavelengths, as will be fabricated in Chapter 5. The device has been simulated initially without the additional process to obtain necessary implantation profiles, and find the electric field distribution and depth of the depletion region to determine a suitable isolation trench depth. Subsequently, single-defect type modelling was added using an approach by D. Logan and adapted for use with the surface-illumination in this device [47]. This incorporated a custom c-interpreter function to model the photogeneration rate of divacancies in the material, as well as a uniform concentration of trap levels in the Atlas script to model the recombination effects of this distribution. The results of this simulation were then presented and demonstrates the successful enhancement of detector response to a wavelength of 1550 nm and a reduction in the intrinsic response at 633 nm.

Chapter 5

Device fabrication

Fabrication of this design was performed at the Southampton Nanofabrication Centre, following the schematic process flow illustrated in Figure 5.1, with full details included in Appendix A.2. It was undertaken as part of a collaborative effort with the Ion Beam Centre (IBC) at the University of Surrey. All processing steps were carried out in Southampton with the exception of ion implantation, which was performed by staff at the IBC. A total of 6 wafers were produced using this design, with differing high-energy ion implantation conditions for the creation of defects within the devices. The first two did not include a high-energy ion implantation (step g in Figure 5.1) to act as a reference for performance comparison. One included the high-energy B^+ implant and another used the He^+ implantation, as described in the previous chapter. These were unmasked to effect the entire device area and show how this process can be incorporated without the need for an additional lithography step. The final two wafers used a Eu^+ -based implantation process developed at the IBC between steps b and c as a part of the collaborative use of facilities. These were designed and tested in Surrey and do not contribute towards analysis of the implantation processes discussed in this work.

Starting with 4" p-type Si wafers, the first process step deposits a thin layer of SiO_2 as a sacrificial layer for protecting the surface cleanliness and act as an implantation barrier to reduce channelling effects. After alignment marks formation, step b performs an unmasked implantation of P^+ at 35 keV and a dose of $4 \times 10^{13} \text{ cm}^{-2}$, with a beam current of $100 \mu\text{A cm}^{-2}$ and 10° incidence angle. This doped the upper portion of the substrate n-type to create the vertical depletion region. Following this, high-concentration contact doping regions were formed in steps c and d, masked by a standard S1813 (Shipley 1813) positive photoresist, spun on to a thickness of 1.2 - 1.3 μm . The n^{++} region used a 21.2 keV P^+ implantation at a dose of $1 \times 10^{14} \text{ cm}^{-2}$, while the p^{++} region used a 31 keV B^+ implantation at a dose of $2 \times 10^{15} \text{ cm}^{-2}$, with the same beam current and incidence angle as the first implant.

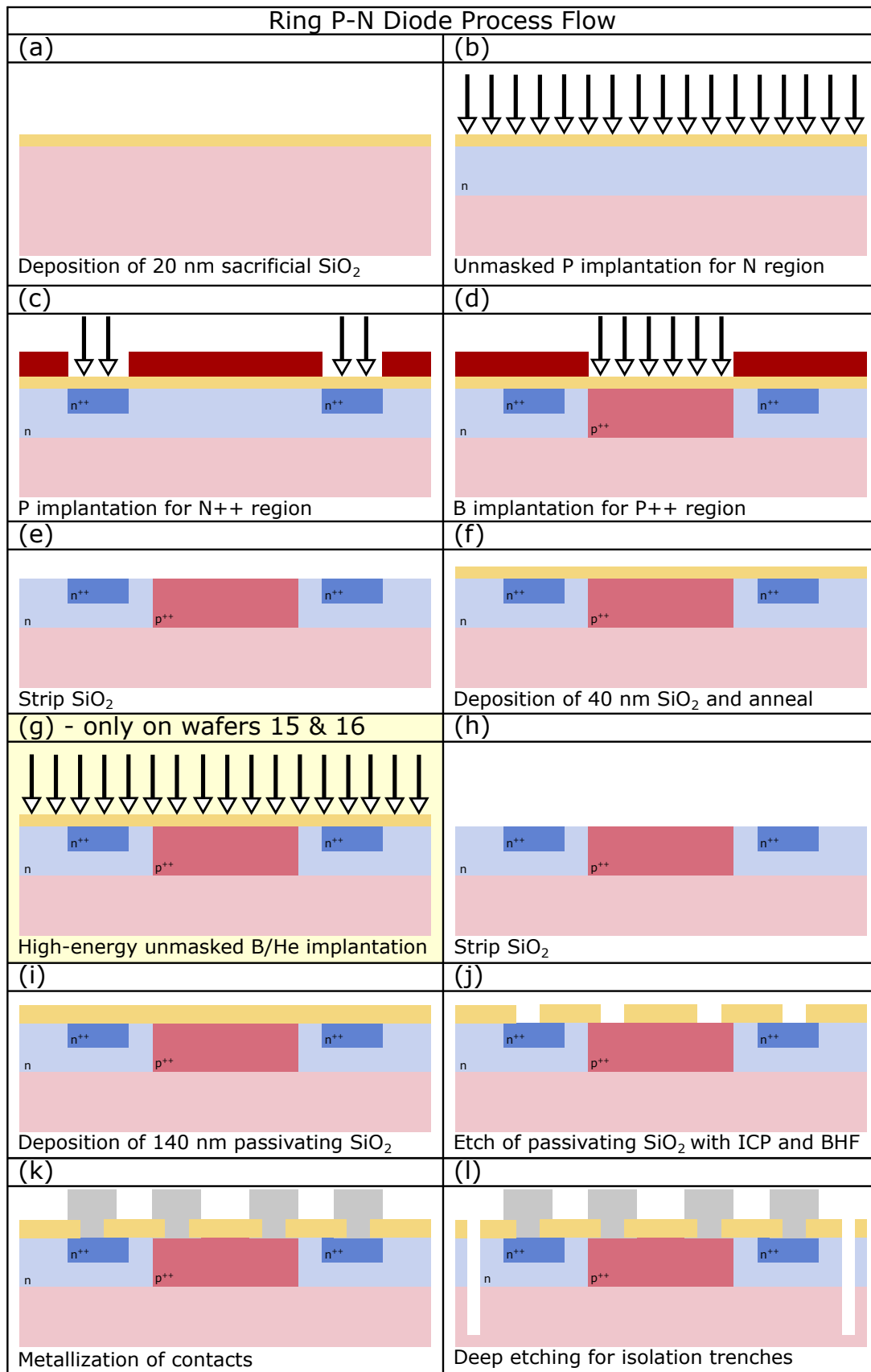


FIGURE 5.1: Free-space device showing key fabrication steps, with alignment marks formed between steps a and b (not to scale)

Dopant drive-in was performed by removal of the SiO₂ layer in 20:1 buffered hydrofluoric acid (BHF) and depositing a fresh layer of 40 nm thick plasma-enhanced chemical vapour deposition (PECVD) SiO₂ as an anneal cap, followed by dopant activation during a rapid thermal anneal at 1050 °C for 1 minute in steps e and f. This annealing process required a slower temperature ramp time than standard tool recipes to account for the slower response of the temperature sensor positioned below the sample when a 4" wafer is mounted onto a 6" carrier, and heated from the lamps above. At this stage, select wafers received the defect-inducing implantation in step g, followed by removal of the SiO₂ layer in BHF diluted with deionized water to a ratio of 20:1 on all wafers in step h. There is no anneal following this implant to ensure defects remain in the device, since activation not only migrates dopant atoms into host lattice sites but also removes unwanted damage from the wafer.

Surface passivation is performed in steps i and j by depositing a layer of 140 nm PECVD SiO₂, followed by etching holes to allow connection between the active region and metal contacts. 120 nm of this depth used a reactive ion etch (RIE) process, with the final 20 nm removed in 20:1 BHF. This two-stage approach prevents etching of the Si below and reduces surface contamination, which would have a detrimental effect on the metal-semiconductor interface quality. Next, aluminium was deposited onto a patterned photoresist layer by e-beam evaporation for contact metallization, followed by lift-off in a 1.5 hr sonic NMP bath to leave the desired patterning. This used AZ2070 negative photoresist spun-on to a thickness of 5.5 μm . It was chosen since the slight undercutting which occurs during development reduces the amount of material deposited on the side walls, enabling a cleaner lift-off. This photoresist requires an additional post-exposure bake to cross-link the exposed areas and prevent removal in a developer solution, which was not needed for the previous S1813. After development, another brief BHF dip was performed to remove native oxide from the surface to aid a good ohmic contact, prior to Al deposition using a BAK600 e-beam physical vapour deposition (PVD) evaporator to a thickness of 437 nm. Metal sintering was not performed at this stage since the high temperature needed would anneal out any previously created defects.

The final three process steps form the isolation trenches using a 1.3 μm thick S1813 barrier and inductively-coupled plasma (ICP) etching of the SiO₂ passivation layer and silicon substrate. Whilst originally intended to be 1.5 μm deep, some of the devices came to have a trench depth over 3 μm , as shown in the scanning electron microscope (SEM) image of Figure 5.2. Unfortunately, this overetching did remove some of the protective barrier in places and a small portion of the device surface, as shown in the upper right of the SEM image in Figure 5.3, but it only had a small detrimental effect on performance, as discussed in Chapter 6.

As a result of a PECVD sample stage temperature of 325 °C during deposition of the passivating SiO₂ layer, there is little room for investigating the effect of post-implantation annealing. However, this temperature is not too high to significantly anneal out the

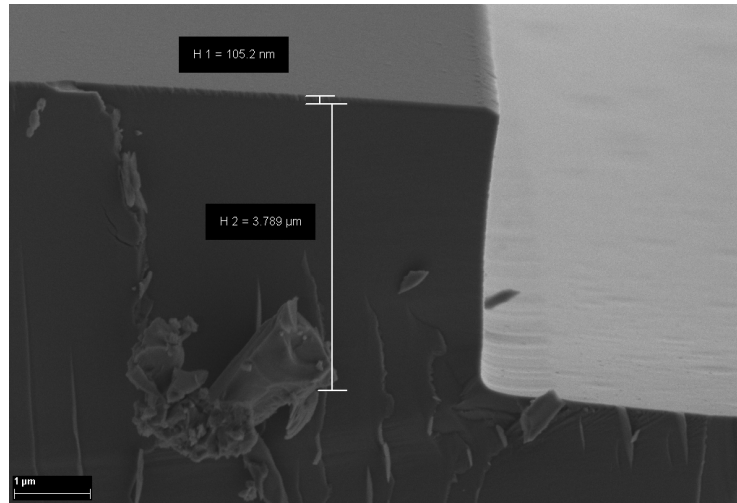


FIGURE 5.2: Cross-sectional SEM image of finished device isolation trench, with passivation SiO_2 thickness and trench depth indicated. Image taken by F. Gardès.

introduced divacancies, but is sufficient to have removed much of the smaller defects which would have reduced the external quantum efficiency. This means a reasonable concentration of key defects will be present and are the majority species remaining for the chosen implantation parameters used. To better demonstrate the influence of post-implantation annealing on device characteristics, it would have been better to ensure all high-temperature process steps were performed prior to implantation. This would reduce the maximum temperature experienced by the defect-containing device to 115°C during the pre-exposure bake of later photolithography stages. This would only then require a higher ion energy to account for the increase in distance from the surface to the target depth during implantation. Incremental isochronal annealing can then be used after fabrication to tailor the device for maximum photocurrent response.

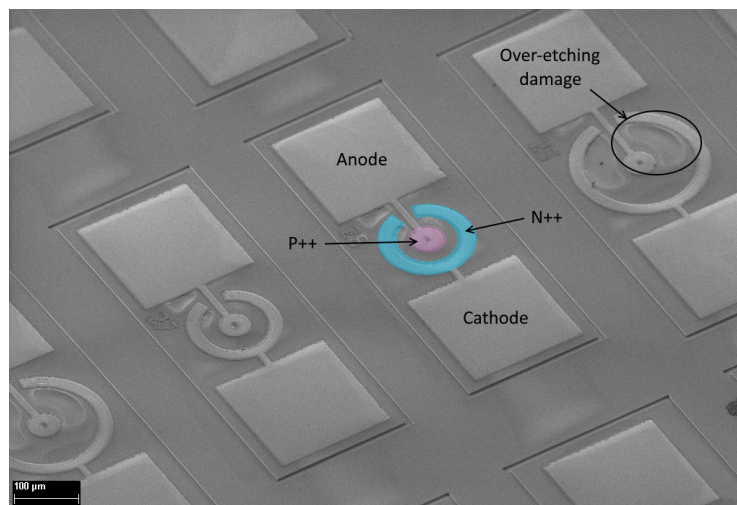


FIGURE 5.3: SEM image of the surface of finished devices showing false-colour p^{++} and n^{++} regions, contacts pads, and the damaged surface as a result of over-etching the trenches. Image taken by F. Gardès.

5.1 Summary

This chapter has described the fabrication of a free-space silicon photodetector at the Southampton Nanofabrication Centre, with the services of the Surrey Ion Beam Centre to perform ion implantation processes. This design has used an existing photolithography mask set with 4" p-type silicon wafers, incorporated with an additional n-type surface layer doped region and high-energy ion implantations as described in the previous chapter. Whilst an overetching has occurred during the formation of isolation trenches, subsequent characterisation in the next chapter has shown this not to be significantly detrimental to the devices.

Chapter 6

Characterisation results and discussion

To characterise these devices, a Keithley 2401 source-meter was used to apply a bias between electrical contacts, while an optical fibre with $10\text{ }\mu\text{m}$ wide core illuminated the sample from above at an angle approximately $6.3\pm 1^\circ$ off-normal, as shown in Figure 6.1. This was the closest to normal incidence that could be achieved while still being able to see the fibre tip from a microscope mounted above. The fibre's light source was either an Agilent 8163B lightwave multimeter tunable from 1520 to 1630 nm, or a HeNe laser operating at a wavelength of 633 nm. With the tunable laser set to a power of 5 mW to match the HeNe output, the response to wavelengths around 1550 nm could be compared to the intrinsic response of silicon which uses a band-to-band absorption transition. To

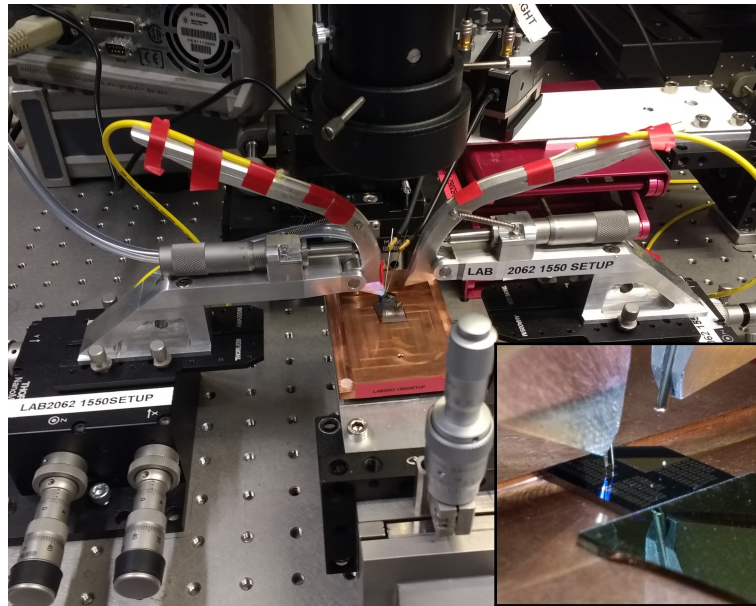


FIGURE 6.1: Photograph of experimental set-up used to characterise free-space photodetector (inset - close-up of chip under test)

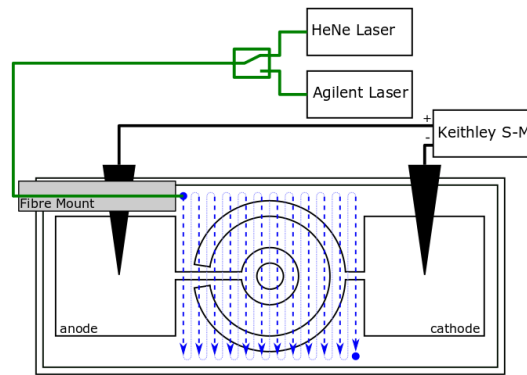


FIGURE 6.2: Schematic of experimental setup, indicating the scanning path of an optical fibre across the device surface where the “Fibre Mount” aligns the incident beam at the desired angle from the surface. “HeNe Laser” is the 633 nm fixed wavelength laser source, “Agilent Laser” is the 1520 to 1630 nm tunable laser, “Keithley S-M” is the Keithley source meter.

demonstrate the variation of photocurrent across the device, the fibre was manually scanned through up to 270 data points (18 x 15 on the larger unimplanted device) to make up the desired $10\text{ }\mu\text{m}$ grid over the surface. This is illustrated in Figure 6.2 and produces a heat map of total electrical current response over the device ring’s area. After this, a few points were selected to establish: IV curves, responsivity, linearity of laser power response and spectral response.

6.1 Current response heat maps

For each implantation condition tested two devices were chosen for analysis, one with a $20\text{ }\mu\text{m}$ gap between inner and outer ring heavily-doped contact regions, and the other with a $50\text{ }\mu\text{m}$ gap. These were then scanned at both wavelengths of 633 nm and 1550 nm under a reverse bias voltage of 2 and 7 V, over an approximately $10\text{ }\mu\text{m}$ grid. This step size was then scaled post-measurement to match the expected dimensions in the mask file, thus accounting for sample stage inaccuracy (the largest device area tested is roughly $210\times 194\text{ }\mu\text{m}$). This data was then filtered against noise and interpolated for double the resolution to better illustrate the variation in current response for the measured area.

The response of a smaller-gap unimplanted device, is shown in Figure 6.3. The current flow due to illumination at a wavelength of 633 nm is much stronger at up to 172.8 nA at a reverse bias of 2 V (or 221.8 nA at 7 V reverse bias) compared to 75.4 nA (or 141.9 nA) for absorption at 1550 nm. Whilst minimum dark current for this device is 15.32 nA at 2 V (or 49.91 nA). Using Equation 2.8, this corresponds to a SNR of 20.2 dB (10.8 dB) at 633 nm, and 11.8 dB (5.3 dB) for 1550 nm. This response is as expected since this device can only absorb primarily using a band-to-band mechanism. There are some data points on the 1550 nm wavelength plot with a very strong response (up to 503.4 nA at 2 V or

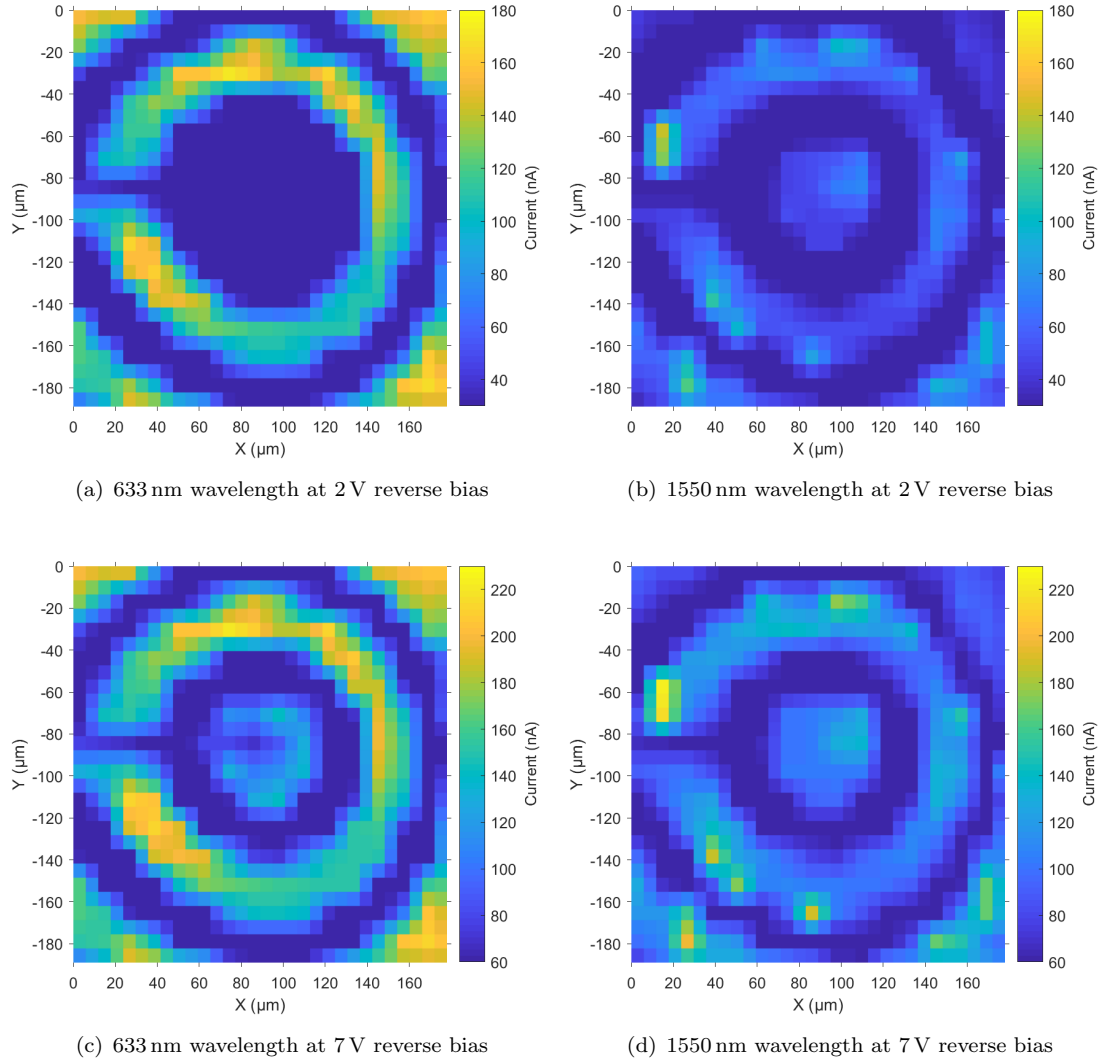


FIGURE 6.3: Heat map current response of unimplanted device with a $20\ \mu\text{m}$ gap between rings, at wavelengths of 633 and 1550 nm and a reverse bias of 2 or 7 V

711.8 nA at 7 V in the unfiltered data set). These are likely due to surface defects or interface effects since they mostly occur at the metal layer edge.

The $50\ \mu\text{m}$ gap device for the same implantation conditions, as shown in Figure 6.4, has a stronger current response of 206.3 nA at 2 V (260.1 nA at 7 V) for 633 nm band-to-band and 106.9 nA (182.7 nA) for 1550 nm trap-assisted absorption. This is primarily due to the extra dark current of 36.1 nA (86.6 nA) accumulated over the larger device area. Evidenced by the reduced SNRs of this device, which are 13.5 dB (6.0 dB) at a wavelength of 633 nm, and 5.8 dB (0.80 dB) at 1550 nm.

The response of the He^+ -implanted, smaller-gap device is shown in Figure 6.5. The current produced at a wavelength of 1550 nm is now higher than that at 633 nm as a result of the added implantation. These are up to 225.8 nA at 2 V (319.3 nA at 7 V) for 1550 nm, and 135.7 nA (167.8 nA) for 633 nm. This device has a minimum dark current

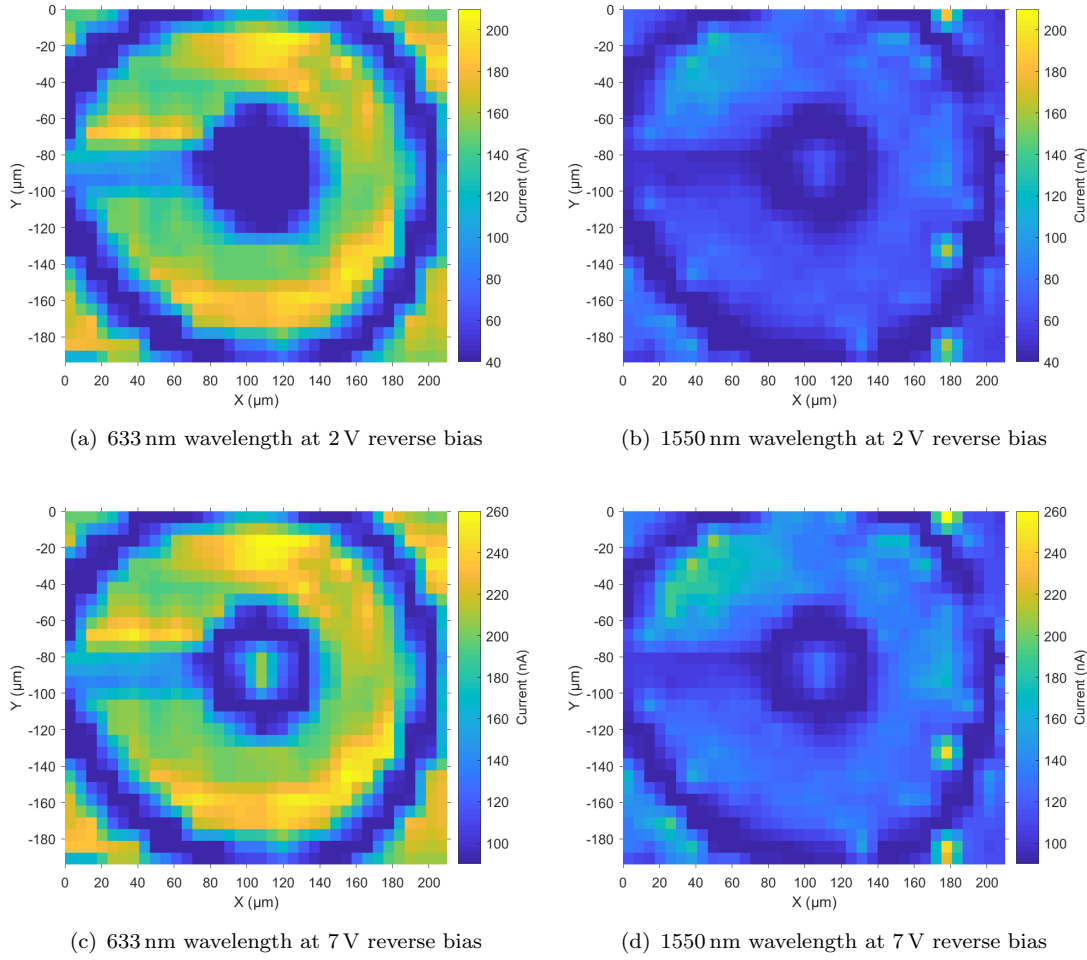


FIGURE 6.4: Heat map current response of unimplanted device with a $50\ \mu\text{m}$ gap between rings, at wavelengths of 633 and 1550 nm and a reverse bias of 2 or 7 V

of 10.6 nA (39.3 nA), corresponding to SNRs of 26.2 dB (17.1 dB) at a wavelength of 1550 nm, and 21.4 dB (10.3 dB) at 633 nm. Which clearly shows a shift towards more trap-assisted photon absorption.

The larger-gap device after this implantation is shown in Figure 6.6, and only has a stronger response to intrinsic absorption at 633 nm compared to the smaller device. The maximum current is 169.9 nA at 2 V (246.1 nA at 7 V) for 633 nm, and has reduced to 146 nA (255.5 nA) at 1550 nm. This suggests that over the larger area, there may not be sufficient defects available to generate enough current to overcome the increase in both intrinsic response and a larger dark current, which here is 32.8 nA (99.2 nA). However, most of this increase at 633 nm is due to the dark current since the SNR has dropped to 10.8 dB (4.0 dB) at 1550 nm and 12.4 dB (3.4 dB) at 633 nm. It is also apparent that while the 1550 nm response is lower than the 633 nm response at 2 V, it is only just starting to produce a higher current under a bias of 7 V. Therefore the greatest quantity of defects have likely stopped deep within the substrate due to the implantation energy of 600 keV forcing the He^+ ions to a depth of $2.23\ \mu\text{m}$ according to SRIM calculations [107].

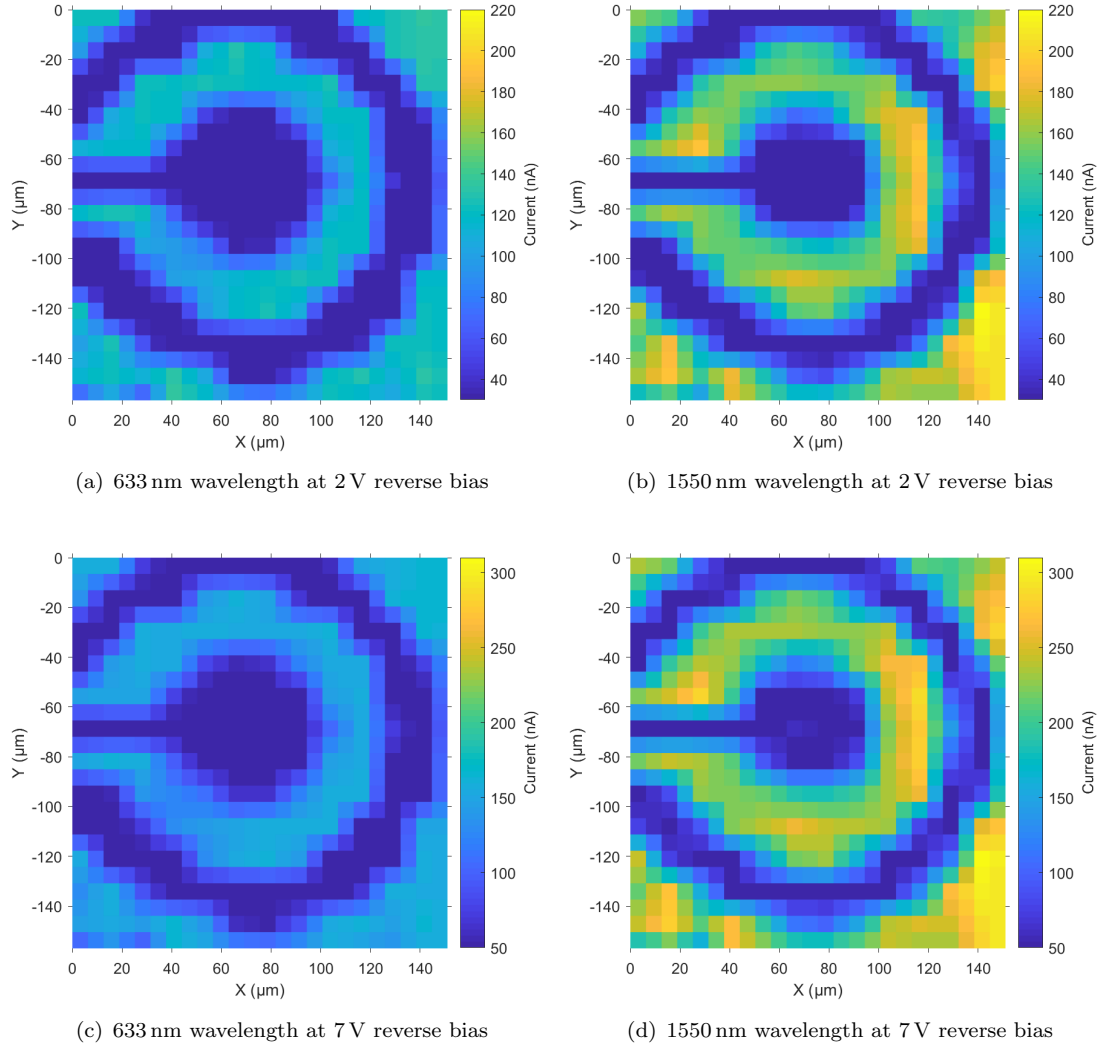


FIGURE 6.5: Heat map current response of He^+ -implanted device with a $20\ \mu\text{m}$ gap between rings, at wavelengths of 633 and 1550 nm and a reverse bias of 2 or 7 V

Although there are still a sufficient quantity for an increased trap-assisted response when compared to the larger reference device, alongside a reduction in the intrinsic response.

To consider only the photocurrent which flows as a result of carrier generation by illumination, the dark current is defined as the lowest 10% of the colour bar. The minimum dark current is then subtracted from the average of the remaining data, to determine the average photocurrent. For the smaller-gap device on the reference wafer, this amounts to 101.7 nA at a reverse bias of 2 V (106.3 nA at 7 V) for a wavelength of 633 nm, and 66.3 nA (79.6 nA) at 1550 nm. Whilst for the bigger-gap device, this is 113.8 nA (116.0 nA) and 48.2 nA (58.6 nA) respectively. For comparison, the smaller-gap He^+ -implanted device achieved photocurrent averages of 105.1 nA (108.7 nA) at 633 nm, and 136.7 nA (170.2 nA) at 1550 nm. While the larger-gap device achieved 104.0 nA (104.9 nA) at 633 nm, and 80.7 nA (103.7 nA) at 1550 nm. From these results, and both inter-ring gap sizes, the response of the He^+ -implanted devices to a wavelength of 1550 nm is greater

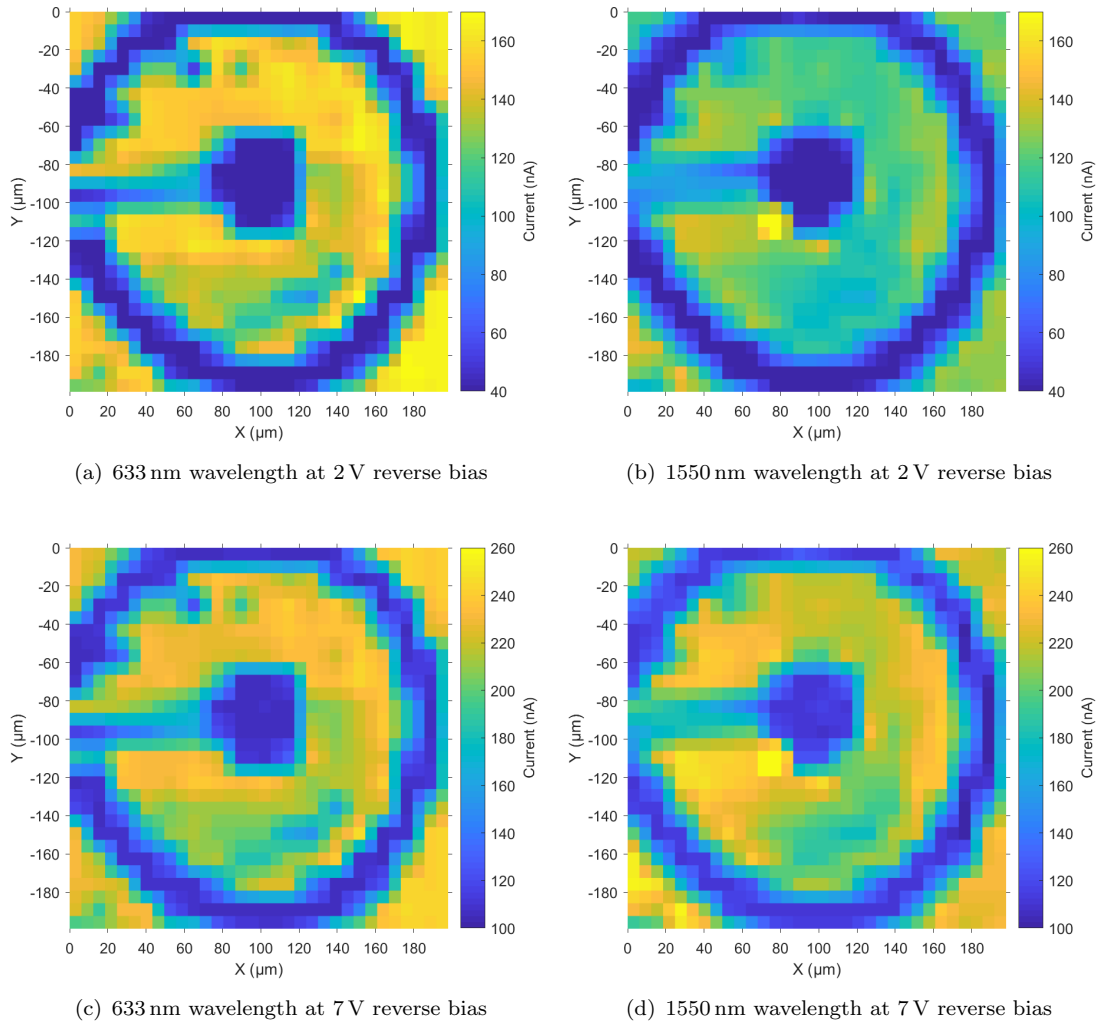


FIGURE 6.6: Heat map current response of He^+ -implanted device with a $50\ \mu\text{m}$ gap between rings, at wavelengths of 633 and 1550 nm and a reverse bias of 2 or 7 V

than that at 633 nm. The smaller devices perform better than the larger ones, roughly doubling the photocurrent due to trap-assisted absorption as opposed to a 67–77 % increase. For comparison the intrinsic absorption increases by 2–3 % and 9–10 % for the small and large devices respectively.

The response of the B^+ -implanted, $20\ \mu\text{m}$ gap device is shown in Figures 6.7a and b. Devices tested on this wafer show noisier and less uniform electrical results since they come from near the edge of the wafer, where the implantation process is less uniform across the surface area. They are taken from here since the majority of the devices over the wafer display a more Schottky-like contact response, likely due to the high-energy implantation pushing heavily-doped regions a little deeper into the substrate and reducing the surface concentration. Even though B^+ is implanted to an ion range of $2.55\ \mu\text{m}$, there may still be a sufficient concentration at the surface to compensate the now reduced n^{++} region surface concentration and cause a near intrinsic semiconductor

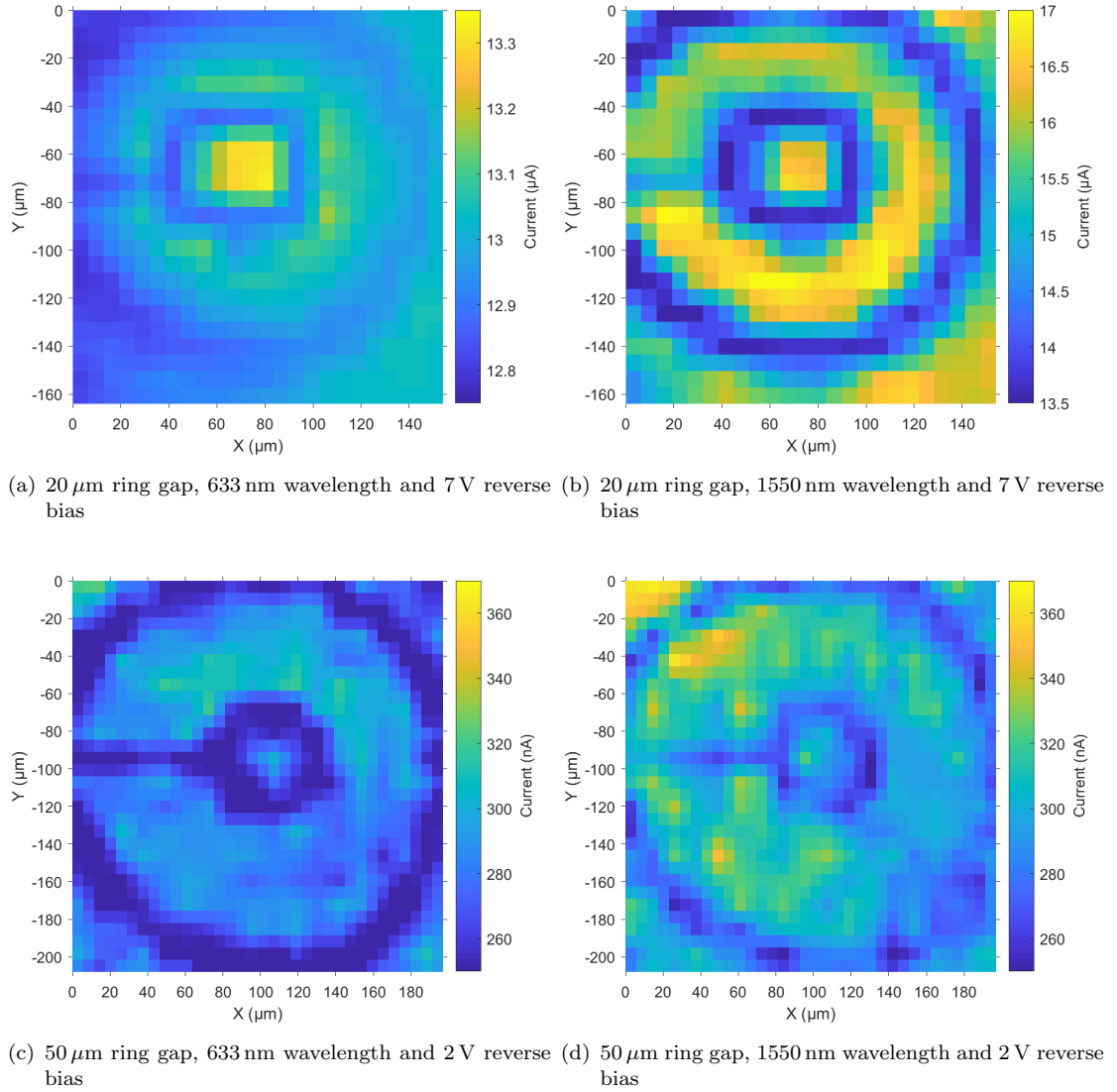


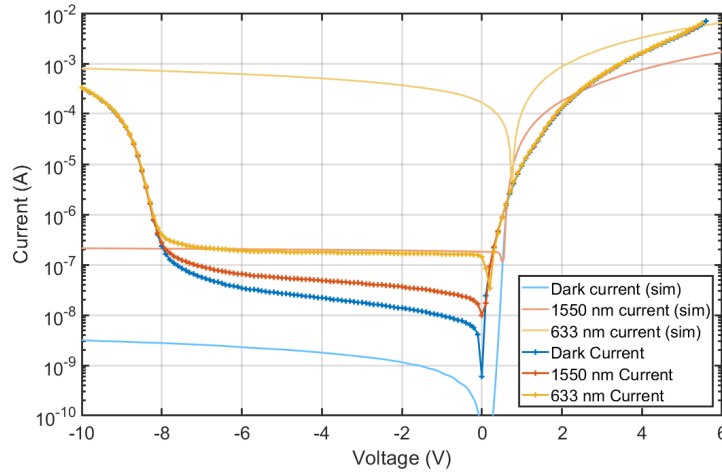
FIGURE 6.7: Heat map current response of B^+ -implanted devices, with a ring gap, wavelength and bias as described

contact to be formed. There does however appear to be a general increase in current at 1550 nm under a reverse bias of 7 V for the small-gap device, similar to the He^+ -implanted devices. This produces a current of up to $17.0 \mu\text{A}$ at 1550 nm and $13.3 \mu\text{A}$ at 633 nm, however the very high dark current of $12.8 \mu\text{A}$, reduces the SNR to -9.6 dB and -28.2 dB respectively. Unfortunately this makes the current response of the small device unreliable at 2 V, and current flow for both devices at 7 V is excessively elevated due to reverse breakdown. However, the 2 V response of the larger device shown in Figures 6.7c and d, has a small increase in current at 1550 nm compared to 633 nm (maximum 498.5 nA vs 376.5 nA). With a minimum dark current of 226.1 nA, this results in SNRs of 1.6 dB and -3.6 dB. Average photocurrent was calculated to be 300.4 nA and 287.0 nA for each wavelength respectively, resulting in a 54 % increase at 1550 nm and 47 % decrease at 633 nm compared to the reference wafer. Although lower dark current

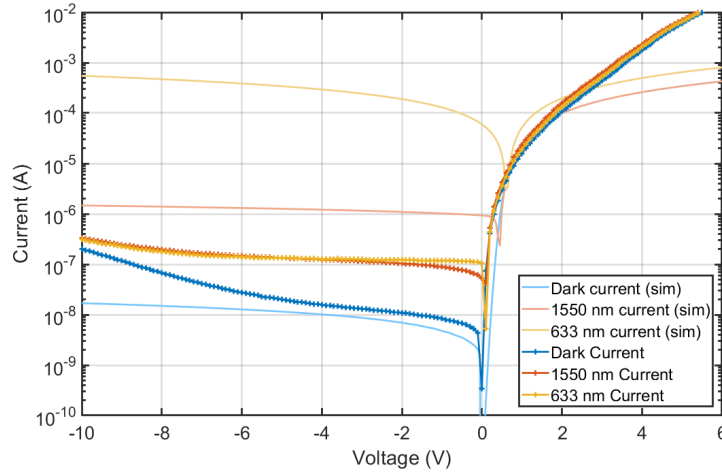
devices are needed, this does suggest the B^+ induced damage has had the intended effect as well as reducing the intrinsic response.

6.2 Photodetection characteristics

Positioning the fibre between concentric rings, a voltage sweep was performed to obtain the experimental IV curves shown in Figure 6.8 for the $20\ \mu\text{m}$ inter-ring gap devices. These are overlaid on the previous simulation results obtained in Section 4.2.3. From these results the minimum dark current produced was $13.77\ \text{nA}$ for the unimplanted wafer, and $10.87\ \text{nA}$ for the He^+ -implanted wafer. The current produced under $633\ \text{nm}$ illumination experiences a small reduction (26-27 % over a reverse-biased range of 2-5 V), while the $1550\ \text{nm}$ response has a significant gain (142-188 % over the same range). This increases further when considering only the photo-generated proportion



(a) Unimplanted wafer device

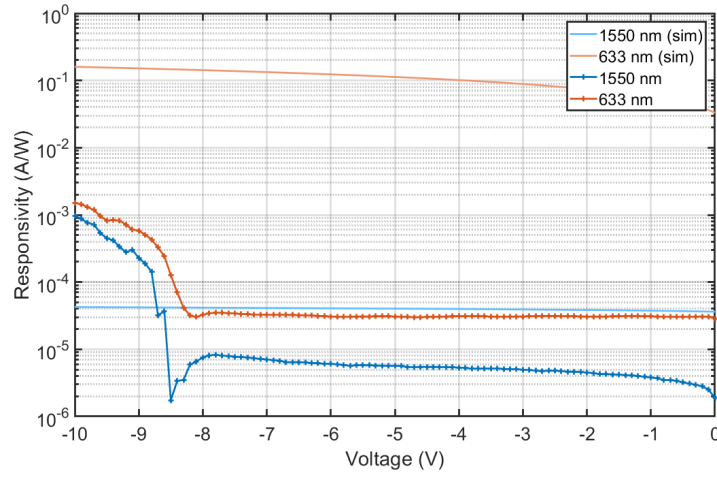


(b) He^+ -implanted wafer device

FIGURE 6.8: IV response of unimplanted and He^+ -implanted devices with a $20\ \mu\text{m}$ gap between rings and laser power of $5\ \text{mW}$, overlaid on prior simulation results.

of the current flow to 301-316%. The highest SNR is achieved for low reverse bias voltages, reaching 10.8 dB and 10.5 dB for each device respectively under 633 nm light, then 4.2 dB and 9.8 dB under 1550 nm light.

The experimental responsivity of these devices is shown in Figure 6.9, also overlaid on prior simulation results. Whereby the He^+ implantation has increased the 1550 nm response from an average of $5.26 \mu\text{AW}^{-1}$ in the unimplanted device to $20.65 \mu\text{AW}^{-1}$, considering only the region from -7.8 to 0 V where both devices are below breakdown voltage. Meanwhile the 633 nm response has decreased from $31.26 \mu\text{AW}^{-1}$ to $22.23 \mu\text{AW}^{-1}$ over the same range. This is a 3.9x increase in response to the longer wavelength due to the introduced deep-level defects, and a 29% reduction in band-to-band response due to greater carrier recombination. Whilst the intrinsic response is nearly independent of voltage, the defect-mediated responsivity has a noticeable dependence on voltage due to



(a) Unimplanted wafer device

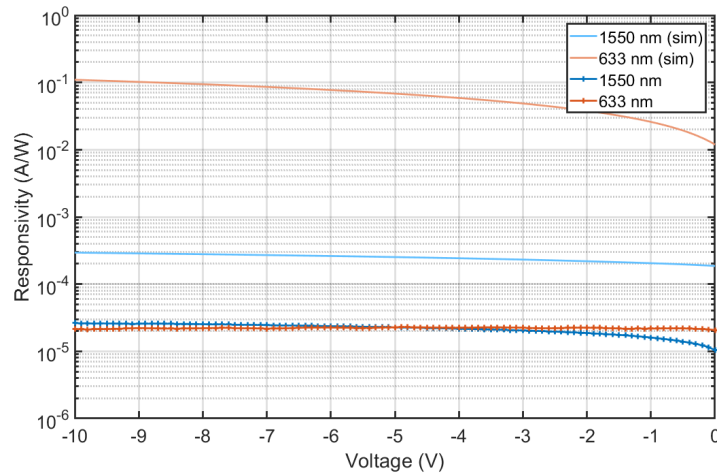
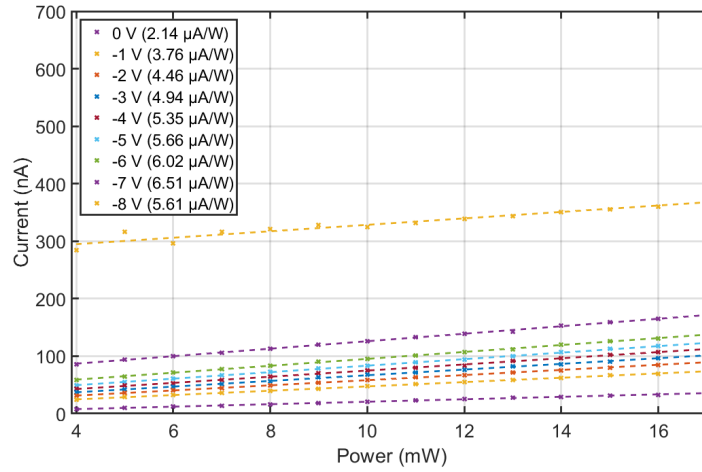
(b) He^+ -implanted wafer device

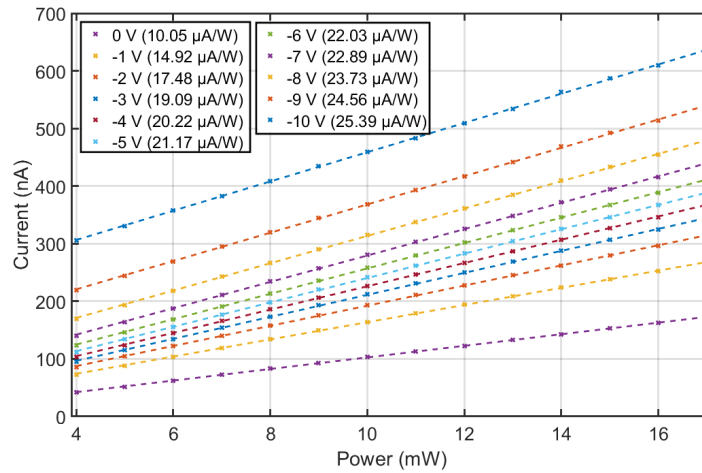
FIGURE 6.9: Responsivity of unimplanted and He^+ -implanted devices with a $20 \mu\text{m}$ gap between rings and laser power of 5 mW, overlaid on prior simulation results.

an increasing defect concentration towards the ion stopping range. It is expected that a lower-energy He^+ implantation will further increase device performance.

To ensure the photocurrent response produced is due to a linear process rather than nonlinear effects, the graphs of Figure 6.10 were measured. Each voltage has a linear fit performed with the gradient of each line indicated in the corresponding data legend, giving the responsivity at that voltage. As can be seen, the increase in current with laser power agrees well with expectation when operating below the breakdown voltage. For the case of the unimplanted device, breakdown starts to occur at around -7.8 V so the linearity begins to deteriorate for the measurement at -8 V as well as experiencing a loss in responsivity. At higher voltages, the primary cause of current flow is no longer due to photogenerated carriers but is instead an avalanche mechanism, with a significant negative dependence on laser power due to carrier accumulation reducing the strength of the electric field and thus reducing device gain [108].



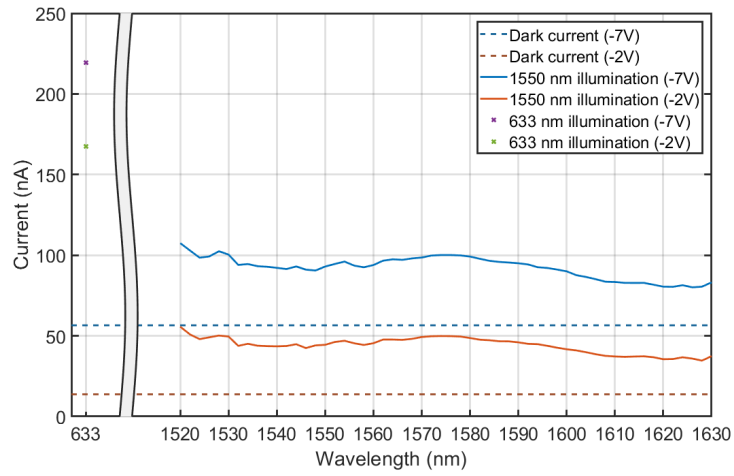
(a) Unimplanted wafer device, at 0 to 8 V reverse bias



(b) He^+ -implanted wafer device, at 0 to 10 V reverse bias

FIGURE 6.10: Sweep of 1550 nm laser power on unimplanted and He^+ -implanted devices with a $20\text{ }\mu\text{m}$ gap between rings, at a range of reverse bias voltages

To observe the spectral response of the devices, the Agilent laser source was scanned from 1520 to 1630 nm at a power of 5 mW while the devices were under a reverse bias of 2 and 7 V. This was then combined with data from the IV sweeps to produce the graphs in Figure 6.11. For the unimplanted case, the 1520 to 1630 nm range shows a minimal response compared to that at 633 nm. A reduction by only 24.3 nA at a 7 V reverse bias from 1520 to 1630 nm suggests the peak response wavelength is much closer to 633 nm, as would be expected for intrinsic silicon. While the He^+ -implanted device shows a much stronger response within this wavelength range, surpassing the 633 nm current flow. Combined with the larger 59.0 nA reduction in current across the 1520 to 1630 nm range would suggest the peak response has shifted closer to 1520 nm. Additional spectral broadening may have also occurred since both 633 and 1520 nm are closer in magnitude, and higher, as compared to the unimplanted device



(a) Unimplanted wafer device

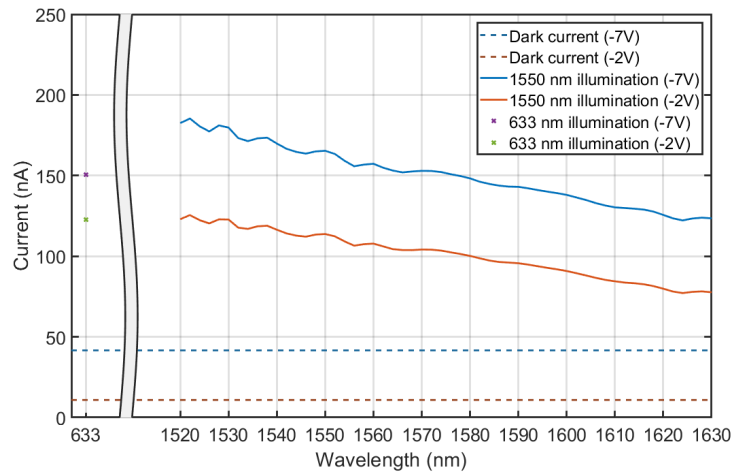
(b) He^+ -implanted wafer device

FIGURE 6.11: Spectral response of unimplanted and He^+ -implanted devices, with a $20\ \mu\text{m}$ gap between rings at -2 and -7 V and a laser power of 5 mW

6.3 Environmental lighting effects

Since the devices tested here are for free-space operation, further IV sweeps were performed to investigate the influence of different environmental lighting scenarios on the strength and clarity of the optical response. The effect of additional lighting is to add background noise to the incident spectrum, which may make the intended signal wavelength harder to detect. Tests were performed using the room, table or microscope light on during measurement, with the last scenario using power levels of 10, 30, 60 and 100 %. The room lighting consists of 3 fluorescent bulbs with a colour temperature of 4000 K, giving spectral peaks around 555, 630 and 440 nm wavelengths. The table light is a single fluorescent bulb with a temperature of 3500 K, it emits a lower luminosity but is positioned much closer to the sample. While the microscope uses white LEDs with a temperature of 6400 K, spectral peaks around wavelengths of 455 and 515 to 615 nm, and has an adjustable power input. For these tests power levels of 10, 30, 60 and 100 % are used for the microscope light.

Shown in Figure 6.12 is the IV curve for a He^+ -implanted device subjected to each of the environmental conditions, and without any optical signal stimulus. It highlights how much the background illuminance increases the dark current as a result of each source emitting at wavelengths which are intrinsically absorbed by silicon. Higher currents are present for the sources producing a greater intensity of light on the sample's surface. The table light has the lowest intensity, increasing flow to 43.8 nA at a reverse bias of 1 V (up from 32.5 nA in the dark), followed by the room light at 55.7 nA. Then the highest is produced by the microscope, varying from 108.1 nA at 1 V and 10 % power, up to 1.33 μA at 100 % power. This current still increases under the 633 and 1550 nm lasers, but less than in an otherwise dark room, leading to a poorer SNR as illustrated in Figure 6.13. At a reverse bias of 1 V, the SNR for this device is 3.0 dB and 11.9 dB for wavelengths of

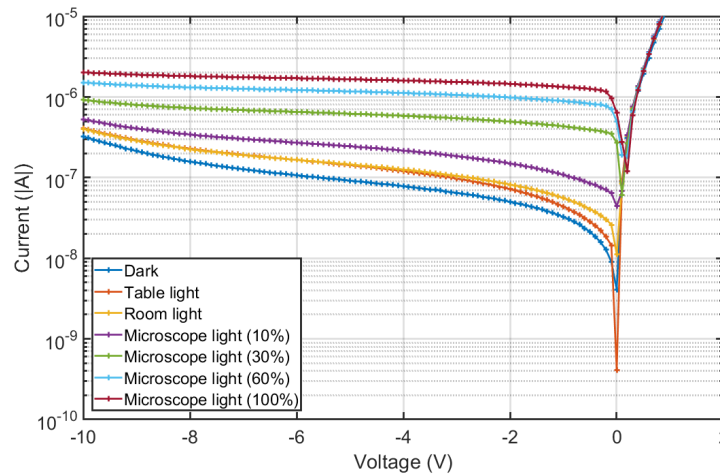
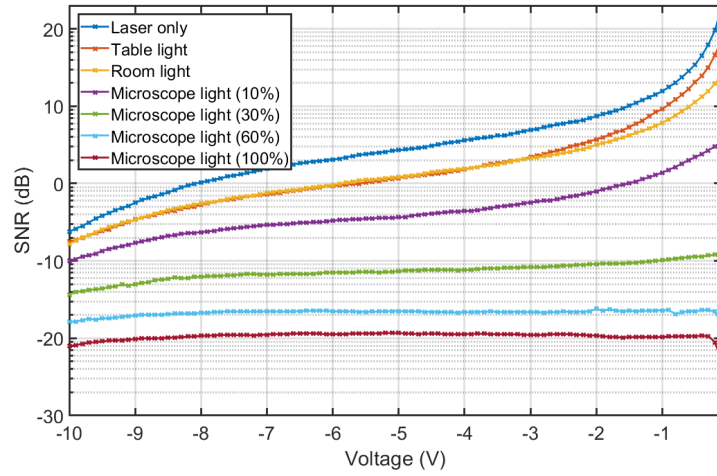
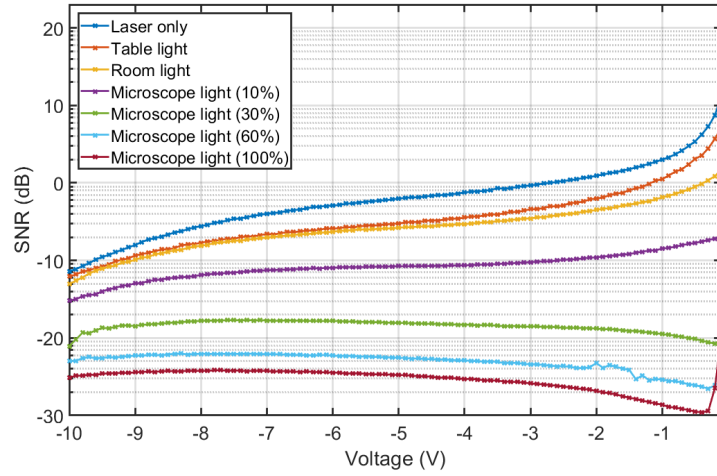


FIGURE 6.12: IV response of He^+ -implanted device subjected to various environmental lighting scenarios without optical signal stimulus



(a) SNR under 633 nm laser illumination



(b) SNR under 1550 nm laser illumination

FIGURE 6.13: SNR of He^+ -implanted device subjected to various environmental lighting scenarios

1550 and 633 nm respectively. These drop to -1.9 dB and 7.9 dB respectively when the room light is on, and almost match the current flow without signal stimulus under the microscope light with 60 % power, dropping to -25.4 dB and -16.4 dB respectively. From these results, it is evident that these devices are prone to poorer performance if the light incident upon the surface is not restricted to only contain the intended detection signal.

6.4 Summary

This chapter has provided the optical characterisation of the device designed in Chapter 4 and fabricated in Chapter 5. It covers manually scanning an optical fibre over the device surface area while measuring the terminal current, to produce heat maps illustrating the variation in response over the silicon surface, and shadowing over metallic contacts. These clearly show an increase in current in response to 1550 nm wavelength light upon

integration of the high-energy implantation process. Whilst the He^+ implant works well for the small-gap device, unfortunately B^+ has partly compensated the existing doping in the structure, in addition to pushing all regions a little deeper, making the contacts behave in a more Schottky manner. This has rendered most devices unusable, however those which work display a similar strong response to 1550 nm for the small-gap device. The He^+ implanted device is further analysed to determine the IV curves, responsivity, power linearity and spectral response. These show an increase in trap-assisted absorption which is now similar to the intrinsic response; a good linearity against optical power at reverse voltage magnitudes below breakdown; and a shift with broadening in spectral response towards longer wavelengths. An additional investigation into the detrimental effect of background lighting is also performed.

Chapter 7

Waveguide-integrated photodetector with defect-mediated absorption

7.1 Theory and design

As discussed in Chapter 6, the quality of photodetection is highly dependent upon the source of noise present which may mask the current generated by the intended signal wavelength. As such, the photoresponse can be improved by confining the propagation medium into an optical waveguide, where it can be isolated from external light sources and concentrated on a small cross-sectional area. This enables greater photon density within the absorbing region to increase the chance of capture at a defect site, faster extraction by moving contacts closer together to reduce the carrier drift length, and decoupling of drift and absorption lengths. This chapter demonstrates a metal-semiconductor-metal (MSM) junction across a waveguide to create a waveguide-integrated photodetector. It incorporates the simulation of different ion species for the implantation process, settling on the use of Ar^+ as a heavier noble gas than He^+ to increase defect concentration within the detector. Titanium nitride (TiN) is also used as a barrier enhancement layer when combined with substrate biasing, to reduce dark current flow through the device.

7.1.1 The metal-semiconductor-metal junction

Most integrated photodetectors use a p-i-n structure across the waveguide to create an electric field under reverse bias for carrier extraction. However, they suffer from a slow response due to diffusion-limited current flow of carriers generated just outside the depletion layer, and a heavy doping contrast between contacts [42]. This can be

improved by reducing the junction layer thickness so long as series resistance is not significantly increased, and lowering doping concentration [42]. Additionally, defect-mediated detection requires a long device to counteract the characteristic low absorption coefficient ($8 - 200 \text{ dBcm}^{-1}$), which further restricts high-speed operation [10, 83]. A MSM photodetector allows a faster speed since the junction field does not extend as far into the substrate, and there is no contact doping to create a high contrast. This creates a lower capacitance per length, and since it is a majority carrier device they are inherently faster with a few picoseconds response time, corresponding to a higher bandwidth in excess of 100 GHz [10, 39, 109]. The capacitance can be roughly half the value for a similarly efficient p-i-n device, and overcomes the diffusion limitation by reducing junction width whilst maintaining low series resistance [42, 110]. The lack of doped regions also enables a much simpler fabrication process since there are fewer photolithography stages needed, making production faster and cheaper.

The MSM concept is based on two back-to-back Schottky contacts, which requires a voltage bias to extract carriers generated in the semiconductor and restricts dark current with the barrier height across each junction [10]. Current flow increases with voltage due to the expansion of both reverse-biased contact depletion regions, until saturation when they fully span the gap between them. At this point the electric field at the anode is zero, which occurs at the flat-band voltage as illustrated in Figure 7.1a and defined by Equation 7.1 [110].

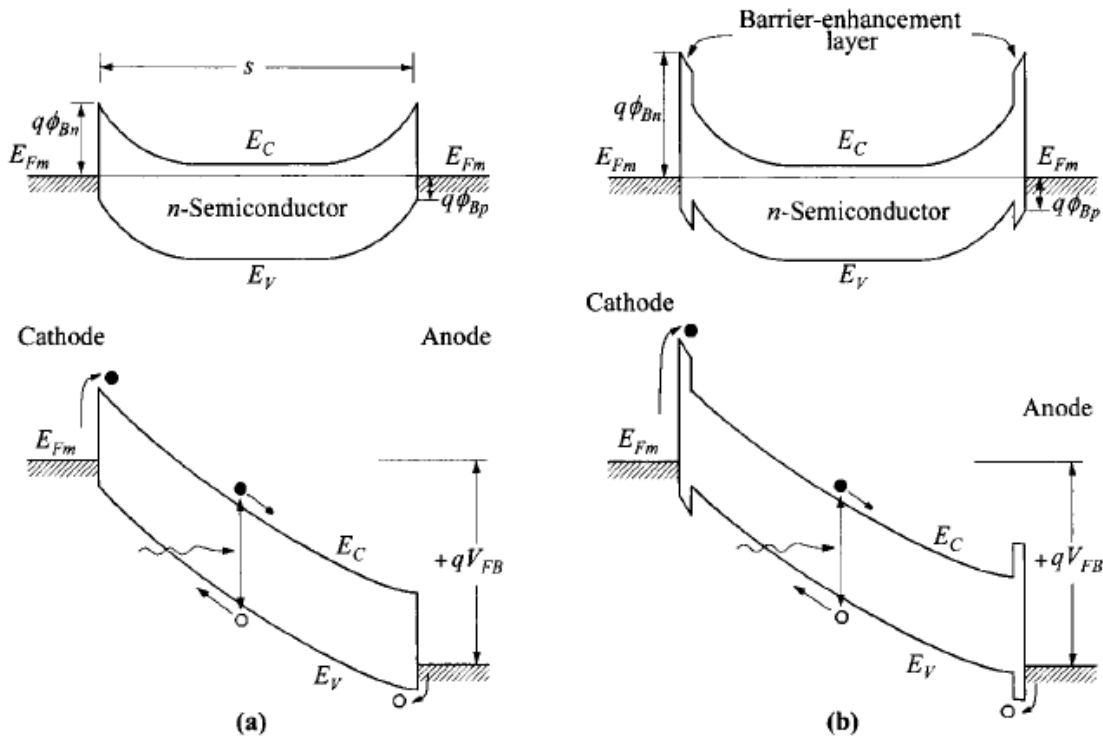


FIGURE 7.1: Energy band diagrams of MSM structure under equilibrium and forward flat-band bias (V_{FB}), a) without and b) with barrier-enhancement layer, sourced from [110]

$$V_{FB} = \left(\frac{qN}{2\epsilon_s} \right) s^2 \quad (7.1)$$

Where N = semiconductor doping concentration, ϵ_s = semiconductor permittivity, s = contact separation, $\Phi_{Bn/p}$ = barrier height for electron/hole injection, E_{Fm} = fermi level. The speed of MSM devices are likely to be transit or recombination time limited. The transit time can be improved with a shorter contact spacing and a high-purity semiconductor, while a faster recombination time needs a higher concentration of centres but tends to produce a lower responsivity and is less compatible with field-effect transistor (FET) integration [111]. It is expected that defect-mediated devices will be recombination time limited due to the presence and use of trap states for carrier generation, however this will also reduce the intrinsic carrier mobility and increase transit time [112]. Whichever process is the limiting factor, they can both be reduced by use of a shorter spacing between contacts, with the transit-time limited bandwidth defined as in Equation 7.2, where V_s = saturation velocity [110]. Reduction of this distance also allows the maximum transit speed to be reached at lower voltages [112].

$$f_{tr} = \frac{0.44 V_s}{\sqrt{2} s} \quad (7.2)$$

It has been reported that a reduction in metal width leads to smaller capacitance [113]. For the devices investigated here however, contact pads are large enough that the electric field is confined to the edges of the pad and any effect of reducing their width would only be noticeable for very thin metal strips [100]. This width reduction would then cause a significant increase in series resistance of the pad for the device lengths used here, offsetting the gain from a lower capacitance, and increasing the RC time constant unless a significantly thicker metal thickness is employed [112]. It would also make electrical probing difficult. Additionally, the speed of carrier extraction could be improved by using an n-type semiconductor region to create a majority of electrons in the device, since these have a higher saturation velocity compared to holes. It can be further improved by removing the current due to holes, since the initially electron-dominated impulse current response experiences a tailing-off due to slower moving holes [112].

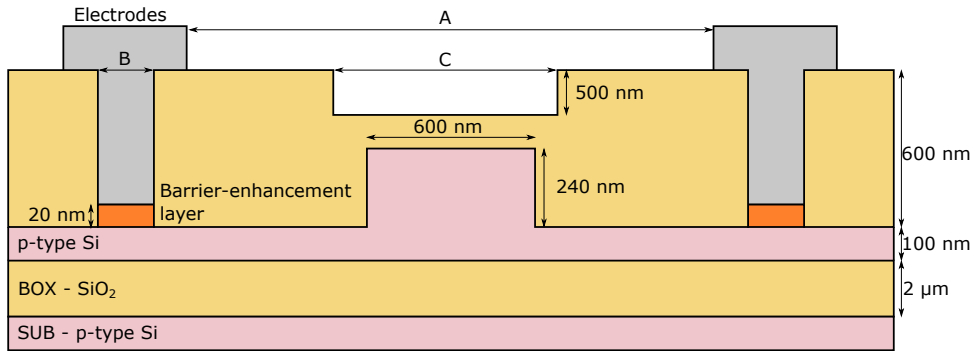
The main downside to a MSM design is a higher dark current than a p-i-n device due to the thermionic emission of carriers over the Schottky barrier and those thermally generated in the semiconductor. Comparatively, the reverse-biased potential barrier of a p-i-n structure is what prevents a similar flow [110]. This thermionic emission is more of an issue when operating with smaller band gap materials, such as germanium, due to the lower barrier [110]. Integration of a thin (<100 nm), wider band gap barrier-enhancement layer between metal and semiconductor helps to reduce this current, by enlarging the potential barrier seen by the carriers at the contact as illustrated in Figure 7.1b. Use of this layer has achieved low dark currents in the nanoampere range and at a high

speed of 150 GHz, at the expense of quantum efficiency [110]. It can also be reduced by using a smaller metal-semiconductor junction area to reduce carrier injection, which also reduces capacitance [114].

7.1.2 Device description

The design proposed here uses a rib waveguide structure on a SOI platform as comparable to literature, but with a 340 nm device layer and a larger cross-sectional area to support the mode size of a laser operating at a wavelength of $2\ \mu\text{m}$, rather than $1.55\ \mu\text{m}$. A p-type substrate is used here since they are more readily available than n-type, while the structure provides good optical confinement with electrical contact possible by connecting to electrodes above the thinner slab regions on each side. The slab is etched down to a height of 100 nm, for sufficient horizontal confinement without causing a high series resistance through the device. It has been shown that thinner wings can reduce the parasitic loss by better confining the optical mode, allowing for a shorter contact separation, as well as reducing capacitance at the expense of series resistance, but is set to this height to aid fabrication [83]. A thin barrier-enhancement layer is added at the metal-semiconductor interface to reduce dark current as shown in Figure 7.2, and discussed in Section 7.1.5.

The device is passivated using a 600 nm thick SiO_2 layer, which is thin enough to allow pattern etching using ZEP e-beam resist, whilst still being thick enough to block sufficient high-energy implantation damage. Over the waveguide rib, a 500 nm deep window is removed to allow damage creation within the detector region only. Key dimensions are illustrated in Figure 7.2, with contact separation (A), contact hole width (B) and defect window width (C) varying in the mask layout as listed. Positioning contacts on either side of the waveguide improves upon some published alternatives which mount metal



$$A = 1.0/1.5/2.0/4.0\ \mu\text{m} \quad B = 3/4/6/10\ \mu\text{m} \\ C = 0.4/0.6/1.2/1.6/2.0/2.4\ \mu\text{m}$$

FIGURE 7.2: Cross-section of MSM design, detector length varies from $250\ \mu\text{m}$ to $3\ \text{mm}$. Variable device dimensions A, B and C are indicated, listed in microns and described in the text, not to scale.

contacts vertically, since interaction of the optical mode with metal creates additional parasitic losses and reduces responsivity [115, 116]. The horizontal confinement of this structure prevents this from occurring. A strip waveguide outside of the detector regions would provide better confinement, but add additional lithography and etching steps and suffer from an increased sidewall roughness, increasing propagation loss. Therefore the rib structure is used throughout to simplify fabrication and improve losses.

To maintain wavelength independence when coupling light on/off chip, an end-fire approach is used to allow operation at both 1.55 and 2 μm . Compared to gratings, this also couples more power into the waveguide to aid measurement of the lower current produced by a low absorption coefficient and short detector. For this, the 600 nm cores were tapered out to 10 μm at the edges of the device cell to match the width of the optical fibres to be used during testing. These tapers need to be of sufficient length to create a small enough taper angle to minimize loss, as defined by Equation 7.3 [117].

$$L_{\text{taper}} = \frac{W_{\text{input}} - W_{\text{output}}}{2\tan(\theta)} \quad (7.3)$$

Where $W_{\text{input/output}}$ is the input/output waveguide width and θ is the angle between the direction of propagation and taper side wall [117]. A length of 1 mm is used here, producing an angle of 0.27° for near full transmission.

Each chip contains 16 device sets over a 1.5×2 cm area, arranged in 2 columns as shown in the mask layout in Appendix B.3. With each set including devices of length: 250 μm , 500 μm , 1 mm, 2 mm and 3 mm. A total of 9 device chips were patterned on each wafer to reduce the e-beam writing time compared to a full wafer of cells. This allowed 3 for each metal stack as described in Section 7.1.5, using an additional cell for test structures, and common alignment marks and wafer layout as used within the research group. When not otherwise set to the values listed in Figure 7.2, contact separation is 3 μm , contact hole width is 8 μm and defect window width is 600 nm. Contact pads are 100 μm wide to aid the use of probes during characterisation. In addition to the above 14 dimensional tests, there is an additional set with standard dimensions as a backup, plus a set tailored with dimensions for high-speed operation. This consists of 1 μm separation and 3 μm holes, with a 600 nm wide defect window to match the rib width. This may only be close to optimal window width if the mode profile does not extend 133 nm into the ribs to match the lateral distance travelled from the edge of the mask by the implanted ions. This value is calculated as the sum of the lateral range and straggle for 300 keV Ar^+ ions implanted into Si, as discussed in Section 7.1.4, using SRIM simulations [107].

7.1.3 Waveguide geometry simulations

Using a fixed waveguide height, the rib width was designed using modal analysis simulation in Lumerical's MODE solver software, and literature by L. Chrostowski [71, 118]. Simulations were targeted at a wavelength of $2\ \mu\text{m}$ as similar work on defect-mediated devices have operated around this value, with additional consideration for operation at $1.55\ \mu\text{m}$ as this was the standard laser available for characterisation. Dimensions were designed for single-mode operation as this will ensure sufficient optical confinement to reduce modal dispersion loss into higher-order modes, and reduce the susceptibility to cross-talk [47]. The fundamental TE mode is used as it has the highest signal power compared to other modes, and can easily be isolated to prevent multimode propagation. If this occurred each mode would travel with a different propagation constant, leading to intermodal dispersion [47]. The fundamental TM mode is also allowed to propagate since this allows for a dual-polarization multiplexed system, and since they are perpendicular will propagate independently with negligible power transfer between the two.

Firstly, a SOI slab waveguide was modelled to plot the effective refractive index (n_{eff}) and confinement factor (CF) over a range of wavelengths using a 1D finite difference eigenmode (FDE) solver, as shown in Figure 7.3. The n_{eff} method of designing waveguide geometry collapses a 3D structure into a 2D cross-section for subsequent solving by a finite difference approach [71, 118]. This method is an efficient means of calculating modal profiles of planar structures intended for single-mode operation, indicating confinement when n_{eff} is above the refractive index of the cladding [71]. Since polarization is not completely bipolar in reality, true fields are quasi-TE or quasi-TM, with the TE polarization fraction in MODE Solutions indicating how close to TE (near 1) or TM (near 0) a mode is.

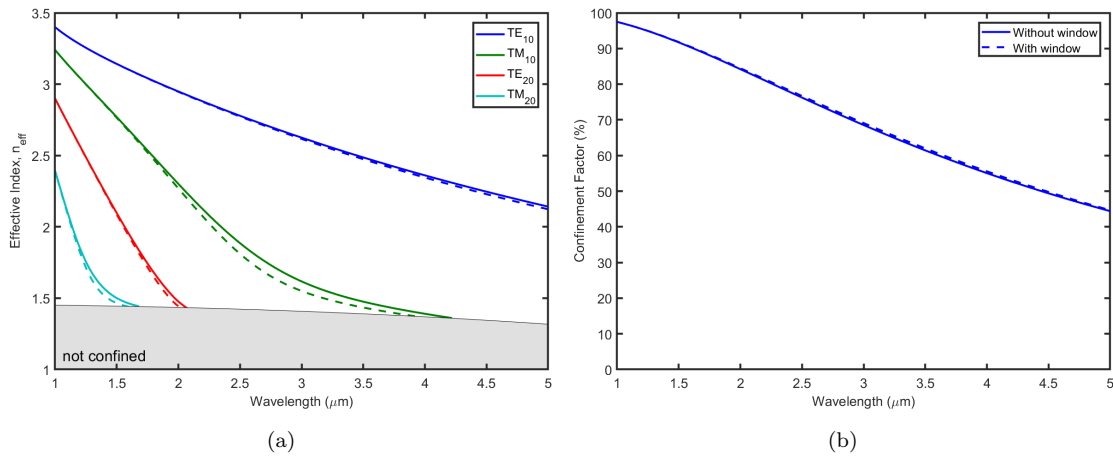


FIGURE 7.3: Simulation of a) Effective refractive index and b) Confinement factor against wavelength, of SiO_2 on SOI slab waveguide for a Si height of 340 nm. Solid line is model without etched opening in top SiO_2 layer, dashed line is with opening.

The CF is calculated as the percentage of power contained within the fundamental TE mode (TE_{10}) of the waveguide. For comparison, the common platform of a 500×220 nm Si core within a cladding of SiO_2 , and operating at a wavelength of $1.55 \mu m$ has a CF of 90 % [71]. There is a reasonable n_{eff} for the TE_{10} mode in this 340 nm platform for a wavelength of $2 \mu m$ as seen in Figure 7.3a, whilst CF in Figure 7.3b is 84 % and both will improve upon adding horizontal confinement. This shows the suitability of this platform for this wavelength regime since higher-order modes are not significantly confined as they fall below the refractive index of the cladding (SiO_2).

Also illustrated, is the difference between a slab with 100 nm and 600 nm of upper cladding, as would be the case with or without an etched implantation window in the top surface respectively. It can be seen that the window has minimal effect on the fundamental TE_{10} mode at $2 \mu m$ and $1.55 \mu m$, increasing the CF by 0.3 % and less than 0.14 % respectively. Towards longer wavelengths the larger mode interacts more with the upper cladding, and experiences a greater index contrast against air ($n = 1$).

Subsequently, horizontal confinement was created by restricting the lateral dimension to the rib width, and adding a thin 100 nm high slab centered on the rib and aligned to it's bottom edge. Simulations were performed with and without the addition of a $2.4 \mu m$ wide and 500 nm deep etched window in the upper SiO_2 cladding to observe the effect of the defect-inducing implantation window. A sweep of rib width was performed, tracking each mode, to produce the graphs shown in Figure 7.4 at a wavelength of $2 \mu m$ and in Figure 7.5 at $1.55 \mu m$

Note that at certain crossing points between a TE and TM mode (see Figure 7.4a at 850 nm wide, between TE_{20} and TM_{10}), the waveguide polarization is strongly mixed over a small range (eg. approximately 800 to 900 nm). These anti-crossing points occur

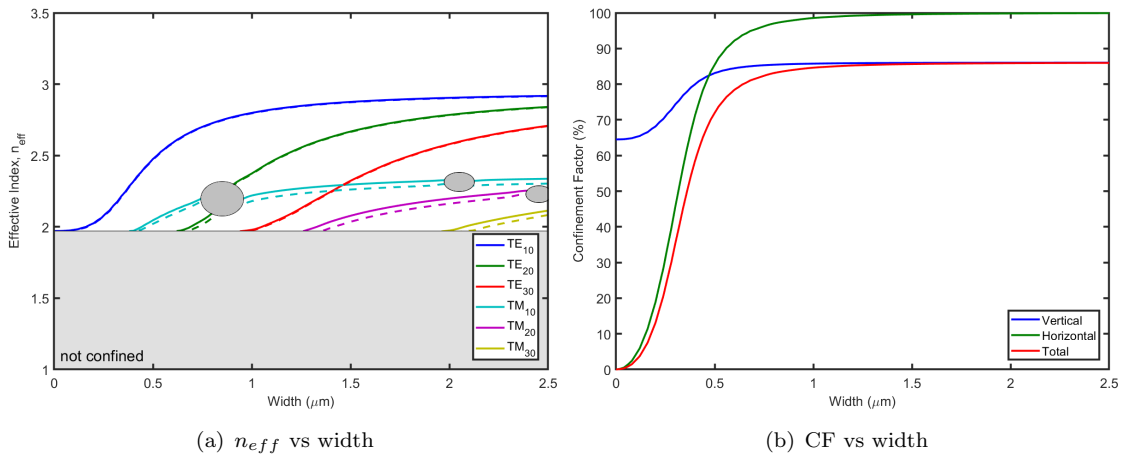


FIGURE 7.4: Simulation of n_{eff} and CF against rib width of a SiO_2 on SOI rib waveguide at a wavelength of $2 \mu m$. Solid line is model without etched opening in top SiO_2 layer, dashed line is with opening, and circles indicate anti-crossing points. Si height is 340 nm, and slab height is 100 nm.

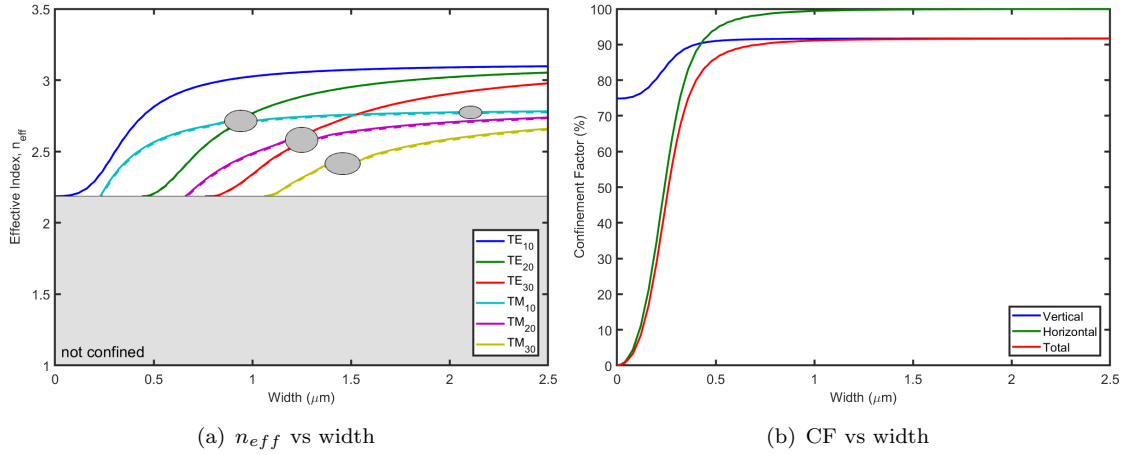


FIGURE 7.5: Simulation of n_{eff} and CF against width of a SiO_2 on SOI rib waveguide at a wavelength of $1.55 \mu\text{m}$. Solid line does not include etched opening in top SiO_2 layer, dashed line does, and circles indicate anti-crossing points. Si height is 340 nm , and slab height is 100 nm

as a result of antisymmetric mode interaction where they cannot have the same effective index, which does not occur for symmetric modes [119]. The effected points are indicated on the graph as a greyed-out circle, and each curve tracks the relevant polarization as indicated in the legend. The corresponding dimensions can not be used for passive waveguiding due to the high likelihood that the propagating mode will change and cause modal losses, however this hybrid polarization could be useful for polarization conversion tapers.

Single mode operation occurs for widths up to 620 nm without, or 661 nm with window at a wavelength of $2 \mu\text{m}$, or up to 449 nm and 452 nm respectively at $1.55 \mu\text{m}$. The corresponding maximum confinement at the upper limit of the single-mode region is 79% without and 80% with the window at $2 \mu\text{m}$, and are both 84% at $1.55 \mu\text{m}$. The oxide window has the effect of increasing the upper widths which can maintain single mode operation since the region of air introduced has a lower refractive index than SiO_2 , improving vertical confinement. However, since this reduction is not matched horizontally, there is a slight modal expansion in this direction and therefore a wider core is needed to contain it. From these results, an upper limit of 620 nm is imposed to maintain single mode operation for both cases at a wavelength of $2 \mu\text{m}$.

Operating at $1.55 \mu\text{m}$ with this width may cause a small excitation of the second TE mode, as seen in Figure 7.5a. However, a wider core will experience lower propagation losses from sidewall scattering, which has a greater effect than modal loss. From this a rib width of 600 nm was chosen, corresponding to a n_{eff} of 2.59 and a total CF of 78.4% . To achieve a similar CF to the previously mentioned $500 \times 220 \text{ nm}$ Si platform, a taller rib would be needed as the vertical confinement is the limiting factor in these devices. A wavelength sweep of the device at the chosen rib width is shown in Figure 7.6, which shows single TE mode operation is maintained down to a wavelength of around

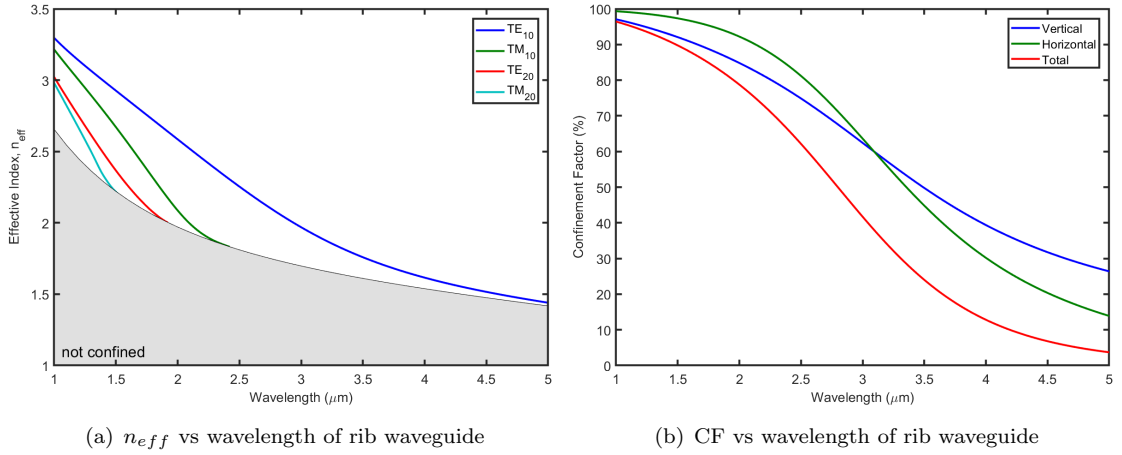


FIGURE 7.6: n_{eff} and CF against wavelength of a SiO_2 on SOI waveguide with a rib width of 600 nm, and including an etched opening in top SiO_2 layer. Rib height is 340 nm and the slab is 100 nm high.

1.9 μm , and the second TM mode only becomes confined below a wavelength of 1.52 μm . For operation towards longer wavelengths, this cross-section can maintain at least 50 % confinement up to a wavelength of 2.8 μm . The field profile for the designed structure is illustrated in Figure 7.7a at a wavelength of 1.55 μm and Figure 7.7b at 2 μm . Both have the structural dimensions superimposed over the profile, and show good optical confinement of the fundamental TE mode.

Regarding parasitic loss to electrodes due to modal interaction with the metal, the minimum distance between rib and contact is 400 nm. At this distance the optical mode

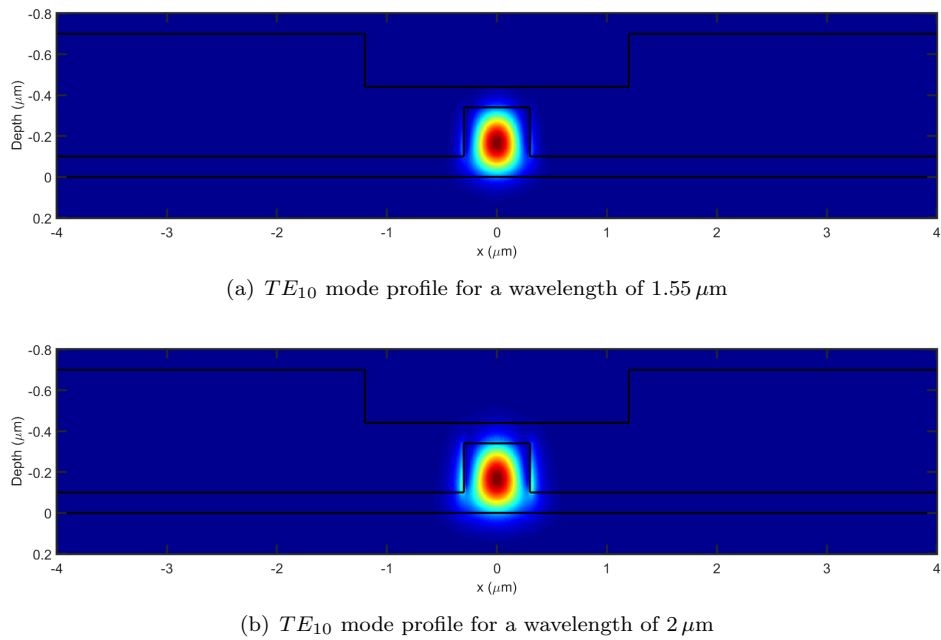


FIGURE 7.7: Optical mode profile superimposed upon cross-section of designed SOI detectors

intensity drops to -23 dB of its maximum value when operating at a wavelength of $2\ \mu\text{m}$, or -34 dB at $1.55\ \mu\text{m}$, with most devices having a drop of -73 and -102 dB respectively due to a larger contact separation. This ensures negligible loss for most devices, with only the smallest experiencing a minimal amount of interaction with the metal electrodes.

7.1.4 Ion implantation simulations

From the previous design in Chapter 4, it is apparent that a greater concentration of defects is needed to stimulate a stronger current response. As such, multiple different ion species have been simulated using SRIM calculations for the implantation process in this design [107]. SRIM uses a series of monte carlo statistical algorithms to perform a quantum mechanical analysis of interactions between incident ions and host material atoms, to establish their stopping range, energy loss and damage creation [107]. This allows for tailoring of the peak, vacancy-type, defect concentration to a depth which overlaps optimally with the rib waveguide center where the highest intensity optical mode will propagate. This also coincides with the center of the depletion region since each contact is at an equal distance from the rib on each side. The defect species modelled is single vacancies since this is the primary defect created independent of incident ion species, and the concentration of any compound defects created is likely dependent upon the strength and distribution of vacancies. Those defects with the highest concentration, and which will produce the greatest photocurrent in a detector, would likely be vacancy-based. When designing a process, it is important to consider not only the damage created by a given ion species but also any additional doping it may create.

Numerous published results have used Si^+ since it matches the host crystal atoms and therefore ensures any absorption changes observed are entirely due to the defects created [34, 49, 62, 68, 83, 96, 98, 120–126]. Whilst B^+ has also been used elsewhere as it's small size requires a lower implantation energy to reach a desired depth [5, 24, 35, 48, 56, 102, 103, 122, 123, 127]. More recent work has used Ar^+ ions since it is larger than Si, so creates greater damage, as well as being inert which prevents the introduction of additional doping within the device [52, 81, 82, 128]. The ion species simulated here focus on group IV elements (Si^+ , Ge^+ and Sn^+) and the noble gases (He^+ , Ne^+ , Ar^+ and Kr^+), since they will either match the number of outer-shell electrons in the host material of silicon, or have a complete outer shell making them inert. B^+ and P^+ ions have also been simulated as they are readily available in CMOS foundries, and phosphorous may produce some favourable vacancy compounds which will have different absorption characteristics to the divacancy (see Figure 3.3) [80].

Although more damage than the previous design is required, consideration is needed to avoid the amorphization threshold and prevent breakdown of the crystalline structure. If this is considered as the point at which 10 % of lattice sites are occupied by defects,

this would mean a maximum concentration of $5 \times 10^{21} \text{ cm}^{-3}$ in silicon [46,129]. However, it is generally agreed that around 90-99% of damage that would be created by ion implantation at zero thermal energy (as is the case in SRIM simulations) would be lost for a sample at room temperature. This implies that the maximum concentration becomes $5 \times 10^{22} \text{ cm}^{-3}$, assuming that the high number of point defects introduced do not significantly conglomerate into more thermally stable complexes before they are able to diffuse out of the target material [130].

A 1D structure consisting of 100 nm SiO_2 on 340 nm Si on $2 \mu\text{m SiO}_2$ was used to simulate the implantation of various ion species at energies ranging from 20 to 900 keV. The calculated ion range against implantation energy is illustrated in Figure 7.8, with the depth of the waveguide core indicated. The simulation conditions used were: monolayer collision steps for full damage cascades, 2000 ions for reasonable monte-carlo distributions, a plotting window depth of 800 nm and 0° incidence angle. The default material parameters (density, stoichiometry and activation energies) were used throughout. While the energy range ensured that all ions (except Sn^+) reached a minimum range of 440 nm, corresponding to the bottom of the core. Some ions are not simulated for the entire range since they are already very deep into the material structure at much lower values. These show a near-linear relationship for atoms greater than or equal to silicon in mass, and an increasing polynomial relationship towards smaller ions. As illustrated, larger ions require an increased implantation energy to reach a target depth, which is shown in Figure 7.9 to be a decaying exponential relationship. Ions larger than Sn^+ may become impractical for defect generation in this structure due to the significantly higher energy

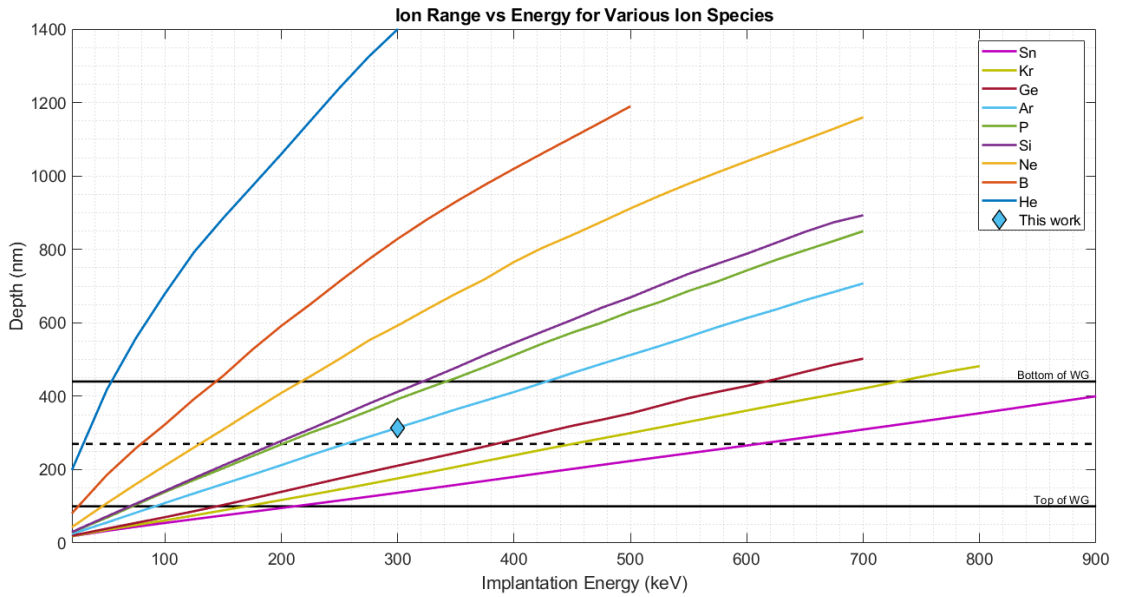


FIGURE 7.8: SRIM calculations of ion range against implantation energy for various elements. The material layer stack consists of: 100 nm SiO_2 on 340 nm Si on $2 \mu\text{m SiO}_2$. Superimposed over the graph is the waveguide top, bottom and center line.

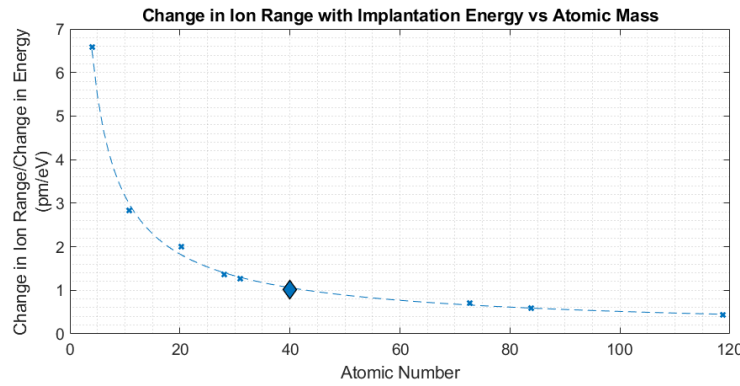


FIGURE 7.9: Rate of change in range with energy plotted against atomic number for each element simulated, showing the decaying effect of implantation energy on range for heavier ions (curve is a guide for eye only)

that would be required, in addition to a potentially amorphousizing amount of damage that would result.

Simulations have shown that only a very small proportion of energy (up to 4.7%) contributes to the creation of vacancies, as illustrated in Figure 7.10. This is primarily due to recoiling atoms near the surface, and ions deeper into the substrate. It can be seen that lower energy implantations have a greater proportion of energy transferred to vacancies, and as such, shallower structures which require a reduced peak concentration depth would benefit from more efficient defect creation. The recoil loss for ions significantly smaller than the target material (He^+ and B^+) is very low since there is insufficient momentum to knock atoms out of the crystal lattice, and therefore most energy is lost to ionization by ions. These were demonstrated to have a small response to illumination in Chapter 4. To create greater damage, more energy needs to be lost by recoiling atoms

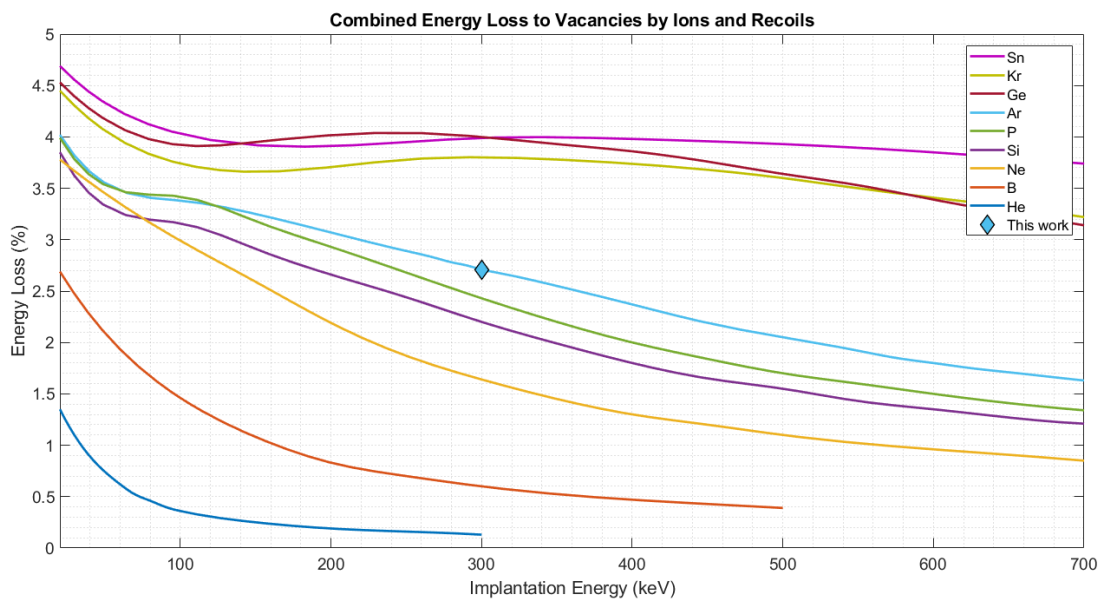


FIGURE 7.10: Combined energy lost to vacancies by ions and recoiling atoms

than by ions leading to the use of elements larger than the host material of silicon. An additional benefit of larger ions is a greater energy tolerance during fabrication since there is a slower rate of change than smaller ions at the energies required to align to the waveguide core. Since the increasing loss by ions at higher energies is likely due to channelling effects, more vacancies may also be created by reducing this to increase the energy lost by recoiling atoms instead of the incident ions. This can be achieved by using a wafer with an alternative crystal orientation, an off-normal implantation angle, or a SiO_2 barrier which is thin enough to not significantly reduce the benefit of near-surface damage from lower-energy implantation.

An optimal response from the created defect distribution is obtained when the peak vacancy concentration depth coincides with the center of the waveguide core. To tailor this depth, the 2D vacancy distribution obtained from SRIM simulations is summed over the planar dimension to emulate the cumulative effect upon a point of implanting over an area. This 1D distribution is then smoothed to reduce the random variation from monte-carlo simulations and better illustrate a device with much higher number of ions implanted, the depth of peak concentration is then plotted against implantation energy to produce Figure 7.11. A similar relationship is seen as with ion range in Figure 7.8, but at a steeper gradient since the vacancy distribution peaks at a shallower depth than ions. Near the deeper Si-to- SiO_2 interface at the bottom of the waveguide, additional energy is required to overcome the barrier and extend into the substrate as a result of silicon having a higher density, since ions will collide with more target atoms than when travelling through the surrounding SiO_2 .

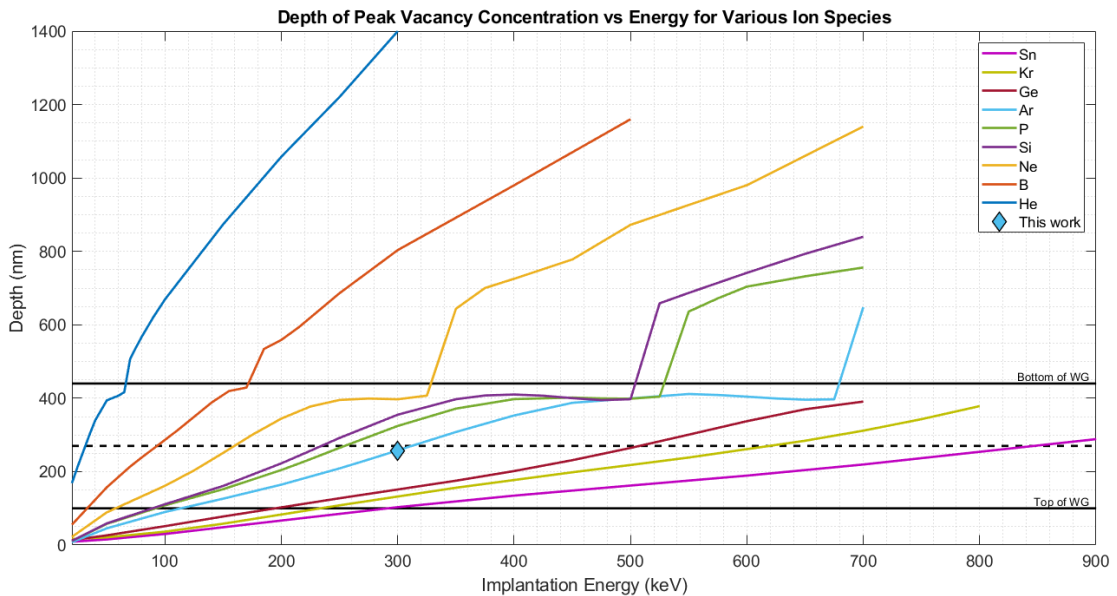


FIGURE 7.11: Depth of peak vacancy concentration calculated from the 3D damage distribution of SRIM calculations, plotted against implantation energy for various elements. The material layer stack consists of: 100 nm SiO_2 on 340 nm Si on 2 μm SiO_2 .

Superimposed over the graph is the waveguide top, bottom and center line.

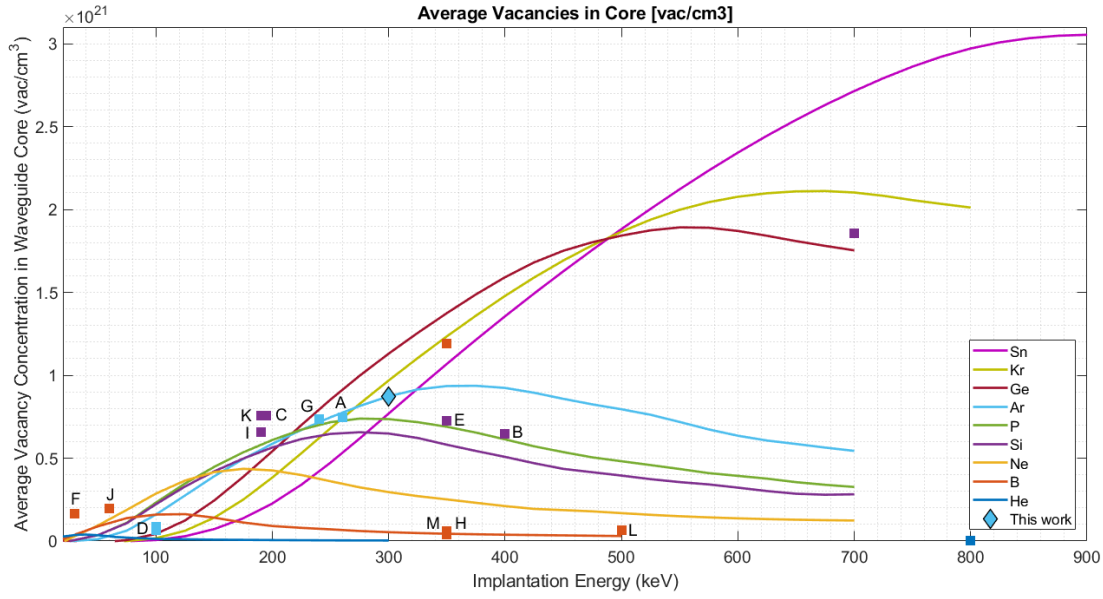


FIGURE 7.12: Average vacancy concentration in waveguide core with an ion dose of $1 \times 10^{13} \text{ cm}^{-2}$, also shown are published results (squares) under the conditions as listed in Table 7.1 and Appendix B.2, and the chosen Ar^+ energy (diamond)

To identify the energy needed to maximise the defect concentration within this structure, a cumulative total number of vacancies with depth was calculated and used to determine those present within the core region, averaged over its height. This resultant average vacancy concentration is plotted in Figure 7.12, along with those calculated for the simulated SRIM profiles under conditions used within the published literature. Each point with information about the device responsivity is noted by a letter, with the values given in Table 7.1. See Appendix B.2 for full simulation conditions and a summary of the photodetector characteristic results obtained.

TABLE 7.1: Responsivity of published results. An asterisk indicates values calculated from provided results or measured from graphs.

	Responsivity (mA.W^{-1})	Wavelength (μm)	Anneal ($^{\circ}\text{C}$)	Dose (cm^{-2})	Voltage (V)	Source
A	281	1.55	350	7×10^{12}	5	[82]
B	160	1.55	300	10^{13}	20	[122]*
C	101.5	1.55	-	10^{13}	5	[34]
D	71	1.56	300	10^{12}	5	[81]*
E	53.6	1.55	350	10^{13}	5	[121]
F	35.5	1.52	-	10^{13}	5	[24]*
G	21.3	2.25	300	7×10^{12}	5	[52]
H	11	1.55	-	10^{13}	10	[102]*
I	6.2	1.54	325	10^{13}	5	[120]
J	6.03	2.02	-	10^{13}	5	[5]*
K	5.2	2.2	350	10^{13}	5	[49]
L	2.17	1.56	200	5×10^{12}	5	[56]*
M	1.06	1.55	350	10^{13}	5	[122]*

Generally, the best response is seen by devices with a higher average vacancy concentration in the waveguide core, with most points roughly coinciding with the curves for this device (given a wide design variability). With the exception of the highest B^+ point at 350 keV and Si^+ at 700 keV which use a much higher implantation dose of $3 \times 10^{14} \text{ cm}^{-2}$ and $7.5 \times 10^{13} \text{ cm}^{-2}$ respectively. If calculated with an equivalent dose, these points match the curves. As such Ar^+ is to be used in this chapter for a better response than Si^+ as in most studies. Note that all studies using Ar^+ in Figure 7.12 use a lower dose, and if scaled to $1 \times 10^{13} \text{ cm}^{-2}$ increase even more than shown here. Also, in one of these previous studies by B. Souhan, it was suggested that divacancies produced by this ion have an increased thermal stability of up to 400 °C [82]. While another by the same author, proposed that positioning both vacancy and ion concentration peaks near to the core center would improve this effect and therefore device response [52]. To find this depth, the graph in Figure 7.13 was produced to show the average combined offset, with respect to the percentage of waveguide height (a 50 % offset indicates either the waveguide top or bottom interface). From this 300 keV was selected from the higher energy end of the minimum offset for argon, so more energy is available for the creation of defects and to slightly favour the vacancy contribution.

This chosen energy is added to all the figures in this subsection, and corresponds to an average vacancy concentration of $8.72 \times 10^{20} \text{ cm}^{-3}$. The peak concentration is within the waveguide core and amounts to $1.24 \times 10^{21} \text{ cm}^{-3}$, which falls below the required 10 % of lattice sites to prevent amorphous silicon forming. After including room-temperature thermal effects this will result in approximately $1.24 \times 10^{19-20} \text{ cm}^{-3}$ remaining. From

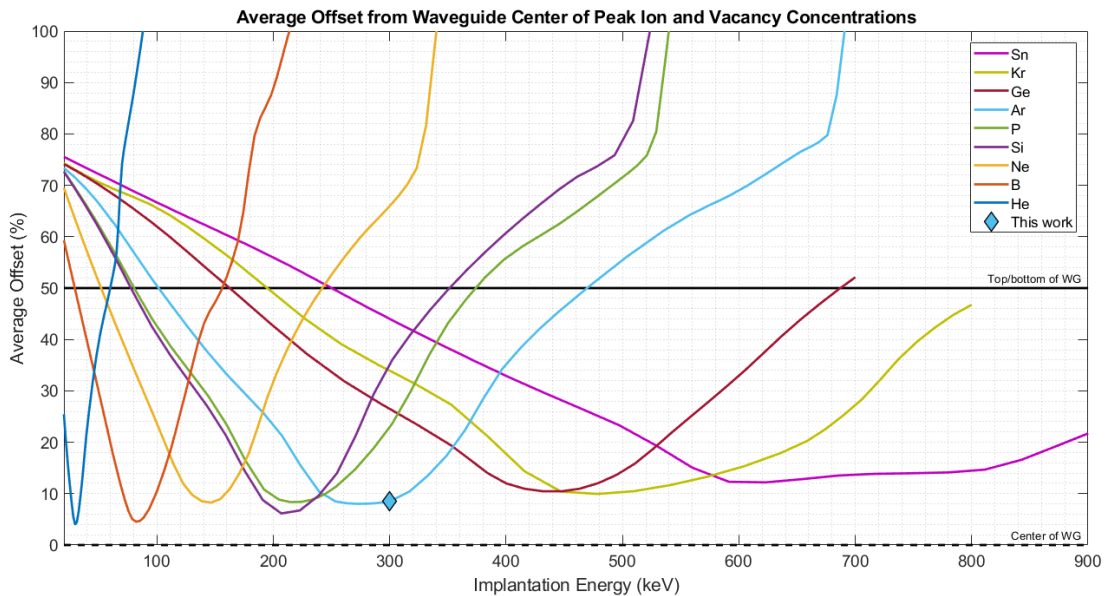


FIGURE 7.13: Average offset of peak ion and vacancy concentrations from the center of waveguide core for various ion species, given as a percentage of waveguide height. The material layer stack consists of: 100 nm SiO_2 on 340 nm Si on $2 \mu\text{m}$ SiO_2 . Superimposed over the graph is the waveguide top/bottom interface with cladding, and the center line.

this, up to a maximum dose of $4 \times 10^{13} \text{ cm}^{-2}$ could be used to further increase defect concentration, however $1 \times 10^{13} \text{ cm}^{-2}$ is maintained for consistency with literature (see Table 7.1). From Equation 4.1 for vacancy-type defect concentration at half ion range (as used in Section 4.2), it is expected that $4.16 \times 10^{18} \text{ cm}^{-3}$ of primarily divacancies will remain in the device (up to 34 % of room-temperature vacancy concentration).

7.1.5 TiN barrier-enhancement layer

To reduce the flow of dark current by thermionic emission over the Schottky barrier, a barrier-enhancement layer is incorporated between the silicon substrate and bulk metal of the contact layer. Against an n-type Si substrate, an abrupt junction with a high work function material creates a high Schottky barrier against current flow, therefore an enhancement layer would need a higher work function than the primary contact material. The inverse is true for a p-type substrate. The primary material used here is Al which has a relatively low work function of 4.06 - 4.26 eV (variable as a result of crystallinity and orientation) [131]. Therefore, a high work function barrier is to be desired in combination with substrate biasing to invert the majority carrier concentration between contacts to emulate n-type Si. Three different metal stacks were investigated, as listed in Table 7.2, each with an Al thickness of 300 nm. The first stack does not contain an enhancement layer, to act as a reference for comparing the effectiveness of the barrier-enhancement layers.

TABLE 7.2: Metal contact layers used for MSM device fabrication

Metal Stack	Thickness (nm)	Description
Al	300	Reference device without additional layers
TiN - Al	20 - 300	TiN is a barrier enhancement layer Al is the bulk metal contact
Ti - Au - Al	20 - 20 - 300	Ti is an adhesion layer Au is the barrier-enhancement layer Al is the bulk metal contact

The second stack adds titanium nitride (TiN), which has a tunable work function of 4.55 - 4.77 eV (for a 7 - 20 nm thick layer [132]) and increases with layer thickness. If this layer is too thick, a high bias voltage will be needed during operation, whilst if too thin, tunnelling will be possible through the barrier negating any gain by increasing it's height. As such this layer is 20 nm thick to provide a reasonably high barrier, while preventing the need for a high bias voltage.

Deposition of this layer is performed using a magnetron sputterer with a DC source target, and process conditions as published by S. Ohya et. al. [133]. A 15 sccm high-purity Ar flow is used with a Ti target and N₂ gas flow, combining into TiN on the sample surface, which is rotated during deposition for improved uniformity with no substrate heating or bias applied. Control of the N₂ flow rate at a constant pressure allows

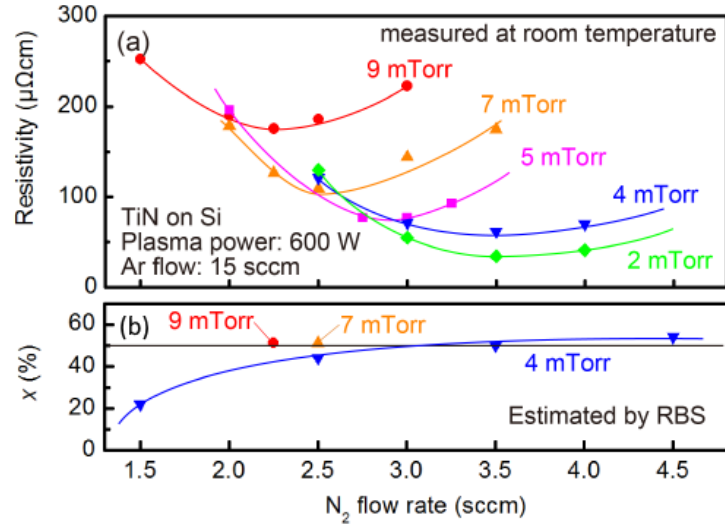


FIGURE 7.14: Dependence upon N₂ flow rate of a) resistivity and b) composition of TiN films, sputter-deposited at 2, 4, 5, 7 and 9 mTorr at room temperature. Ar flow is 15 sccm and DC plasma power is 600 W with a 4" Ti target and Si substrate. Sourced from [133]

tailoring the process for stoichiometric Ti_xN_{1-x} ($x = 0.5$), aiming for a minimum in the graph of Figure 7.14a, as confirmed by S. Ohya. et al. by Rutherford Back Scattering (RBS) composition measurements in Figure 7.14b [133]. In this work, a pressure of 2 mTorr is used for a higher quality film and reduction in contaminants. Based upon the range of 2 - 9 mTorr in the literature, this corresponds to a rate of 3.5 sccm [133]. Additionally, DC plasma power is reduced from 600 W in the published data to 330 W, since the target diameter has reduced from 4" to 3", therefore maintaining a power density of approximately 7.3 Wcm^{-2} .

The third stack uses Ti (work function of 4.33 eV) as an adhesion layer against Si, as is common practise, but with an additional Au (5.31 - 5.47 eV) layer since it has a higher work function [131]. These could therefore both improve the quality of the contact and increase the barrier enhancement. The thickness of these two layers were matched to the previous stack at 20 nm to ensure any difference observed was a result of the different material rather than a change in thickness.

7.2 Device fabrication

This design was also fabricated at the Southampton Nanofabrication Center, following the schematic process flow illustrated in Figure 7.15, with full details included in Appendix B.1. The starting wafer was a SOI wafer with 400 nm thick device layer and 2 μm thick buried oxide (BOX) layer. Since a 340 nm high waveguide is required, fabrication began by reducing the upper layer by a process of thermal oxidation followed by 20:1 BHF etching. Based on preliminary tests and multi-point ellipsometry measurements

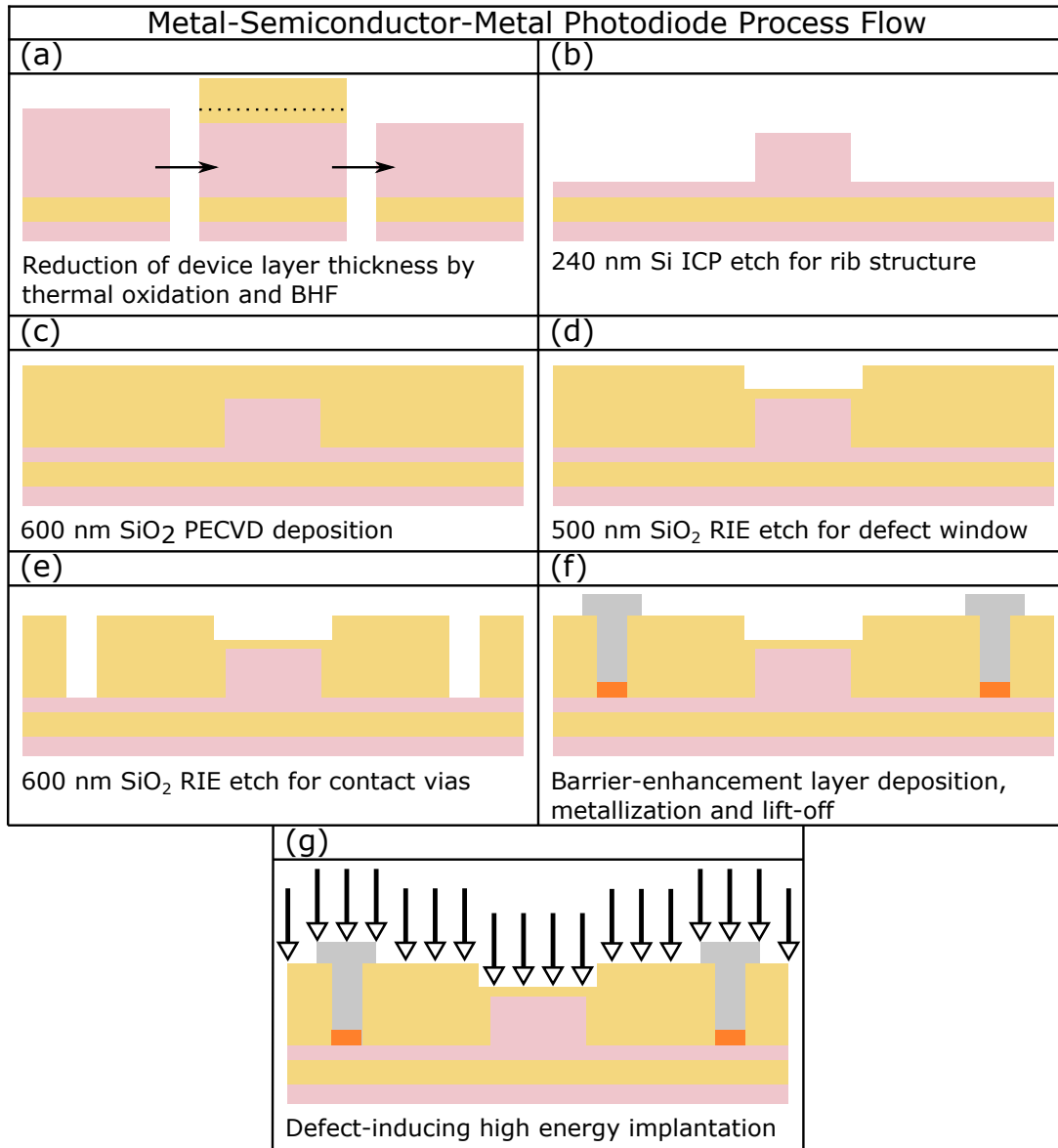


FIGURE 7.15: MSM process flow showing key fabrication steps (not to scale)

across the wafer area, growth of SiO₂ by dry oxidation consumed Si at a rate of roughly 4.55 nm per 10 nm of SiO₂. This resulted in 340 - 344 nm of Si remaining after 9 hr 33 m on the device wafers. This was then followed by formation of alignment marks using photolithography with 1.2 μ m of spin-on Shipley S1813 photoresist, and a 1.34 μ m RIE etch through the device layer and into the BOX. This is not illustrated in Figure 7.15.

Next, an inductively coupled plasma (ICP) Si etch was used to form the waveguide ribs. This creates a lower side wall roughness than a conventional RIE process, and is desirable for reducing waveguide scattering loss. This was lithographically defined using ZEP 520A positive e-beam resist and exposed by e-beam operator Dr A. Khokhar using a Jeol JBX 9300 e-beam writing system for direct-write patterning of small feature sizes. The thickness of this resist was 450 nm, and a thin layer of espacer was spun on

top to prevent negative charge build-up which would cause beam deflection and pattern distortion. This etch removed 272 nm, leaving wings with a height of 68 nm which is slightly thinner than designed but not expected to be detrimental to device performance. This is then followed by deposition of a 600 nm PECVD SiO₂ passivation layer.

Defect-inducing implantation windows were formed next using a RIE process, with the pattern defined using e-beam exposure. Since the required depth is 500 nm of SiO₂, and the process chemistry had a near 1:1 etching selectivity against a ZEP barrier, initial trials used a thicker 700 nm ZEP layer. This resulted in cracking at feature corners due to the effect of stress created by thin features with a high aspect ratio, as shown in Figure 7.16. Chamfering the feature corners did reduce this effect, but not remove it. Therefore 450 nm was used with a HDMS (hexamethyldisilazane) primer layer underneath, and resulted in the removal of approximately 517 nm of SiO₂ to leave 83 nm above the waveguide rib. This was followed by a 580 nm SiO₂ RIE etch with the same ZEP barrier to form contact vias in the passivation layer for connection to metal electrodes. The final 20 nm was removed using BHF 20:1 to prevent over-etching into the silicon underneath.

Metal layers were initially deposited on top of a PMMA-on-MMA bilayer resist structure to a total estimated height of 600 nm. This would provide a roughly 100 nm undercut to aid lift-off due to the faster development rate of MMA. It would be difficult for sputtered PVD processes to completely coat this bilayer and prevent separation of the material on top of the resist. An effect which is not present in evaporation due to the strong directionality of deposition. The shorter overall height of the bilayer may also aid stability of the 1 μ m wide resist strips between contact pads for the smallest photodetectors. However, subsequent lift-off overnight in NMP followed by up to 25 min

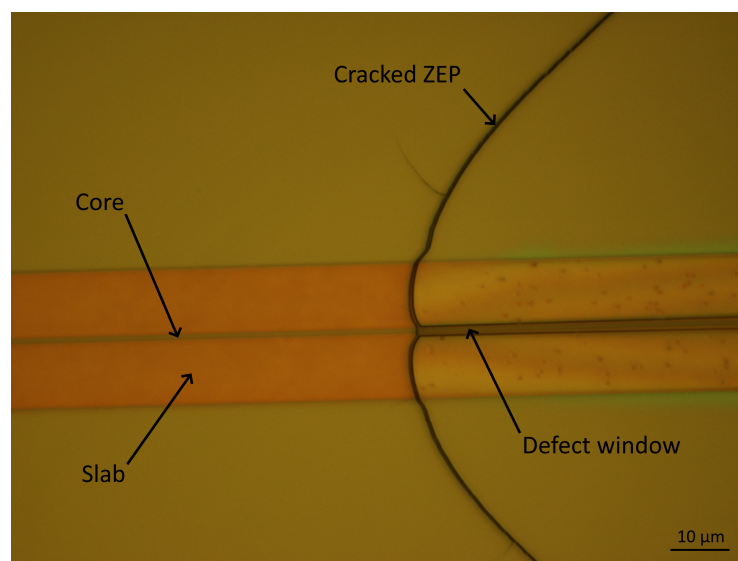


FIGURE 7.16: Optical microscope image of 700 nm ZEP after development of defect window, showing significant cracking from the corners of the developed hole. This feature was present for the majority of corners and devices.

sonication at 40 °C failed to separate the unwanted areas of the metal film, as shown in the SEM image in Figure 7.17. After focussed ion beam (FIB) milling was used to expose a cross-section of the device, this revealed the bilayer was actually 74 nm PMMA on 385 nm MMA with an undercut of 87 nm and a 30° off vertical slope in the lower layer sidewall. The similarity of this bilayer to the height of the metal layer (up to 430 nm during testing) demonstrates why lift-off failed, and subsequent work used the much thicker AZ2070 resist to satisfy the 3x height requirement for this type of process.

Upon resist conversion, metal contacts needed to be inverted in the mask file to account for the change from positive to negative tone resist. AZ2070 was spun on the wafer at 6000 rpm and baked at 110 °C for 15 s to produce a thickness of 4.5 μm . Prior to development, wafers were separated using an automated scribing tool with assistance from J. Reynolds so particles from separation could be washed off in DI water and not contaminate the defined structures. There is also a small undercut profile in this resist once developed to help lift-off, but is not measured here. An additional 30 s dip in 20:1 BHF was used to remove the native SiO_2 immediately prior to metal deposition to improve interface quality.

Deposition of barrier-enhancement layer(s) were performed in an AJA magnetron sputterer. With this process it is easier to control the chemistry of the resultant film during TiN deposition than in a evaporator where nitrogen mostly escapes. Process parameters developed for the sputterer were based upon pre-existing recipes, with Ti and TiN deposited from a DC source target, and other parameters as given in Table 7.3.

Multiple trials targeting a layer thickness of 20 nm gave the coat times listed, although there were issues with measuring step height from the substrate using a stylus-type

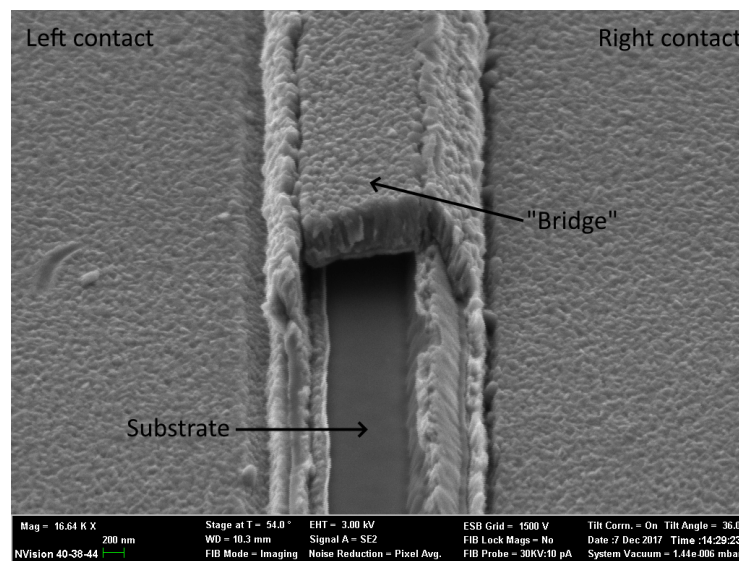


FIGURE 7.17: FIB SEM image showing only partial lift-off from PMMA-MMA bilayer resist. Side wall coverage here has bridged the gap between contacts after resist removal.

TABLE 7.3: AJA magnetron sputterer recipes for deposition of Ti, TiN and Au

Material	Titanium	TiN	Gold
Pressure (mTorr)	3	2	3
Ar (sccm)	30	15	15
O ₂ (sccm)	0	0	0
N ₂ (sccm)	0	3.5	0
Source	DC	DC	DC
Delay time	0	0	0
Ramp time	120	120	20
Power STPT	300	330	300
Shutter delay	30	30	10
Coat time	150 s	215 s	30 s

surface profiler. Surface roughness varied from 6.7 to 63 nm for very thin Ti film thicknesses of 9.7 - 32.2 nm, as shown in Figure 7.18a, creating a very high margin for error. This made single-scan measurements with the low resolution stylus unreliable, and were subsequently replaced with measurements from a ZeScope optical profilometer, over a 1 mm² area positioned roughly half way between the top and bottom surfaces of a step. This tool uses white light interferometry to generate a 3D surface profile of the sample area without needing multiple scan paths. The results had similar average thickness and rate, but with significantly improved surface roughness to 0.38 - 3.89 nm, as shown in Figure 7.18b. The final deposition rates were 8, 5.58 and 40 nm/min for Ti, TiN and Au respectively.

After barrier-enhancement layers, Al was deposited in a BAK600 e-beam PVD evaporator for faster deposition of the thicker layer and a poorer step coverage compared to sputtering for easier lift-off. The final Al thickness was measured as 636 nm since the thicker resist allowed for a larger metal layer, to better fill contact holes and improve

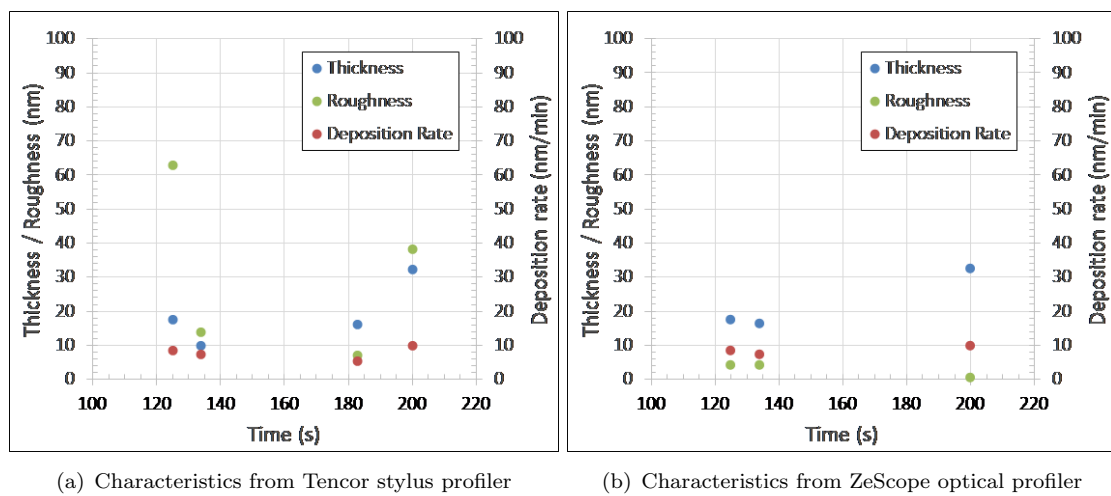


FIGURE 7.18: Thickness, rate and roughness from step profiling sputtered Ti deposition, showing improved average surface roughness of ZeScope compared to Tencor profiler

the connection between substrate and contacts. An overnight static NMP bath with an added 3 - 6 min sonication at 35 °C in fresh NMP provided lift-off.

The final step performs a 300 keV Ar^+ ion implantation at a dose of $1 \times 10^{13} \text{ cm}^{-2}$ as described in Section 7.1.4, to introduce defects into the waveguide core region defined by the pre-etched window. Photolithography could replace the e-beam for contact holes and metallization due to larger minimum feature sizes, but the waveguide rib and defect window definitions are below the minimum feature size of $2 \mu\text{m}$ required for this process.

7.3 Electrical characterisation

Electrical characterisation of the dark current used a Keithley 2401 digital source-meter connected over GPIB to a lab pc running a Matlab script. This performs voltage sweeps across device terminals whilst measuring current flow, with the sample stage held at a temperature of 25 °C. Measurements were performed on each device both before and after implantation, and additionally with the substrate biased between -20 V and 20 V to adjust the Schottky barrier height, as indicated in Figure 7.19. Primarily, dark current is caused by thermionic emission over the barrier in MSM devices, and as such an increase in junction temperature can cause a non-negligible increase in current flow, contrary to diffusion theory which is more sensitive to the applied voltage [110, 134]. For the first voltage sweep of each device, from -10 V to 10 V, the sudden current flow at the start causes an increase in power passing through the junction, and subsequently raises the temperature. Therefore each sweep is performed 3 times for more reliable and repeatable results, with only the first effected by the applied power since it dissipated over a very short period of time. It is expected that improvements in fabrication, and operating the device at lower voltages as is typical in integrated systems, would reduce the influence of junction heating such that this approach would not be needed.

Of the dimensional parameters varied within the mask layout, both the contact separation distance and defect window width have no effect on the dark IV characteristics since it is the size and quality of the metal-semiconductor interface which influences

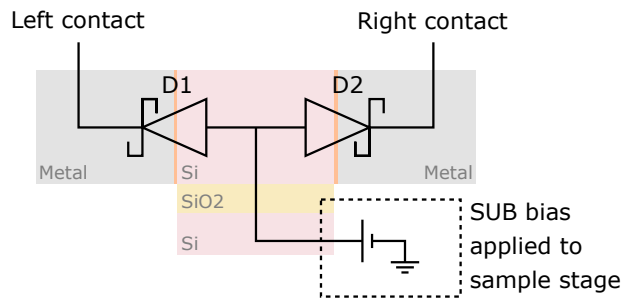


FIGURE 7.19: Representative schematic of schottky diodes created across the MSM structure illustrated in Figure 7.2, overlaid upon material regions

this. The area is determined by the device length and contact hole width, while the interface is influenced by the metal layers and cleanliness during fabrication. IV curves for each contact stack are shown in Figure 7.20 for each detector length, and illustrates the increase in current for longer devices due to a greater contact area. From these, the differing Schottky barrier height for each case is clearly visible as a reduction in current flow with an increasing barrier, with Al having the lowest height, followed by Ti/Au/Al and then TiN/Al (as listed in Table 7.4).

Since each contact within a single device uses the same materials to form the Schottky barrier, the difference in height between them is sufficiently small to form a near-symmetric IV curve. This allows the forward and reverse bias regions of each curve to be considered separately to determine the diode parameters of each contact [135]. Referring to Figure 7.19, holding the right contact at 0 V and applying a positive voltage to the left puts this junction (diode 1) under reverse bias (electrons flow from metal to semiconductor) and controls the current flow. While a negative voltage applied to the left side, puts the junction under the right contact (diode 2) into reverse bias and it controls the current flow. The portion of each curve which is closest to an ideal diode is used for

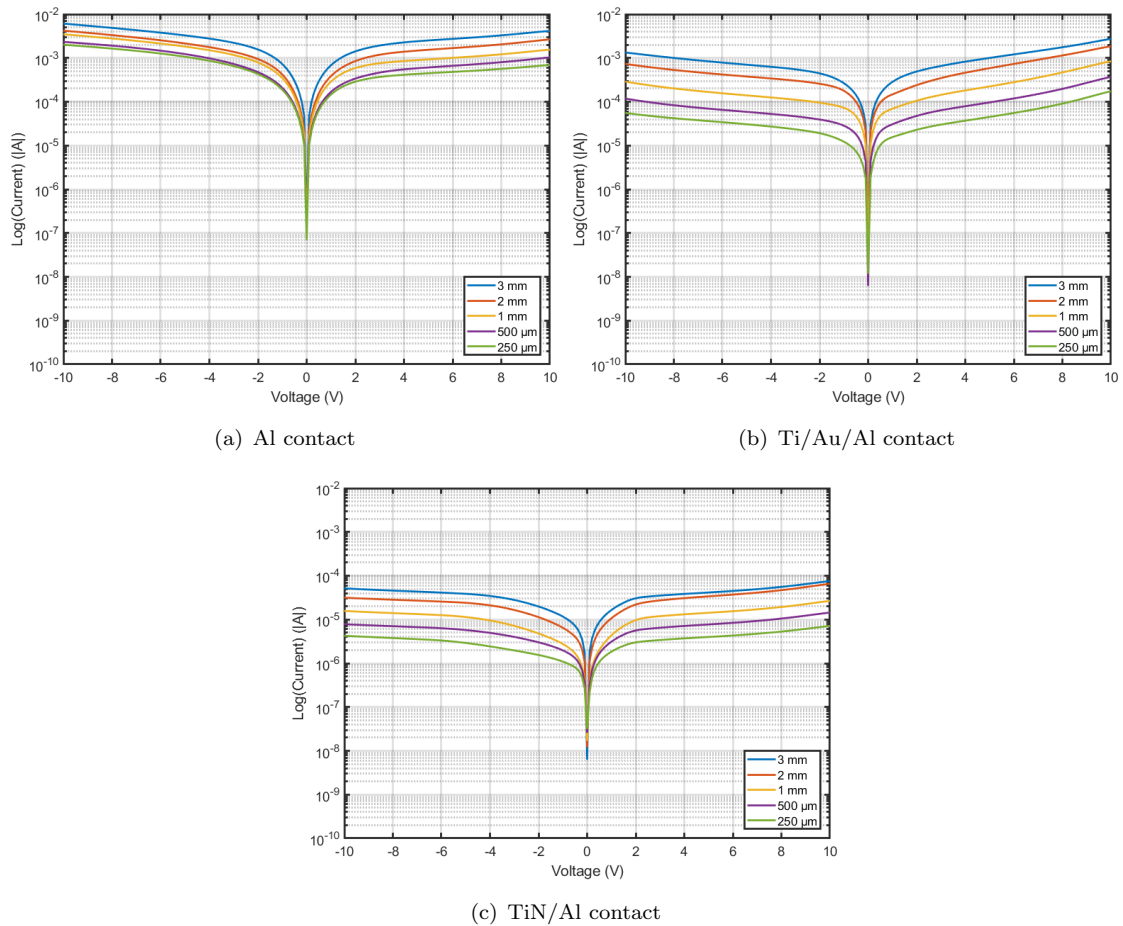


FIGURE 7.20: Dark IV curves for MSM detector with a) Al, b) Ti/Au/Al and c) TiN/Al metal contacts, for each device length

calculations, defined as the voltage range over which the plot of $\ln(I)$ vs V is linear. After removal of resistance effects, this is also where the ideality factor n is closest to the ideal value of 1. Towards higher absolute voltages, effects such as tunnelling through the barrier and Poole-Frenkel emission, cause an increase in dark current but these are not investigated here and as such only the first linear region is considered [136,137].

The resistances present in the device are the shunt resistance R_{sh} which indicates an alternative route for current to flow between contacts, and series resistance R_s which indicates the difficulty of flow through the junction. A good device will have very high R_{sh} for any voltage, with low R_s under forward bias, and high R_s under reverse bias. R_{sh} is calculated as the maximum value on a curve of dV/dI against V for forward and reverse bias independently. Since higher-voltage effects prevent extraction of R_s from the downwards deviation from linearity, a plot of $dV/d(\ln I)$ vs I is used instead. The gradient of this curve represents the resistance according to Equation 7.4, as has been used previously by A. Selçuk et. al. [138].

$$\frac{dV}{d(\ln(I))} = n \frac{kT}{q} + IR_s \quad (7.4)$$

Where q = electron charge (1.602×10^{-19} C), n = ideality factor, k = Boltzmann's constant (1.381×10^{-23} J.K $^{-1}$) and T = absolute temperature in Kelvin (298.15 K) .

Ideality factor and saturation current (I_o) have been calculated using Equation 7.5, which describes the current flow through a reverse biased Schottky contact by thermionic emission over the barrier [136].

$$I = I_o \exp\left(\frac{qV}{nkT}\right) \left[1 - \exp\left(\frac{-qV}{kT}\right)\right] \quad (7.5)$$

Multiplying each side of this equation by $\exp(qV/kT)/[\exp(qV/kT)-1]$ and assuming the case where $V \leq -0.5$ V and $T \leq 370$ K, it can be rewritten in the form of Equation 7.6 [136].

$$I \exp\left(\frac{qV}{kT}\right) = I_o \exp\left(\frac{qV}{nkT}\right) \quad (7.6)$$

Taking the prior graphs in Figure 7.20, n is found from the gradient of the defined linear portion of this curve as in Equation 7.7.

$$\ln(I) = \ln(I_o) + \frac{qV}{kT} \left(\frac{1}{n} - 1\right) \quad (7.7)$$

Extrapolating the tangential line of this linear region allows calculation of I_o as the intercept with the y-axis. This is further described by Equation 7.8, and allows extraction of the Schottky barrier height (ϕ_B) [136].

$$I_0 = AA^*T^2 \exp\left(\frac{q\phi_B}{kT}\right) \quad (7.8)$$

Where A = contact area and A^* = Richardson's constant ($120 \text{ Acm}^{-2}\text{K}^{-2}$). The calculated parameters for the plots displayed in Figure 7.20 are listed in Table 7.4.

TABLE 7.4: Diode parameters before implantation for the devices with measured IV curves shown in Figure 7.20, where R_{sh} / R_s = shunt and series resistances, I_0 = saturation current, ϕ_B / ϕ_{B0} = Schottky barrier height (with/without image-force lowering). Plus the barrier height after implantation without lowering.

		Pre-Implant					Post-Implant
		R_{sh} ($k\Omega$)	R_s (Ω)	I_0 (μA)	ϕ_B (V)	ϕ_{B0} (V)	ϕ_{B0} (V)
Al	Length						
	Diode						
	3 mm	1	4.80	121	1500	0.1324	0.1458
		2	2.03	26.0	781	0.1488	0.1622
	2 mm	1	7.52	14.7	877	0.1354	0.1492
		2	4.28	16.1	578	0.1461	0.1623
	1 mm	1	14.2	65.6	577	0.1283	0.1416
		2	5.44	2.5	565	0.1289	0.1477
	500 μm	1	20.6	39.2	326	0.1252	0.1399
		2	8.34	87.7	507	0.1139	0.1320
Ti/Au/Al	250 μm	1	34.5	202	306	0.1090	0.1231
		2	9.02	62.7	380	0.1034	0.1217
	3 mm	1	6.14	193	176	0.1872	0.1968
		2	14.1	195	350	0.1694	0.1816
	2 mm	1	12.7	1500	77.3	0.1978	0.2034
		2	28.2	472	202	0.1732	0.1847
	1 mm	1	29.9	2160	37.8	0.1984	0.2045
		2	73.8	1100	72.2	0.1818	0.1931
	500 μm	1	67.8	1520	20.1	0.1968	0.2075
		2	168	2520	29.3	0.1871	0.1991
TiN/Al	250 μm	1	151	3260	10.4	0.1959	0.2066
		2	298	2020	13.8	0.1886	0.2015
	3 mm	1	335	6720	27.0	0.2353	0.2471
		2	457	3390	28.3	0.2340	0.2495
	2 mm	1	335	2810	19.0	0.2338	0.2467
		2	772	957	17.9	0.2354	0.2514
	1 mm	1	903	3420	8.74	0.2360	0.2493
		2	1460	6120	8.49	0.2368	0.2527
	500 μm	1	1730	20300	4.81	0.2335	0.2464
		2	3060	37300	4.45	0.2356	0.2518
	250 μm	1	3660	57900	2.64	0.2311	0.2430
		2	5040	51800	2.18	0.2361	0.2521

Also included is the barrier height adjusted to account for the image force lowering effect (ϕ_{B0}). This is dependent upon the strength of the electric field in the device, and causes a gradual increase in current with voltage, becoming more significant when a device is

under reverse bias due to the stronger electric field [136]. It has been calculated using Equation 7.9 [110].

$$\phi_{B0} = \phi_B + \sqrt{\frac{q|V|}{4\pi\epsilon s}} \quad (7.9)$$

Where $\epsilon = \epsilon_r\epsilon_0$ = material permittivity, $\epsilon_{r/0}$ = relative and free-space permittivity (11.7 and 8.85×10^{-12} F.m⁻¹ respectively), and s = contact separation which for this device is $3.4 \mu\text{m}$. The average barrier height calculated without image-force lowering across all devices on each chip are: 145 and 136 mV with Al contacts, 207 and 199 mV with Ti/Au/Al, then 247 and 254 mV with TiN/Al for diodes 1 and 2 respectively.

As well as reducing overall dark current, an interface layer also appears to allow greater control over the flow by adjustment of device length. This can be seen by the greater spread and improved distinction between each length in Figures 7.20b and c, compared to Figure 7.20a. Assuming SNR is constant between the range of lengths implemented, dark current can be reduced to within system tolerance by using a shorter device length; this level of control is not possible for the Al-only contact. The low barrier heights seen in the experimental results of Table 7.4, is believed to be due to a high quantity of surface states. This makes the effective barrier height independent of metal work function and dependent only upon the surface properties at the interface [110]. Whereby the Fermi level is pinned by surface states to a neutral level position with an energy of $q\phi_0$ above the valence band at the interface edge, which increases with surface state density [110]. Even with this, it can be observed that the Ti/Au enhancement layer creates a small increase in barrier height (up to 84 mV for the $250 \mu\text{m}$ long device), while the TiN layer increases the barrier by up to 130 mV. This demonstrates the potential of this material for reducing dark current in this type of device, and is expected to improve further by adjusting sputtering conditions for a more resistive film (see Section 7.1.5).

After high-energy ion implantation, the dark IV curves are now as illustrated in Figure 7.21. The implantation was performed at a 7° off-normal tilt and as such has created an asymmetric device with diode 1 slightly closer to the center of the spread of defects, and therefore influenced the most by them. The effect of implantation is an apparent increase in barrier height for the device under a positive bias (diode 1 reverse biased), as given in the left-most column of Table 7.4, while diode 2 is effectively unchanged. This is more significant for the Al-only contacts (with an average increase of 74 mV in Table 7.4), significantly reducing for a thick interface layer (68 mV for TiN/Al, and 18 mV for Ti/Au/Al), suggesting the cause primarily lies within the semiconductor side of the junction. The barrier layer appears to separate the effect of the change in semiconductor properties, in the case helping to maintain the earlier device symmetry. It also maintains the controllability of current, since the Al contacts no longer have any significant variation with length for negative voltages. The reduction in current flow is believed to be caused by a reduction in carrier mobility or an increase in recombination

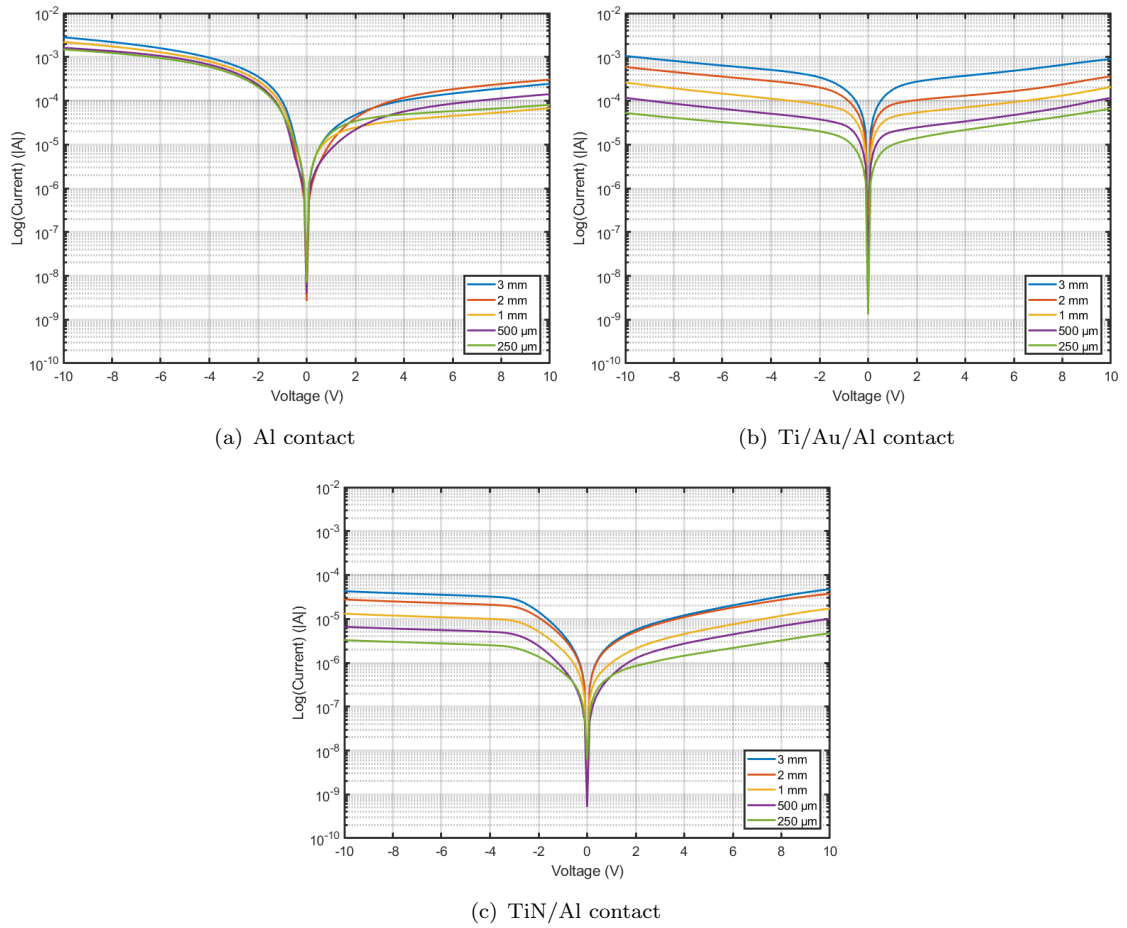


FIGURE 7.21: Dark IV curves for MSM detector after high-energy Ar^+ implantation with a) Al, b) Ti/Au/Al and c) TiN/Al metal contacts, for each device length

as a result of defect introduction. Low mobility would cause majority carriers to require a greater voltage to pass over the barrier, or reach a sufficient energy level to tunnel through it. While increased recombination reduces the number of carriers available for collection at the contacts.

To further reduce dark current in these devices, a range of bias voltages were applied to the substrate to control the quantity of majority carriers present within the semiconductor. Since MSM devices rely primarily on majority carriers, a p-type substrate should experience a reduction in dark current flowing between contacts when a strong negative bias is applied to draw out holes. The result of sweeping the substrate voltage on a $250\ \mu\text{m}$ long device between $-20\ \text{V}$ and $20\ \text{V}$ in increments of $5\ \text{V}$, and for each contact type, is shown in Figure 7.22. The most significant change with substrate voltage is the shift in OCV, as caused by the removal of free carriers. Applying a positive substrate voltage draws out electrons to the surface of the BOX and increases the OCV, while a negative voltage removes holes and reduces the OCV. A positive substrate voltage has a more significant effect since it acts on the minority carrier concentration in this device. For the case of Al when the contact bias magnitude is higher than the OCV

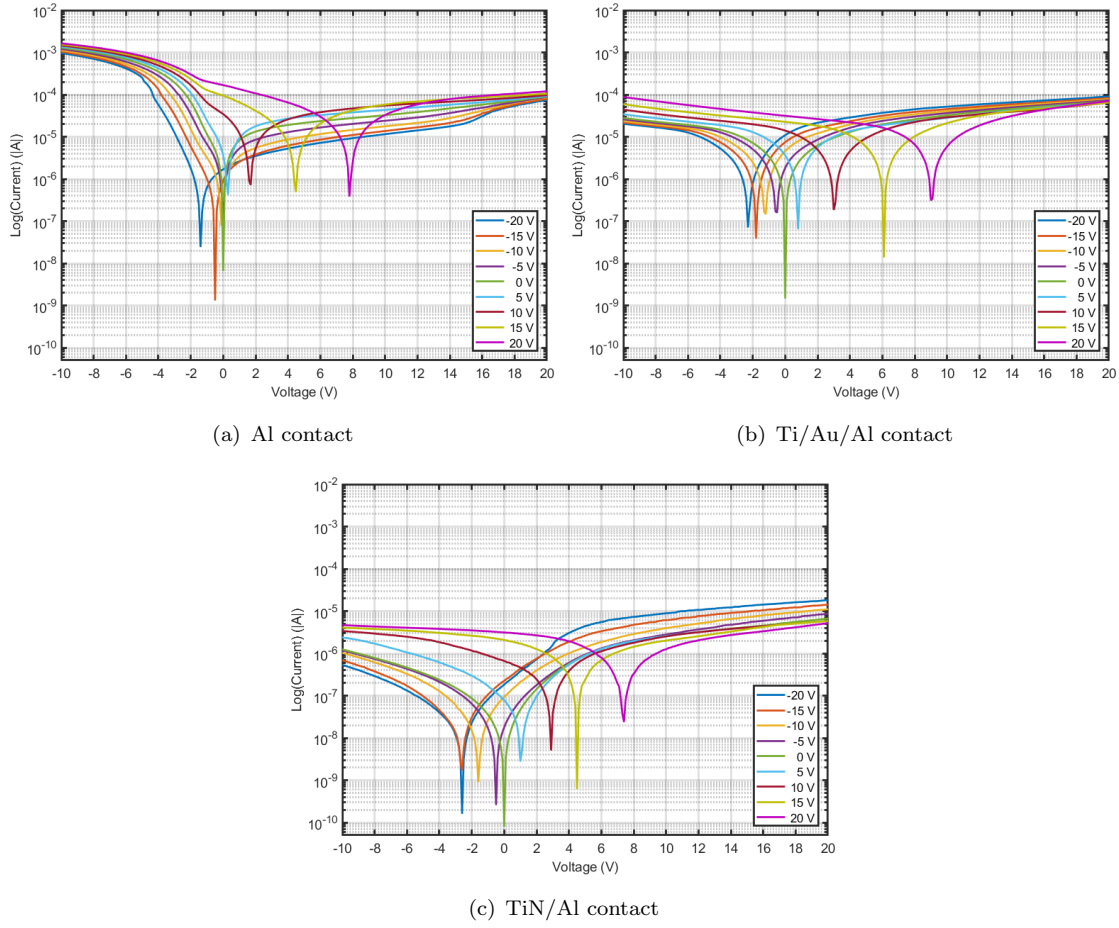


FIGURE 7.22: Dark IV curves for MSM detector after high-energy Ar^+ implantation with a) Al, b) Ti/Au/Al and c) TiN/Al metal contacts, at a range of substrate bias voltages

range ($V \leq -2 \text{ V}$ and $V \geq 14 \text{ V}$), the expected reduction is seen from 514.6 to $250.8 \mu\text{A}$ (51 %) at a substrate voltage of -20 V and a contact bias of -5 V . The opposite effect is seen at 20 V , increasing to $816.7 \mu\text{A}$ (59 %), and with a similar response occurring at $+15 \text{ V}$ contact bias. For the Ti/Au/Al and TiN/Al devices outside of the OCV range, a negative bias of -5 V has a reduction from 17 to $8.9 \mu\text{A}$ and 339 to 78.7 nA for equivalent substrate voltage (a 48 and 77 % change respectively). However, this behaviour is reversed when under a positive contact bias, suggesting that current flow through the junction of diode 1 has a significant component from tunnelling through the barrier instead of being largely dominated by thermionic emission [100]. This can be explained by the introduced divacancies acting as double acceptors, which increases the concentration of holes near the interface. As suggested in previous work by Logan et. al., the use of a n-type silicon substrate would counteract this effect since the divacancies would be compensated for by the majority of electrons [68].

7.4 Summary

This chapter improves upon the environmental lighting SNR limitation in the previous chapter, and explores defect-mediated absorption in an integrated platform, by incorporating defect creation into a MSM structure across a photonic waveguide. It covers theory of the metal-semiconductor-metal junction, and design of the waveguide-integrated photodetector. Then rib waveguide upon which it is based, was simulated using Lumerical's MODE Solutions software. Then SRIM analysis was performed for a range of ion species and energies to determine new implantation parameters for an improvement in trap-assisted response. This considered both the amount of damage produced and an optimal depth for the ion and vacancy distributions, resulting in using Ar^+ at 300 keV with a dose of $1 \times 10^{13} \text{ cm}^{-2}$. A description of the inclusion of TiN or Ti/Au as barrier enhancement layers below the Al contacts to reduce the device dark current is then covered, followed by fabrication. A custom mask design and e-beam patterning is used for the definition of feature sizes down to 200 nm. This is then concluded with experimental measurement of the electrical characteristics for the fabricated devices. These show the desired reduction in dark current for the Ti/Au enhancement layers compared to Al only, which is even greater for the TiN layer. Presently, there are no optical performance measurements due to difficulties with coupling light into the end facets of the fabricated waveguides.

Chapter 8

Conclusions

The photodetector demonstrated in Chapters 4 to 6 has successfully shown the ease of integrating a defect-inducing ion implantation into an existing process flow, and it's beneficial effect of shifting the absorption spectrum of silicon beyond it's intrinsic capability. Whilst the original structure was altered to increase the volume of depleted silicon, within which generated carriers can be extracted, this implantation step has been performed after the formation of dopant regions and prior to passivation. The ion species chosen were He^+ and B^+ since they are light and therefore require little energy to reach the desired depth. Microfabrication of this device is generally uncomplicated, however an overetching of the isolation trenches did occur and removed some material from the surface of the devices. This at most had the effect of partly removing the depletion region between rings, which would slightly reduce the current response for this area in the heat maps in Section 6.1. A small degradation since the deeper region formed with the substrate is unaffected by this fault.

Incorporation of the He^+ implantation has increased the response to 1550 nm light in the smaller-gap devices from 75.4 nA to 225.8 nA at a reverse bias of 2 V, and reduced the response to 633 nm light from 172.8 nA to 135.7 nA at the same bias. The photo-generated portion of this current experiences a 2x increase at a wavelength of 1550 nm and only a 3 % change at 633 nm. Coupled with a dark current as low as 10.6 nA, the SNR also increases from 6.9 dB to 13.3 dB for the same bias. This demonstrates an increase in photogenerated current due to deep-level traps within the band gap from the introduced defects. While an incident signal is still detectable in the presence of background lighting, it noticeably hinders the SNR which becomes near unity when under a high intensity of visible wavelength light. A similarly beneficial performance increase was not observed in the larger area devices, partly due to an increased dark current of 32.8 nA, but also a high implantation energy pushing the damage peak to a depth of around $2.23\text{ }\mu\text{m}$. Combined with a potentially lower carrier lifetime, this allows the 633 nm response to dominate for reverse biases below 7 V. The B^+ implanted devices however suffer from a mostly Schottky contact behaviour. This is due to a reduction in

the heavily-doped contact regions' surface concentration, combined with the new boron ions compensating the surface n-type regions and reducing the free carriers available at the contact. A few larger devices do still demonstrate the increased 1550 nm response with a SNR of 3.4 dB vs 2.3 dB for 633 nm light, with the smaller one having a stronger increase in current at 1550 nm albeit masked by an elevated dark current. Further measurements resulted in average responsivities of 20.65 and 22.23 μAW^{-1} for the 1550 and 633 nm wavelengths respectively, between -7.8 V and 0 V bias. While the spectral response has shifted towards longer wavelengths, broadening to suggest a range of trap levels may have been introduced.

Whilst the free-space device does have the intended improvement in absorption around 1550 nm, it would be more beneficial to perform the high-energy implantation after passivation to avoid the loss of divacancies from PECVD deposition at 325-350 °C. It would also be better after metallization so the contacts shadow the region below and remove additional defects that can only increase recombination. The observed loss of surface concentration that may have caused the B⁺ implanted devices to display Schottky contact behaviour can also be avoided. Photogeneration could also be increased by reducing implantation energy and using a non-dopant ion species. While dark current can be reduced with a deep SiO₂ barrier for substrate isolation, reducing thermal noise, but consideration is needed to allow sufficient absorption length. Trenches would then need to be etched deep enough to reach this layer for complete isolation of each device from the surrounding substrate. Additionally merging the central ring into a single smaller-area contact, will increase the area between rings which is more sensitive to lower bias voltages, while the central region only responded when at higher voltages.

Following on, a MSM style device was designed in Chapter 7 and based upon a 600 x 340 nm rib waveguide on a SOI substrate. This structure restricts the optical signal to a small cross-sectional area near the surface which can more easily be populated with enough defects for a strong response. It is also isolated from environmental lighting effects which were detrimental to the SNR in the previous design. It is isolated from substrate noise by the buried SiO₂ layer, and the defect-mediated region is confined to the rib to limit recombination and reduce their influence on the contact junction. The rib is suitable for propagating light at wavelengths of 2 and 1.55 μm , achieving a total confinement factor of 78.3 and 85.8 % respectively. It will also allow a much greater absorption length than the vertically-illuminated design previously, without sacrificing carrier drift length and operating speed.

Ions at least as large as the host crystal atoms would work best for vacancy creation to stimulate greater damage by recoiling atoms, which is the primary mechanism for their creation, with the best work in literature using Si⁺ and Ar⁺. Doping by ions can be avoided by using group IV elements or inert noble gases. However experimentation with P⁺ could be considered since it is a similar size to Si⁺ and compensates the divacancy double acceptor behaviour. The He⁺ and B⁺ ions used in Chapter 4 create only

a small amount of damage in the MSM structure, since they are smaller than the host crystal atoms and produce few recoils. Additionally, since very little energy is needed to reach the waveguide core the total energy available for defect creation is limited. Comparatively large Sn^+ requires 600 keV for the same depth, leading to expensive ion implantation energy requirements. The choice of Ar^+ ions have a predicted maximum vacancy concentration of $1.24 \times 10^{19-20} \text{ cm}^{-3}$ leading to a $4.96 \times 10^{18} \text{ cm}^{-2}$ divacancy concentration at room temperature, according to SRIM simulations. Shallower devices may enable more efficient defect creation since the near-surface spike in vacancies is combined with a shallower peak vacancy concentration (see Figures 7.8 and 7.10). Reducing channelling effects with a very thin barrier layer or off-normal implantation angle would also cause more damage by recoils rather than ionization.

An integrated barrier enhancement layer consisting of either Ti and Au or TiN, with each layer deposited to a thickness of 20 nm, has also been demonstrated. The TiN layer outperforms both the Ti/Au interface layer and Al-only contacts. The Schottky barrier height of the TiN devices was calculated to be up to 118 mV higher than with only Al forming the contact, while the Ti/Au layer is 63 mV higher. As well as increasing barrier height, these layers also allow finer adjustment of dark current by control of the device length. The high-energy ion implantation was performed in the final stage of fabrication (after passivation and metallization), defined by a thin window in the upper SiO_2 layer. An off-normal angle created an asymmetry in the IV curves and reduced the barrier height by up to 74 mV (Al-only) for the junction closest to the off-center defect concentration. This is likely due to a reduction in carrier mobility in the semiconductor, with thicker interface layers reducing its impact, as seen in the Ti/Au devices. Dark current was further reduced by applying a negative substrate voltage, by 51 % with an Al contact and 77 % with a TiN layer (-5 V contact bias). This behaviour occurs when a reverse bias is applied between contacts (current determined by junction not effected by implant), and only under a forward bias for the Al-only contacts. For both devices with a barrier, a forward contact bias (junction is effected by implant) experiences the opposite effect primarily due to a tunnelling-based current flow.

Optical characterisation has not been performed due to difficulties with end-fire coupling. Therefore further work is needed to measure the waveguiding performance at wavelengths of 1.55 and $2 \mu\text{m}$. Then to establish the Ar^+ implantation influence on photodetection, followed by isochronal annealing at an increasing temperature to determine the thermal stability and the peak responsivity. The design could be adjusted to reduce dark current by improving processing cleanliness, and depositing a more resistive TiN film. While additionally changing to n-type silicon would allow use of the higher Schottky barrier seen by electrons, and would compensate the additional p-type doping introduced by divacancies to lower the carrier concentration and maintain thermionic emission current.

Appendix A: Free-space photodetector

A.1 C-interpreter function for custom photogeneration rate in Silvaco Atlas

Free-space detector carrierGenModel.c: radiate function for *beam* Atlas statement for a custom photogeneration rate as a function of position

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

/*
 * Generation rate as a function of position
 * Statement: BEAM
 * Parameter: F.RADIATE
 * Arguments:
 * x location x (microns)
 * y location y (microns)
 * z location z (microns)
 * t time (seconds )
 * *rat generation rate per cc per sec.
 */
int radiate(double x,double y,double z,double t,double *rat)
{
    double beam, cargen, vacConcen, V2concen;
    double beamStartX, sx, x2;
    double L8, L7, L6, L5, L4, L3, L2, L1, Rmean, RstdDev;
    double R3, R2, R1, SRIMy, minVal, maxVal, normY;
    double adjDose, dose;
    double lambda, h, c, abs_trap;

    // Adjust (x,y) for -300nm starting depth & laser angle (6.3DEG)
    y = y + 0.3;
    x = x - y*tan(6.3*M_PI/180);

    // Model incident beam profile [unitless, normalized distribution]
    // Position of beam over surface [um]
    beamStartX = 42.5;
    // SM fibre with mode-field diameter 10.4 um [sqrt(((10.4/2)^2)/2)]
    sx = 3.67696;
    x2 = (x-beamStartX)/sx;
    // Assume gaussian-x uniform-z model (default depth is 1 um )
    // Scalar ensures total area underneath = 1 (power=1)
    beam = exp(-pow(x2,2)/2) / (sqrt(2*M_PI)*sx);
    // Model implantation damage profile analytically [cm-3]
    // (Split-plot fitting (poly on left, gaussian on right))
    L8 = -2.21734496678749E-30; Rmean = 26100;
    L7 = 1.64935402131371E-25; RstdDev = 2381.17617995813;
    L6 = -4.74414605734378E-21; R3 = 0.0443672862649949;
    L5 = 6.69512787443923E-17; R2 = -1.46329926442772;
```

```

L4 = -4.78981660350695E-13; R1 = 0.52589368078946;
L3 = 1.60361835058446E-09;
L2 = -1.79223409241068E-06;
L1 = 0.00175126415187797;
SRIMy = y*10000;
if (y <= 2.19) {
    // 7th-order poly fit
    vacConcen = L8*pow(SRIMy,7) + L7*pow(SRIMy,6) + L6*pow(SRIMy,5);
    vacConcen = vacConcen + L5*pow(SRIMy,4) + L4*pow(SRIMy,3);
    vacConcen = vacConcen + L3*pow(SRIMy,2) + L2*SRIMy + L1;
}
if (y > 2.19 && y < 2.25) {
    // Linear interpolation between fits to improve curve shape
    // (causes slight over-estimation of total area)
    minVal = 0.041086391133062; maxVal = 0.0439906260733403;
    vacConcen = minVal + (maxVal-minVal)*(SRIMy-21900)/(22500-21900);
}
if (y >= 2.25) {
    // Gaussian fit (same shape as ions at far end of range)
    normY = (SRIMy - Rmean)/RstdDev;
    vacConcen = R3*exp( -pow((normY-R2)/R1,2) );
}

// Convert vacancy concen. profile to divacancy concen. profile
dose = 1e13; // [cm-2]
adjDose = vacConcen*dose;
V2concen = 2.79e10*pow(adjDose,0.63); // [cm-3]
// Scaled to account for change in area from 2D gaussian to rectangle
// PI*5.2^2 / 10.4*1
V2concen = V2concen*8.168140899;
/////

///// Model carrier generation [W-1.s-1.cm-1]
// (Eq. 3.8 in [44] since surface-illuminated device)
// 1e-4 converts um to cm
lambda = 1.55e-6; h = 6.626e-34; c = 2.98e8;
abs_trap = V2concen/7.7e16; // [cm-1]
cargen = (lambda/(h*c)) * abs_trap * exp(-abs_trap*(y*1e-4));
/////

// Rate = beam [n/a] x generation [W-1.s-1.cm-1] x intensity [W.cm-2]
// [cm-3.s-1]
// Intensity provided by SOLVE statement
*rat = beam*cargen;

return(0);
}

```

A.2 Process flow

Photodetector Process Listing				Ref: RPN
Device Description: Free-space illuminated pn diode, with concentric ring contacts				
Starting Material: 4" Si wafer, p-type				
	Process	Tool	Details	Comments & Outcome
1	Oxide Deposition	Dielectric PECVD	Approx. 7.03 - 7.37 s (20 nm thick)	Recipe "SiO2 20nm" Actual time used 18 s at ~1 nm/s. Machine set for 6" wafers? [~25 nm deposited] [>18 nm 2 nd time]
2	Lithography with G0 (1/7)	EVG 620T(B)	Lithography of alignment marks (DF+) Spin coat S1813 4000 rpm, 30 s Pre-exposure bake 90 s @115°C (1.3 µm) Vacuum contact, 2.5 s Develop with MF-319, 45 s, rinse and dry	[Delay due to i-line filter. 2.5 s appears to be correct. X to be measured after lith, over non-patterned area.] ~1.2 µm layer, 4.8 s exposure
3	ICP Etch	Plasmalab System 100	Etch 20 nm SiO ₂ and 1 µm of Si 1 step for oxide and Si, time from machine (time from thickness measurement)	[10 mTorr needs to be operable Measured step height consistently less than resist height Ran for 20+30 s, {1 min}, 1 min 30 s] [Switched to RIE, av 45-50 s R&Ox&Si + ~1 min 30 s Si. Poly-Si recipe]
4	Strip resist	Plasma Asher	10 - 15 min optionally with acetone/IPA	No pattern on 1 st run, repeat 2-4 ~8 min
5	Implantation	(Surrey Uni Tool)	Implant P for top-covering n region Energy 35 keV Dose 4x10 ¹³ cm ⁻² Beam current 100 µAcm ⁻² Angle of incidence 10°	≤25 day turnaround 2 wafers held back for IBC step
6	Lithography with G1 (2/7)	EVG 620T(B)	Lithography of N++ region (DF+) Spin coat S1813 4000 rpm, 30s Pre-exposure bake 90 s @115°C (1.3 µm) Vacuum contact, 2.5 s Develop with MF-319, 45 s, rinse and dry	Exposure time extended to ~5.5 s, no effect. Will extend development to 60 s. After rework, 2.5 s & 45 s used for E & D.
7	Implantation	(Surrey Uni Tool)	Implant Phosphorous for N++ region Energy 21.2 keV Dose 4x10 ¹⁴ cm ⁻² Beam current 100 µAcm ⁻² Angle of incidence 10°	24 day turnaround 7° angle
8	Strip Resist	Plasma Asher	10 - 15 min optionally with acetone/IPA	
9	Lithography with G3 (4/7)	EVG 620T(B)	Lithography of P++ region (DF+) Spin coat S1813 4000 rpm, 30s Pre-exposure bake 90 s @115°C (1.3 µm) Vacuum contact, 2.5 s Develop with MF-319, 45 s, rinse and dry	Exposure 2.5 s & Development 45 s used from before, but unchecked due to ellipsometer issues
10	Implantation	(Surrey Uni Tool)	Implant B for P++ region Energy 31 keV Dose 2x10 ¹⁵ cm ⁻² Beam current 100 µAcm ⁻² Angle of incidence 10°	7 day turnaround 7° angle
11	Strip Resist	Plasma Asher	10 - 15 min optionally with acetone/IPA	
12	HF 20:1	Wet Room	Strip SiO ₂ (time from thickness measurement)	Maybe 15 s for 25 nm? (18 s for 7:1, 6.5 s for 20:1 from .xml) [15+2 s used]
13	Oxide Deposition	Dielectric PECVD	14.06 - 14.74 s (40 nm target thickness (anneal cap))	39 s standard recipe used for 40-41 nm oxide
14	Anneal	Jipelec RTA (Non-metal)	1050°C for 1 min	Overpowering with training recipe (& carrier wafer warping) (Tool fix needed, some may be bowed?)
15	Implantation	(Surrey Uni Tool)	Implant B/He for Defects Energy 1.8/0.6 MeV Dose 5e12 cm ⁻² Angle of incidence 7°	Thomson2014 & Energy Sims 2 different implants for better use of wafers and time

Detailed process flow

16	HF 20:1	Wet Room	Strip SiO ₂ (time from thickness measurement)	<i>Dbl prev time managed ~20 nm for all bar 12 (12 not annealed) Repeat. 60+2 s needed for post-anneal strip. 15/16 needed 25 s longer.</i>
17	Oxide Deposition	Dielectric PECVD	35.15 - 36.85 s (140 nm target thickness)	<i>~2 min 8 s for 140 nm</i>
18	Lithography with G4 (5/7)	EVG 620T(B)	Lithography for Passivation layer (DF+) Spin coat S1813 4000 rpm, 30s Pre-exposure bake 90 s @115°C (1.3 µm) Vacuum contact, 2.5 s Develop with MF-319, 45 s, rinse and dry	<i>2.5 s & 45 s used as before, but near-empty resist reduce thickness and needed re-apply</i>
19	SiO ₂ Etch	RIE80+	Time from machine Oxide thickness – 20 nm	<i>Over-etched due to measuring StH & Rthick -> almost all removed</i>
20	HF 20:1	Wet Room	Strip SiO ₂ (time from thickness measurement)	<i>Remainder removed in varied amounts</i>
21	Resist strip	Plasma Asher	10 - 15 min optionally with acetone/IPA	
22	Lithography with G5 (6/7)	EVG 620T(B)	Lithography of metal (LF-) Spin coat AZ2070 4000 rpm, 30 s Pre-exposure bake 60 s @110°C (5.5 µm) Vacuum contact, 1.1 s Post-exposure bake 60 s @110°C Develop with AZ726, 120 s, rinse and dry	<i>[Min Distance 10 µm] Used: 4.5 min, 2.5 s, 1 min, 35 s, +extra dev if needed</i>
23	HF 20:1	Wet Room	5 s to remove native oxide	<i>20 s</i>
24	Metal Deposition	AJA Sputterer	Stack Al	<i>Not enough Al -> ~437 nm deposited</i>
25	Lift Off	Yellow Room	Sonic NMP bath	<i>Each ran for ~1 hr 30 min, +1 hr for 3. Still some Al remnants left.</i>
26	Metal Sintering at 300°C in metal RTA skipped			
27	Lithography with G6 (7/7)	EVG 620T(B)	Lithography of isolation trenches (DF+) Spin coat S1813 4000 rpm, 30s Pre-exposure bake 90 s @115°C (1.3 µm) Vacuum contact, 2.5 s Develop with MF-319, 45 s, rinse and dry	<i>Same as before, odd colour due to Al visible underneath</i>
28	ICP Etch	Plasmlab System 100	Etch 140 nm SiO ₂ and 1.5 µm of Si 1 step for oxide and Si, time from machine Determine time from etch depth measurements	<i>Over-etched (~3 µm) according to FIB images, resist etched through and majority of devices damaged.</i>
29	Strip resist	Plasma Asher	10 - 15 min optionally with acetone/IPA	<i>Metal Asher mostly ineffective, Sonic NMP used.</i>

S1813 applied after completion for dicing saw
Step number colouring groups match process flow in report

Detailed process flow continued

Appendix B: MSM

B.1 Process flow

Device Description: Waveguide-integrated metal-semiconductor-metal photodiode Starting Material: SOI (400 nm 2 µm SUB)/p-type Si wafer (2 wafers)				
	Process	Tool	Details	Comments & Outcome
0a	Cleaning	Wet Room	RCA-1 10 min, RCA-2 10 min	
0b	Oxidation	Furnace	Dry O ₂ ~3 hr 40 min / 3 hr 22 min Wet O ₂ ~10 min	9 hr 33 min
0c	HF 20:1	Wet Room	Strip SiO ₂ , approx. 5 min 30 +2 s for 160 nm	7 min +2 s
1	Lithography (optical)		Lithography of alignment marks (DF+) Spin coat S1813 4000 rpm, 30 s (1.2 - 1.3 µm) Pre-exposure bake 90 s @115°C Vacuum contact, 2.5 s (filter out) Develop with MF-319, 45 s, rinse & dry	Switched to old group AM mask
2	Plasma Etch	RIE 80+	Etch 340 nm Si and 1 µm SiO ₂	~40 s poly-Si & 44 min SiO ₂ (old), Use 45 s & 40 - 45 min new
3	Strip resist	Plasma Asher	10 – 15 min	
4	Lithography (GDS 3) (MF 0.6/11.2)	E-beam	Lithography of waveguide core ZEP520A (+ive)(fine beam) (450 nm) Ali's spin & exposure routine (leave in oven)	30 min dehydrate ZEP450nm 2500rpm, 3 min @180°C ZED-N50 2 min 15 s, IPA 30 s, DI 30 – 60 s ??
5	ICP Etch	Plasmlab System 100	Etch 240 nm Si (Si thickness – 100 nm)	W6-4 Si all gone W6-5 ~68 nm remains
6	Cleaning	Wet Room	RCA-1 10 min, RCA-2 10 min	RCA-1 tank out of order
7	Strip resist	Plasma Asher	10 – 15 min + FNA/acetone/IPA clean	
8	Oxide Deposition	Dielectric PECVD	Deposit 600 nm oxide	
9	Lithography (GDS 53) (MF -/0.2)	E-beam	Lithography of defect window ZEP520A (+ive)(fine beam) (700 nm) Ali's spin & exposure routine (leave in oven)	900 nm was too thick for smaller windows, so passivation oxide thinned down to 600 nm to use 700 nm ZEP. 2 nd run showed cracking => Harold suggested acetone/IPA clean then HMDS primer layer before ZEP
10	SiO ₂ Etch	RIE80+	Etch 500 nm	
11	Strip resist	Plasma Asher	10 – 15 min + FNA/acetone/IPA clean	
12	Lithography (GDS 12) (MF 1.4/8)	E-beam	Lithography of passivation/vias ZEP520A (+ive)(coarse beam) (450 nm) 30 min oven, 10 min HMDS Spin with "ZEP400RC", then 3 min @180°C Spin "ESPACER3", no bake Ali for exposure Develop in UPW, 2 min 15 s ZED, 30 s IPA	First attempt used ICP => V poor selectivity, partial etch abandoned
13	SiO ₂ Etch	RIE80+	Etch all – 20 nm SiO ₂ (580 nm) (Alt. all SiO ₂ & Si, skip 13)	
14	HF 20:1	Wet Room	Strip SiO ₂ (approx. 40 + 2 s for 20 nm)	
15	Strip resist	Plasma Asher	10 – 15 min	
16	Lithography (GDS 13) (MF 1/100)	E-beam	Lithography of metal contacts (LF -ive) 30 – 60 min oven, cool Spin AZ2070 6000 rpm "AZ2070" (3400 nm) Bake 60 s @110°C, cool Ali for exposure PEB 60 s @110°C Develop in AZ726 for 4 min, UPW, dry	Was MMA/PMMA first time, but was not thick enough for sufficient lift-off Switched to AZ2070 for devices wafer
	Dice		Manually scribe chips into 9 chips (very deep scribe marks needed)	One wafer as test chips, other as device chips
17	HF 20:1	Wet Room	Strip native SiO ₂ , 20 s	
18a	Metal Deposition	AJA Sputterer	Nothing or TiN or Ti/Pt each approx. 20 nm	Au used instead as Pt target expended
18b	Metal Deposition	BAK600	Al or Au 300 nm	Only Al as Au unavailable due to new precious metal system
19	Lift Off	Yellow Room	NMP overnight then morning sonication	
20	Implantation	Surrey Uni Tool	High-Energy Ion Implantation 300 keV Ar ⁺ at 1x10 ¹³ cm ⁻²	

(MF A/B) = min feature between/across shapes in microns
Step number colouring groups match process flow in report

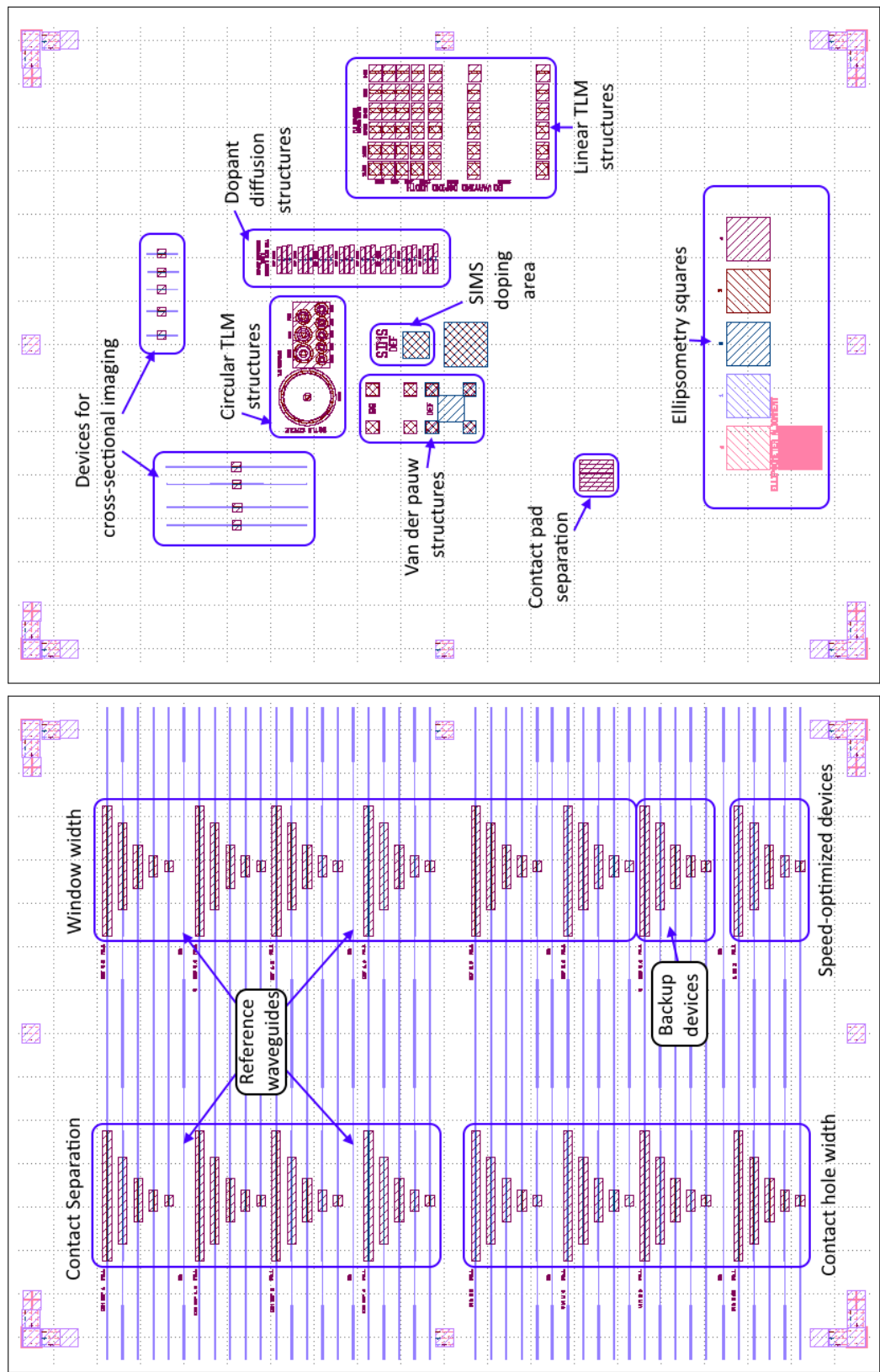
Detailed process flow

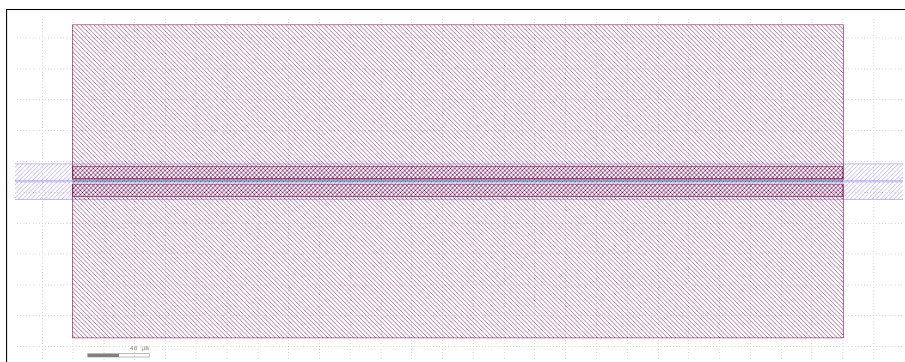
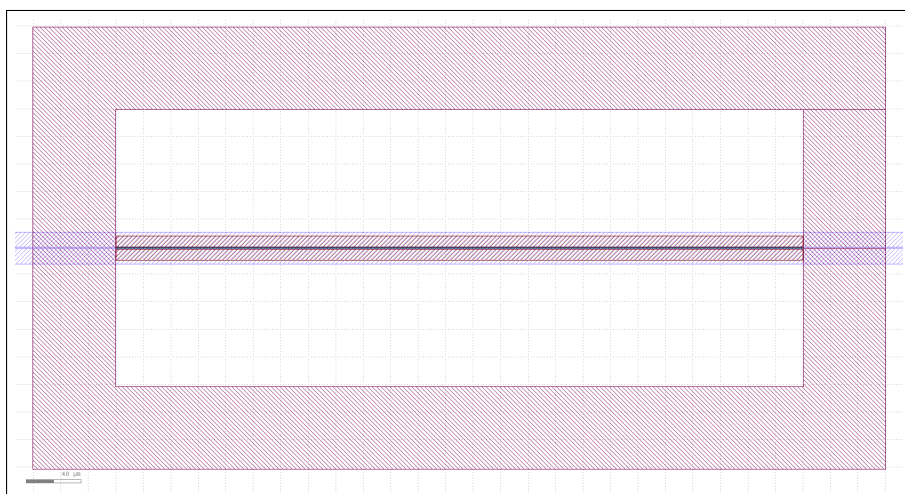
B.2 Implantation data

Results from SRIM simulations using implantation conditions and waveguiding structures as reported in literature, along with the published experimental results. * values are calculated from provided results or measured from graphs.

Source Table 7.1 ref.	[24] F	[5] J	[122] M	[102] H	[56] L	[49] K	[120] I	[34] C	[121] E	[122] B	[81] D	[52] G	[82] A
Implantation species	B	B	B	B	B	Si	Si	Si	Si	Si	Ar	Ar	Ar
Implantation energy [keV]	30	60	350	350	500	190	190	195	350	400	100	240	260
Best implantation dose [cm ⁻²]	1e13	1e13	1e13	1e13	5e12	1e13	1e13	1e13	1e13	1e13	1e12	7e12	7e12
Top SiO ₂ cladding thickness [nm]	0	0	300	200	1000	90	0	90	300	300	0	90	90
Si Waveguide core thickness [nm]	220	220	220	220	220	220	220	220	220	220	200	220	220
Bottom cladding	BOX	BOX	BOX	BOX	BOX	BOX	BOX	BOX	BOX	BOX	BOX	BOX	BOX
Vacancies per ion	357.3	485.2	768.3	767.3	856.5	2134.5	2050.6	2157.7	2848.8	2931.7	1709.5	2921.9	3052.4
Projected ion range [nm]	114.1	203.8	937.5	937.5	1200	264.7	260.0	271.4	487.6	555.7	109.4	253.8	272.6
Energy lost to vacancies from ions [%]	0.36	0.23	0.07	0.07	0.05	0.15	0.15	0.15	0.12	0.11	0.19	0.14	0.13
Energy lost to vacancies from recoils [%]	2.37	1.63	0.45	0.45	0.36	2.51	2.35	2.48	1.82	1.65	3.74	2.75	2.66
Max vacancy concentration [vac/Ang-ion]	0.3119	0.2737	0.1350	0.1618	0.1615	0.9102	0.9341	0.9555	0.8105	0.7706	1.1082	1.3038	1.3140
Vacancy range / Waveguide height [%]	61.82	22.73	-179.55	-225.00	9.09	45.45	15.91	45.45	45.45	13.64	61.00	56.82	40.91
Ion range / Waveguide height [%]	48.14	7.36	-189.77	-235.23	9.09	20.59	-18.18	17.55	14.73	-16.23	45.30	25.55	17.00
Average offset from waveguide centre [%]	6.8	35.0	234.7	280.1	40.9	17.0	51.1	18.5	19.9	51.3	7.8	15.6	21.0
Average vacancies in core [vac.cm ⁻³]	1.63e20	1.94e20	5.81e19	4.80e19	6.37e19	7.59e20	6.57e20	7.59e20	7.27e20	6.47e20	6.67e19	7.34e20	7.46e20
Average ions in core [ion.cm ⁻³]	4.52e17	2.50e17	6.31e15	3.42e15	8.08e16	3.06e17	1.19e17	2.93e17	2.28e17	1.27e17	5.29e16	2.40e17	2.04e17
Vacancies at half ion range [vac/Ang-ion]	0.2555	0.2199	0.0656	0.0421	0.0399	0.7584	0.7462	0.7074	0.3557	0.3437	0.9847	1.0660	0.9805
Adjusted ion dose [cm ⁻²]	2.56e12	2.20e12	6.56e11	4.21e11	2.00e11	7.58e12	7.46e12	7.07e12	3.56e12	3.44e12	9.85e11	7.46e12	6.86e12
V-type defect concn. at half ion range [cm ⁻²]	1.83e18	1.66e18	7.77e17	5.88e17	3.67e17	3.63e18	3.59e18	3.48e18	2.25e18	2.21e18	1.00e18	3.59e18	3.41e18
Maximum dose [ions.cm ⁻²]	1.60e14	1.83e14	3.23e14	3.09e14	3.10e14	5.49e13	5.35e13	5.23e13	6.17e13	6.49e13	4.51e13	3.83e13	3.81e13
Measurement wavelength [μm]	1.46-1.58	2.02	1.55	1.55	1.55	2.2	1.54	1.9	1.55	1.55	1.56	2.2	1.55
Post-implantation anneal [min, °C]	none	none	5, 350	none	5, 200	10, 350	5, 325	none	5, 350	5, 300	30, 300	10, 300	10, 350
Responsivity [mA.W ⁻¹] (V)	35.5* (5)	6.03* (5)	13 (5)	130 (10)	2.17* (5)	5.2 (5)	6.2 (5)	101.5 (5)	53.6 (5)	160 (20)	65 (5)	21.3 (5)	281 (5)
EQE [%] (V)	2.83* (5)	0.37* (5)	2.01* (5)	10 (10)	0.17* (5)	4.5 (5)	0.49* (5)	8.08* (5)	4.27* (5)	11.8* (20)	5.63* (5)	3.7 (IQE)	-
Dark current [nA] (V)	17 (10)	49.6 (5)	-	0.2 (5)	16 (5)	12.7 (5)	1.1 (20)	360 (5)	189	-	0.64 (5)	5 (5)	-
Resonant device	yes	no	yes	yes	no	no	yes	no	yes	yes	no	no	no

B.3 Mask layout



Close up of 500 μm detector, original designClose up of 500 μm detector, final design

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